POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 1041

Elementary Concepts of Electronics

Complete Theory, Practical and Examination Handbook · Semester I · Electronic Circuit Lab Handbook

Practicum | 3:0:3:0 | 90 contact hours | 4.5 credits | REV2026

Course outcomes

1. **CO1:** Basic electronics, electrical quantities, signals and instruments.
2. **CO2:** Passive components and basic circuit construction/analysis.
3. **CO3:** Semiconductor theory and basic devices.
4. **CO4:** PCB technology and soldering practice.

SEMESTER
I

COURSE TYPE
Practicum

THEORY
3 h/week

PRACTICAL
3 h/week

OFFICIAL COURSE INFORMATION

Course Information and Learning Plan

Course identity

Curriculum	**Diploma Curriculum Revision 2026**
Course code	**1041**
Course title	**Elementary Concepts of Electronics**
Semester	**I**
Type	**Practicum**
Teaching scheme	**3:0:3:0**
Contact hours	**90**
Credits	**4.5**

How to use this handbook

1. Begin with the roadmap and course outcomes.
2. Study the theory chapter before the related experiment.
3. Use diagrams and animations to verify the mental model.
4. Work every solved example without looking at the final answer first.
5. Complete record tables in the laboratory, not from memory.
6. Use the question bank only after understanding the chapter.

Official introduction

This course provides foundational knowledge of electronics required for diploma engineering programmes. It covers basic electrical quantities, electronic signals and measuring instruments; introduces passive and active components and semiconductor theory; and develops awareness of electrical safety, electrostatic discharge, printed circuit boards and soldering fundamentals. It prepares learners for advanced electronics subjects and practical applications.

Course objectives

1. Introduce fundamental electronics, electrical quantities, signal types and electronic measuring instruments.
2. Develop understanding of passive components and their configurations in basic circuits.
3. Explain semiconductor devices and their applications.
4. Introduce PCB technology, basic fabrication and soldering techniques.

Prerequisites

Basic Mathematics and Basic Physics from secondary education.

Course outcomes		
CO	**Course outcome**	**Bloom's level**
CO1	Demonstrate basic concepts of electronics, electrical quantities, signal types and electronic measuring instruments in simple electronic systems.	Apply
CO2	Construct and analyse basic electronic circuits using passive electronic components and their configurations.	Apply
CO3	Apply semiconductor theory to demonstrate the operation, characteristics and applications of basic semiconductor devices.	Apply
CO4	Demonstrate Printed Circuit Boards and apply soldering techniques.	Apply

Applicable programmes

- Biomedical Engineering
- Electronics and Communication
- Electronics Engineering
- Electronics Engineering (Embedded Systems)
- Electronics and Computer Engineering
- Integrated Circuit Design & Fabrication
- Instrumentation Engineering
- Micro Electronics

Official major-topic allocation				
Module	**Title**	**Major topics**	**Theory hours**	**Practical hours**
I	Introduction to Electronics, Signals and Measurements	Introduction, applications, electrical quantities, signals, measuring instruments	11	6
II	Passive Electronic Components	Resistors, configurations, capacitors, inductors, transformers, voltage divider, RC timing	12	9
III	Active Electronic Components	Band theory, semiconductors, PN diode, Zener diode, LED, photodiode	12	9
IV	PCB and Soldering	PCB types/materials/fabrication/layout and soldering practice	10	12

> **Official-table clarification:** The cover summary states CIE 40 and ESE 60, while the detailed teaching and examination table separates theory as 20 + 60 and practical as 20 + 40, giving 140 total. Both values are reproduced in their original contexts; this handbook does not invent a replacement figure. The major-topic hour rows also do not alone explain every contact hour because practical examinations and scheduling entries appear separately in the detailed syllabus.

Detailed theory and practical examination split		
Area	**Assessment**	**Marks**
Theory	Continuous Internal Evaluation	20
Theory	End Semester Examination	60
Theory	Subtotal	80
Practical	Continuous Internal Evaluation	20
Practical	End Semester Examination	40
Practical	Subtotal	60
Overall	Total	140

Major equipment and instruments		
Equipment	**Specification**	**Quantity for 30 students**
Digital Storage Oscilloscope	3 MHz, dual channel	15
Signal / Function Generator	3 MHz minimum	15
Power Supply Unit	30 V, dual	15
Multimeter	DMM with capacitor measurement mode	15

Learning resources overview

Textbooks and references listed in the syllabus		
Title	**Author**	**Publication**
Electronic Inventions and Discoveries, Fourth Edition	G. W. A. Dummer	Pergamon
Grob's Basic Electronics	Mitchel E. Schultz	McGraw-Hill
Textbook of Electrical Technology: Part 1	B. L. Theraja and A. K. Theraja	S. Chand
Electronic Devices and Circuit Theory	Robert Boylestad and Louis Nashelsky	PHI
Practical Electronics for Inventors	Paul Scherz and Simon Monk	McGraw-Hill
A Text Book of Applied Electronics	R. S. Sedha	S. Chand
Basic Electronics and Linear Circuits	N. N. Bhargava, Kulshreshtha and S. C. Gupta	McGraw-Hill
Electronic Devices and Circuits	David A. Bell	PHI

INTERACTIVE ROADMAP

From Electrical Quantities to Reliable Hardware

MODULE 1 · CO1 · THEORY 11 H · PRACTICAL 6 H

Introduction to Electronics, Signals and Measurements

AC/DC, signal, frequency, instrument connection [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] measurement [illegible].

Learning target: Explain the theory, interpret diagrams and measurements, apply formulae, perform the listed laboratory work, and answer one-, three- and seven-mark questions.

Core theory chapters (39)

1. Definition and purpose of electronics

Electronics is the engineering discipline that controls electron or charge-carrier flow to sense, process, store, communicate, and act on information.

How it works

An electronic system normally has an input, a processing stage, a power source, and an output. Unlike basic power wiring, electronics often handles low-power signals as well as power-control commands.

Engineering practice

Trace any familiar system—mobile charger, escalator controller, medical monitor—from source to input, processing, output, and feedback.

Common mistake / safety: Do not confuse electronics with electrical power engineering; the two overlap, but the design objectives and signal levels differ.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes [illegible] [illegible]. Mobile carriers-[illegible] fixed dopant ions-[illegible] [illegible] [illegible].

2. Historical evolution of electronics

Electronics progressed from vacuum tubes to discrete semiconductor devices, integrated circuits, microprocessors, embedded systems, and highly integrated system-on-chip devices.

How it works

Each generation reduced size and power while increasing speed, reliability, and functional density.

Engineering practice

Relate the progression to serviceability: discrete boards are easier to repair at component level; highly integrated boards are often replaced as assemblies.

Common mistake / safety: A shorter timeline is not automatically more accurate; preserve the distinction between invention, commercial adoption, and mass production.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□ □□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□ □□□□□□□□□□□□□.

3. Applications of electronics

Electronics enables sensing, amplification, conversion, control, computation, communication, display, and protection.

How it works

Most modern products combine analog sensing, digital processing, power conversion, and human-machine interfaces.

Engineering practice

Classify applications by function rather than by product name.

Common mistake / safety: Do not assume every electronic system is digital; sensors and power stages are often analog.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□ □□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□ □□□□□□□□□□□□□.

4. Communication electronics

Communication systems convert information into electrical or electromagnetic signals, transmit it through a medium, and recover it at a receiver.

How it works

Typical blocks are source, encoder/modulator, channel, receiver/demodulator, decoder, and destination.

Engineering practice

Identify microphone, transmitter, antenna/cable, receiver, and loudspeaker blocks in a public-address system.

Common mistake / safety: Signal quality depends on bandwidth, noise, distortion, and interference—not only on signal strength.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□ □□□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□ □□□□□□□□□□□□□.

5. Industrial automation and control

Industrial electronics acquires sensor data, executes control logic, and drives actuators through PLCs, controllers, drives, relays, and safety circuits.

How it works

A closed-loop system compares feedback with a setpoint and corrects the error.

Engineering practice

In an escalator, speed sensing, safety-chain monitoring, VFD control, braking, and annunciation form coordinated electronic subsystems.

Common mistake / safety: Never treat a safety circuit as ordinary control logic; safety architecture requires validated fail-safe behaviour.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

6. Medical and healthcare electronics

Medical electronics measures biological signals, supports diagnosis, monitors patients, and controls therapeutic equipment.

How it works

Signals such as ECG are small and noise-sensitive, demanding isolation, shielding, filtering, and patient protection.

Engineering practice

Study why isolation and leakage-current limits are critical in patient-connected equipment.

Common mistake / safety: Never connect general laboratory instruments directly to a patient or body.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□ □□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□ □□□□□□□□□□□□.

7. Instrumentation and measurement

Instrumentation converts a physical quantity into a usable signal, conditions it, displays or records it, and may feed it to a controller.

How it works

Accuracy, precision, resolution, range, sensitivity, loading, and calibration are separate performance terms.

Engineering practice

Before measuring, select the correct function, range, terminals, reference point, and expected magnitude.

Common mistake / safety: A digital display does not guarantee correctness; incorrect setup can produce a precise-looking wrong value.

Malayalam explanation: Measurement □□□□□□□□□□□□ □□□□□□ function, range, terminals, probe reference, expected value □□□□□□ verify □□□□□□□. Display □□□□□□□□□ □□□□□□ □□□□□□ measurement □□□□□□□□□□□□□.

8. Automobile electronics

Vehicle electronics covers engine control, battery management, braking, lighting, safety, infotainment, diagnostics, and electric-drive systems.

How it works

Sensors and electronic control units exchange data over robust communication networks.

Engineering practice

Inspect a vehicle fuse-box diagram to distinguish power distribution from low-current control.

Common mistake / safety: Automotive transients can be severe; laboratory assumptions about supply quality may not apply.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□ □□□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□ □□□□□□□□□□□□□□.

9. Consumer electronics

Consumer products integrate power supplies, user interfaces, processors, memory, sensors, audio/video stages, and communication modules.

How it works

Cost, compactness, electromagnetic compatibility, safety, and user experience drive design decisions.

Engineering practice

Block-diagram a television, induction cooker, or washing machine.

Common mistake / safety: Sealed products may retain hazardous voltage; observation does not imply permission to dismantle.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□ □□□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□ □□□□□□□□□□□□□□.

10. Defence and security electronics

Defence and security systems use sensing, communication, navigation, surveillance, identification, and resilient control.

How it works

Reliability, redundancy, cybersecurity, environmental tolerance, and secure communications are major concerns.

Engineering practice

Compare normal commercial requirements with mission-critical requirements.

Common mistake / safety: Avoid discussing or attempting restricted weapon-control details.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□ □□□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□ □□□□□□□□□□□□□□.

11. Computer and IT systems

Computers use electronic switching devices to represent, process, store, and communicate binary information.

How it works

Processors execute instructions, memory stores data, interfaces exchange signals, and power-management circuits maintain stable operation.

Engineering practice

Identify analog and digital sections on a computer motherboard.

Common mistake / safety: A logic '1' is a voltage range, not a universal fixed voltage.

Malayalam explanation: □□□ electronic system-□□ Power Supply → Input/Sensor → Processing/Control → Output/Actuator □□□□ block flow □□□ □□□□□. Safety□□□ signal path-□□ □□□□□□□□□ □□□□□□□□□□□□.

12. Renewable-energy systems

Renewable-energy electronics converts, controls, protects, and monitors energy from solar, wind, storage, and grid interfaces.

How it works

Power electronics performs DC–DC conversion, inversion, maximum-power-point tracking, charging, and grid synchronization.

Engineering practice

Follow the energy path in a small solar system: panel → controller → battery → inverter/load.

Common mistake / safety: High DC voltage and stored battery energy remain hazardous even when the grid is disconnected.

Malayalam explanation: Power energy conversion rate □□□; Energy = Power × Time. W, Wh, kWh, J □□□□□ units □□□□□□ mix □□□□□□□□ conversion □□□□□□□.

13. Electric charge

Electric charge is a fundamental property of matter measured in coulombs (C). Electrons carry negative charge and protons carry positive charge.

How it works

One coulomb corresponds to approximately 6.24×10^{18} elementary charges.

Engineering practice

Use charge as the starting point for understanding current: current is rate of charge flow.

Common mistake / safety: Conventional current direction is opposite to electron drift direction in metallic conductors.

Malayalam explanation: Electric charge □□□□□□ □□□□□□□□□□□ □□□□□□□□ electrical property □□□. Charge □□□□□□□□ □□□□□□□□□ Current; □□□□□□ Q, I, t □□□□□ quantities □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□.

14. Electric current

Electric current is the rate of charge flow through a cross-section.

How it works

$I = Q/t$, where I is amperes, Q is coulombs, and t is seconds.

Engineering practice

Measure current by opening the circuit and inserting an ammeter in series.

Common mistake / safety: Never connect an ammeter directly across a voltage source; its low internal resistance can cause destructive current.

Malayalam explanation: Current □□□□□□ charge flow rate □□□. Ammeter □□□□□□□□ circuit-□ series □□□ connect □□□□□□□; source-□□ across □□□ connect □□□□□□□□□.

15. Voltage

Voltage is electric potential difference: energy transferred per unit charge.

How it works

$V = W/Q$, where voltage is in volts, energy in joules, and charge in coulombs.

Engineering practice

Measure voltage between two points with a voltmeter connected in parallel.

Common mistake / safety: Voltage is always relative to a reference point; 'voltage at a point' is incomplete without the reference.

Malayalam explanation: Voltage □□□□□ points □□□□□□□□□□ potential difference □□□. Voltmeter parallel □□□ connect □□□□□□□□□□ reference point □□□□□□□□□□□□□□□□ □□□□.

16. Resistance

Resistance opposes current and converts electrical energy, commonly to heat.

How it works

For an ohmic conductor at constant conditions, $R = V/I$.

Engineering practice

De-energize the circuit before measuring resistance with a multimeter.

Common mistake / safety: In-circuit resistance can be distorted by parallel paths.

Malayalam explanation: Resistance current flow-□□ oppose □□□□□□□□□□. Value, tolerance, power rating □□□□□□ □□□□□□□□□ □□□□□□□□□□□□□ resistor □□□□□□□□□□□□□□□□□□□.

17. Electrical power

Electrical power is the rate of electrical energy conversion.

How it works

$P = VI$; for resistive circuits $P = I^2R = V^2/R$.

Engineering practice

Check component power ratings with margin, especially resistors and semiconductors.

Common mistake / safety: Using the correct resistance value with an inadequate power rating can still cause failure.

Malayalam explanation: Power energy conversion rate □□□; Energy = Power × Time. W, Wh, kWh, J □□□□□ units □□□□□□ mix □□□□□□□□ conversion □□□□□□□.

18. Electrical energy

Electrical energy equals power multiplied by time.

How it works

E = Pt. Joule is the SI unit; watt-hour and kilowatt-hour are widely used practical units.

Engineering practice

Convert time units before substitution and state the final unit.

Common mistake / safety: Do not mix watts and kilowatts or seconds and hours without conversion.

Malayalam explanation: Power energy conversion rate □□□; Energy = Power × Time. W, Wh, kWh, J □□□□□ units □□□□□□ mix □□□□□□□□ conversion □□□□□□□.

19. AC quantities

Alternating quantities change magnitude and direction periodically.

How it works

A sinusoidal AC voltage can be represented by $v = V_m \sin(\omega t + \varphi)$.

Engineering practice

Observe polarity and frequency using a DSO.

Common mistake / safety: Mains AC is hazardous; use approved isolated training sources and probes.

Malayalam explanation: AC signal □□□□ □□□□□□□□□□ magnitude-□□□ direction-□□□ □□□□□□□□. Frequency, Period, Amplitude, Phase □□□□□□ waveform □□□□□□□□□ □□□□□□□□□□□□ □□□□□□ terms □□□.

20. DC quantities

Direct quantities maintain one direction, though their magnitude may contain ripple.

How it works

An ideal DC source is constant with time; practical supplies may include ripple, noise, or transient variation.

Engineering practice

Measure battery and regulated-supply outputs using the correct DC range.

Common mistake / safety: Reversed polarity can damage polarized components.

Malayalam explanation: DC-□□□□□□ fixed polarity/direction □□□□□. Practical DC supply-□ ripple □□□□□□□□□□ noise □□□□□□□□; meter-□□□□ DC mode □□□□□□□□□□□□□□□.

21. AC versus DC

AC reverses direction periodically; DC has a fixed direction. Their generation, transmission, conversion, measurement, and applications differ.

How it works

AC is conveniently transformed; DC is native to batteries and most electronic circuits.

Engineering practice

Identify whether the instrument is set to AC or DC before taking a reading.

Common mistake / safety: The same numeric voltage does not imply equal heating or peak stress unless waveform and RMS meaning are known.

Malayalam explanation: AC signal □□□□ □□□□□□□□□□ magnitude-□□□ direction-□□□ □□□□□□□□. Frequency, Period, Amplitude, Phase □□□□□□ waveform □□□□□□□□□ □□□□□□□□□□□□□ □□□□□□ terms □□□.

22. Meaning of a signal

A signal is a measurable variation that carries information about a physical condition, command, or data.

How it works

Signals may be electrical, optical, acoustic, or mechanical; electronics converts them into suitable electrical forms.

Engineering practice

State the signal variable, amplitude range, time behaviour, bandwidth, and reference.

Common mistake / safety: Noise is not always random; interference can be periodic and repeatable.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

23. Analog signals

An analog signal can take continuously varying values within a range.

How it works

Analog circuits preserve amplitude information but are sensitive to noise, offset, and component tolerances.

Engineering practice

Temperature-sensor voltage and microphone output are common examples.

Common mistake / safety: Sampling an analog signal does not make the original physical quantity digital.

Malayalam explanation: Analog signal continuous values □□□□□□□□□□□□□□□□. Noise, offset, loading □□□□□□ measurement-□□ □□□□□□□□□□.

24. Digital signals

A digital signal uses discrete states, commonly binary LOW and HIGH levels.

How it works

Noise margins allow reliable state interpretation within defined voltage ranges.

Engineering practice

Use a DSO or logic probe appropriate to the logic family.

Common mistake / safety: Do not assume 5 V logic; modern systems commonly use lower levels.

Malayalam explanation: Digital signal discrete LOW/HIGH ranges □□□□□□□□□□□□□□□. Logic HIGH □□□□□□ □□□□□ circuits-□□□ □□□ voltage □□□□; logic family □□□□□□□□□□ range □□□□□.

25. Sinusoidal AC waveform

A sine wave is a smooth periodic waveform defined by amplitude, frequency, phase, and offset.

How it works

$v(t) = V_m \sin(\omega t + \varphi)$.

Engineering practice

Measure peak-to-peak voltage and period on a DSO, then calculate amplitude and frequency.

Common mistake / safety: Peak, peak-to-peak, average, and RMS are different quantities.

Malayalam explanation: AC signal □□□□ □□□□□□□□□□ magnitude-□□□ direction-□□□ □□□□□□□□□. Frequency, Period, Amplitude, Phase □□□□□□ waveform □□□□□□□□□ □□□□□□□□□□□□□ □□□□□□ terms □□□.

26. Cycle

A cycle is one complete repetition of a periodic waveform.

How it works

Equivalent points one cycle apart have the same magnitude and direction of change.

Engineering practice

Use repeated landmarks such as positive-going zero crossings.

Common mistake / safety: Do not measure a half-cycle and report it as a full period.

Malayalam explanation: □ concept □□□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□□ □□□□□□□□□□□□□□.

27. Time period

Time period T is the time required for one complete cycle.

How it works

$T = 1/f$ and is measured in seconds.

Engineering practice

Use oscilloscope time divisions multiplied by time/div.

Common mistake / safety: Convert milliseconds or microseconds to seconds before calculating frequency.

Malayalam explanation: AC signal □□□□ □□□□□□□□□□ magnitude-□□□ direction-□□□ □□□□□□□□. Frequency, Period, Amplitude, Phase □□□□□□ waveform □□□□□□□□□ □□□□□□□□□□□□□ □□□□□□ terms □□□.

28. Frequency

Frequency f is the number of cycles per second.

How it works

$f = 1/T$ and is measured in hertz (Hz).

Engineering practice

Compare function-generator display with DSO measurement and calculate percentage error.

Common mistake / safety: Trigger instability can make a stable signal appear to change frequency.

Malayalam explanation: AC signal □□□□ □□□□□□□□□□ magnitude-□□□ direction-□□□ □□□□□□□□. Frequency, Period, Amplitude, Phase □□□□□□ waveform □□□□□□□□□ □□□□□□□□□□□□□ □□□□□□ terms □□□.

29. Angular frequency

Angular frequency expresses the rate of phase change in radians per second.

How it works

$\omega = 2\pi f$.

Engineering practice

Use angular frequency when writing sinusoidal equations or analysing reactive circuits.

Common mistake / safety: Angular frequency is not the same as ordinary frequency.

Malayalam explanation: AC signal □□□□ □□□□□□□□□□ magnitude-□□□ direction-□□□ □□□□□□□□. Frequency, Period, Amplitude, Phase □□□□□□ waveform □□□□□□□□□ □□□□□□□□□□□□□ □□□□□□ terms □□□.

30. Amplitude

Amplitude describes the maximum excursion of a waveform from its reference level.

How it works

For a centered sine wave, peak-to-peak value Vpp = 2Vm.

Engineering practice

Confirm whether an instrument displays peak, peak-to-peak, or RMS.

Common mistake / safety: Do not use Vpp directly as Vm.

Malayalam explanation: AC signal □□□□ □□□□□□□□□□ magnitude-□□□ direction-□□□ □□□□□□□□. Frequency, Period, Amplitude, Phase □□□□□□ waveform □□□□□□□□□ □□□□□□□□□□□□ □□□□□□ terms □□□.

31. Instantaneous value

The instantaneous value is the value of a time-varying quantity at a specified instant.

How it works

For a sine wave, $v = V_m \sin(\omega t + \varphi)$.

Engineering practice

Mark the time point and evaluate the phase angle.

Common mistake / safety: An instantaneous value can be zero even while the source is active.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□□□.

32. Peak value

Peak value is the maximum magnitude reached relative to the zero or reference level.

How it works

For an ideal sine wave, $V_{rms} = V_m/\sqrt{2}$.

Engineering practice

Use peak value when checking semiconductor and insulation stress.

Common mistake / safety: A 230 V RMS mains supply has a much higher peak voltage.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

33. Electronic measuring instruments

Measuring instruments extend human observation into voltage, current, resistance, time, frequency, and waveform domains.

How it works

The instrument must have suitable range, bandwidth, category rating, input impedance, and calibration.

Engineering practice

Inspect leads and settings before connecting.

Common mistake / safety: Measurement safety is determined by the entire setup, not just the instrument body.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□ □□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□ □□□□□□□□□□□□□.

34. Voltmeter

A voltmeter measures potential difference and is connected in parallel.

How it works

A high input resistance minimizes loading.

Engineering practice

Connect the common lead to the reference and the voltage lead to the test point.

Common mistake / safety: Never use a damaged probe or exceed the input rating.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□

□□□□□□□□□□□□□.

35. Ammeter

An ammeter measures current and is connected in series.

How it works

Its internal shunt resistance is low, so incorrect parallel connection is dangerous.

Engineering practice

Start with the highest suitable current range and correct terminal.

Common mistake / safety: After current measurement, return the lead to the voltage terminal to prevent future mistakes.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□□□□.

36. Digital multimeter

A DMM combines voltage, current, resistance, continuity, diode, and sometimes capacitance or frequency functions.

How it works

The selector, input jacks, display symbols, and protection fuses must all be understood.

Engineering practice

Use diode mode to identify PN-junction polarity.

Common mistake / safety: Continuity mode is not a substitute for measuring insulation resistance.

Malayalam explanation: Digital signal discrete LOW/HIGH ranges □□□□□□□□□□□□□□. Logic HIGH □□□□□□ □□□□□ circuits-□□□ □□□ voltage □□□□; logic family □□□□□□□□□ range □□□□□.

37. DC power supply

A laboratory DC supply provides adjustable regulated voltage and current limiting.

How it works

Constant-voltage mode controls voltage; constant-current mode limits current when the load demand exceeds the set limit.

Engineering practice

Set voltage and current limit before connecting the circuit.

Common mistake / safety: Current limiting protects the circuit but does not make unsafe wiring acceptable.

Malayalam explanation: Power energy conversion rate □□□; Energy = Power × Time. W, Wh, kWh, J □□□□□ units □□□□□□ mix □□□□□□□□ conversion □□□□□□□.

38. Function generator

A function generator produces test waveforms such as sine, square, and triangle waves with adjustable frequency, amplitude, and offset.

How it works

Its output amplitude can depend on load termination.

Engineering practice

Verify the actual output on a DSO rather than relying only on the panel display.

Common mistake / safety: Excessive offset or amplitude can damage the circuit under test.

Malayalam explanation: Measurement □□□□□□□□□□□□□ □□□□□□□ function, range, terminals, probe reference, expected value □□□□□□ verify □□□□□□□. Display □□□□□□□□□□□ □□□□□□ □□□□□□ measurement □□□□□□□□□□□□□□.

39. Digital Storage Oscilloscope

A DSO displays voltage versus time and can measure amplitude, period, frequency, phase, rise time, and transients.

How it works

Vertical scale, horizontal scale, trigger, coupling, probe ratio, and reference connection determine the displayed result.

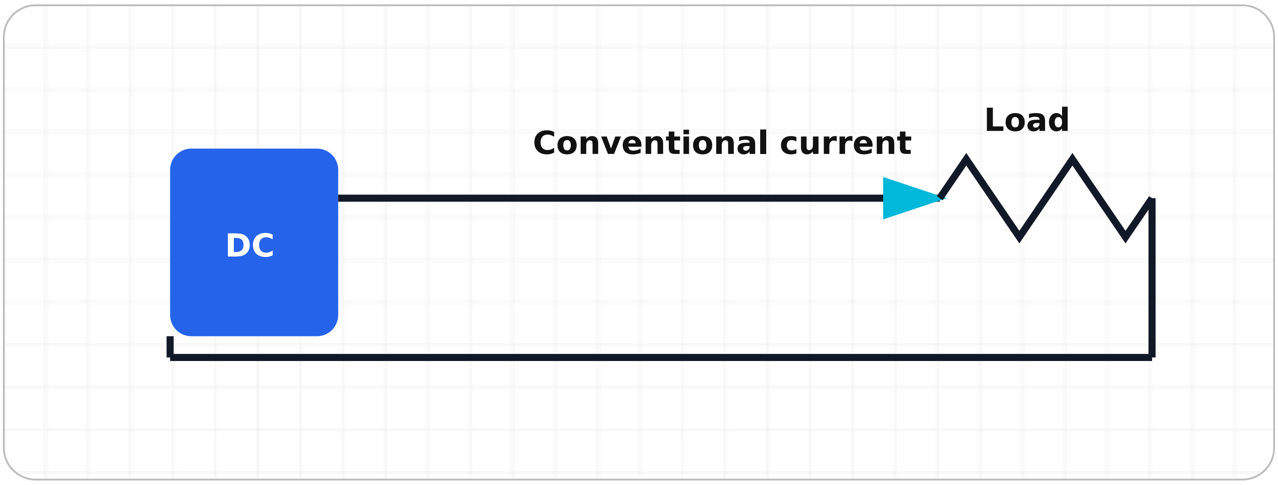
Engineering practice

Compensate the probe and confirm the probe-ratio setting.

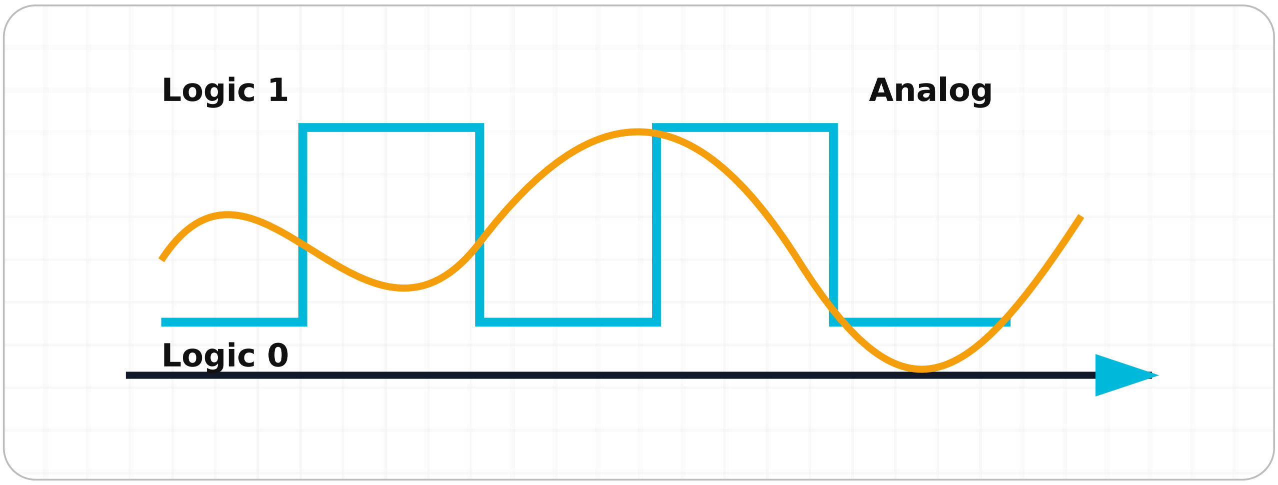
Common mistake / safety: Connecting an earth-referenced probe ground to a live non-isolated point can create a short circuit.

Malayalam explanation: Digital signal discrete LOW/HIGH ranges □□□□□□□□□□□□□□□. Logic HIGH □□□□□□□ □□□□□□ circuits-□□□ □□□ voltage □□□□; logic family □□□□□□□□□□□□ range □□□□□□.

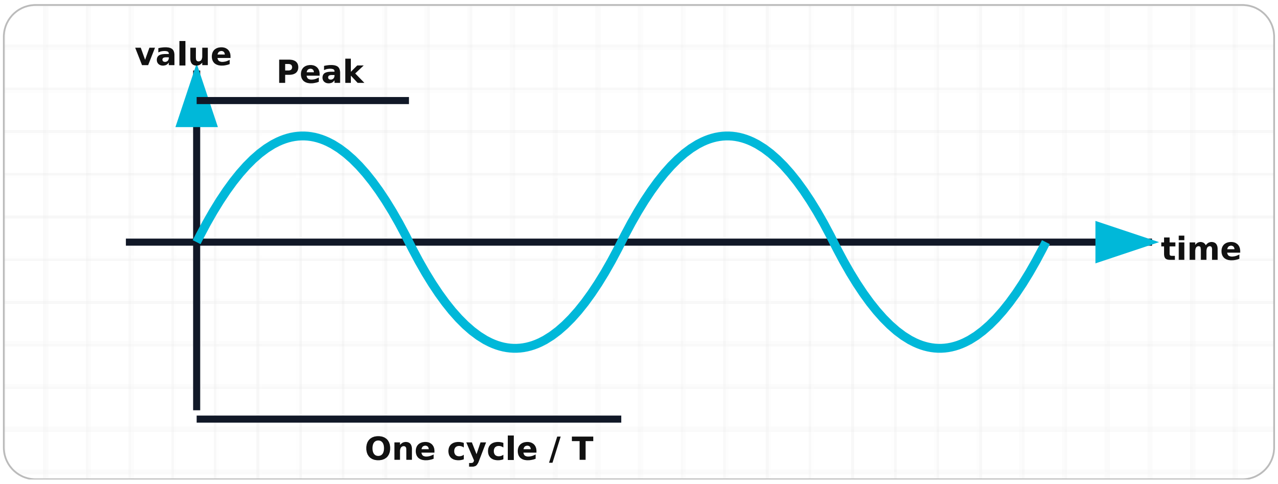
Diagram reference



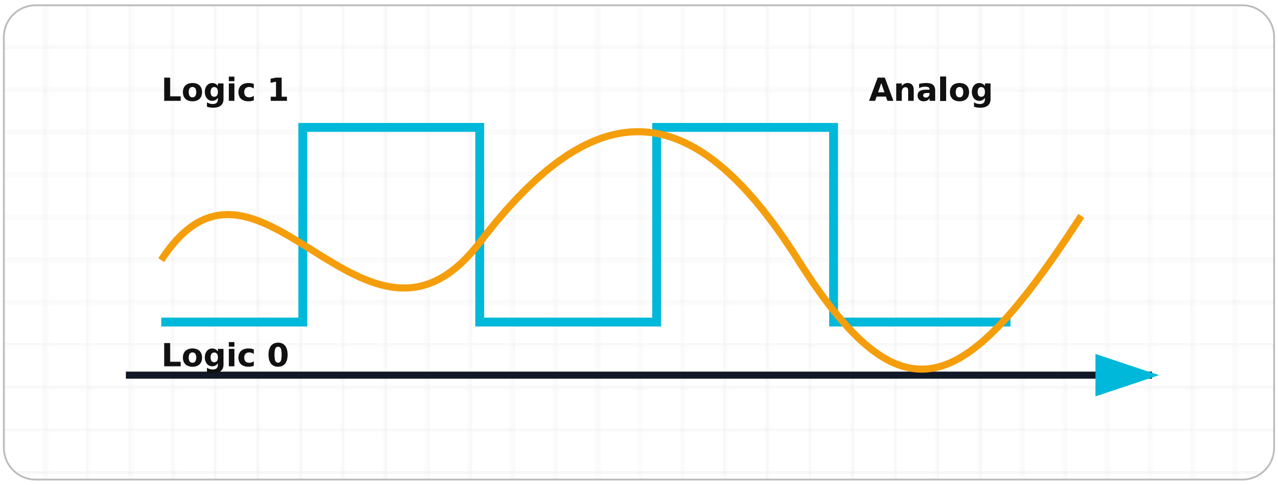
Electric-current flow. Conventional current leaves the positive terminal, passes through the load, and returns to the source.



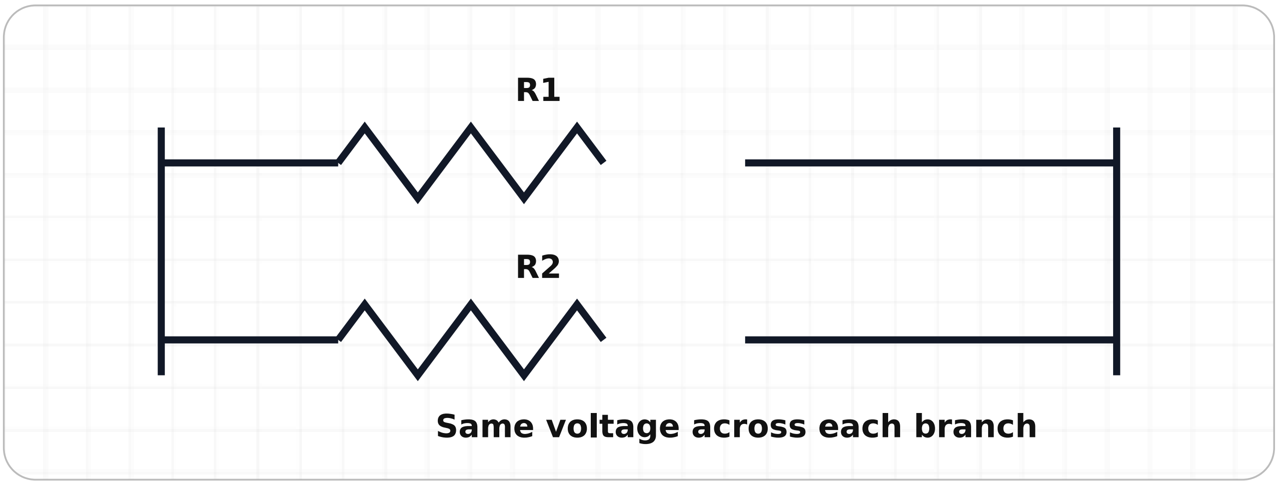
AC and DC comparison. DC remains in one direction; AC alternates and repeats periodically.



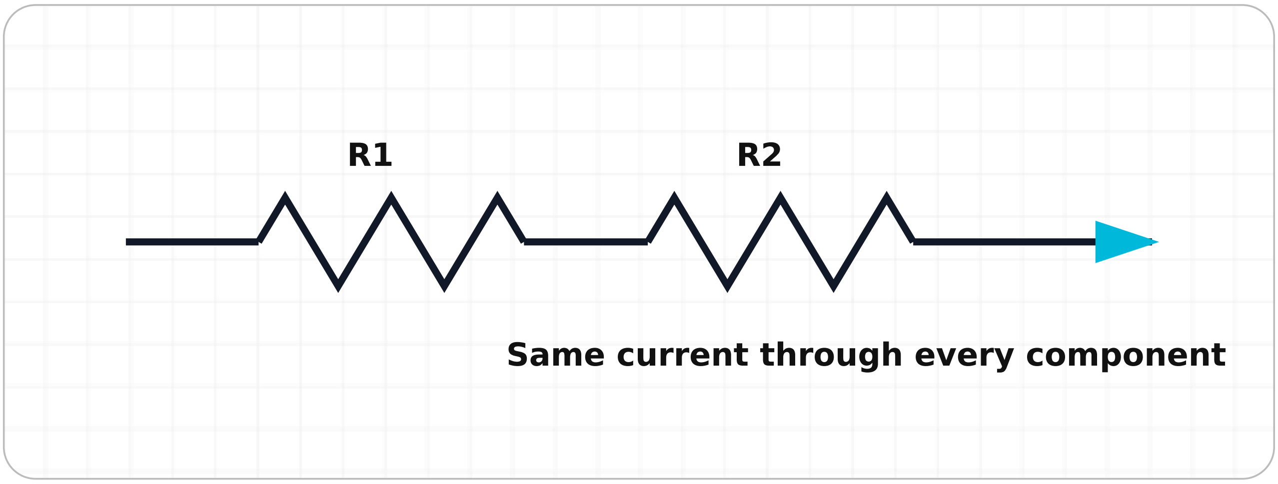
Labelled sine wave. A sine wave is labelled with axis, peak value, and one full time period.



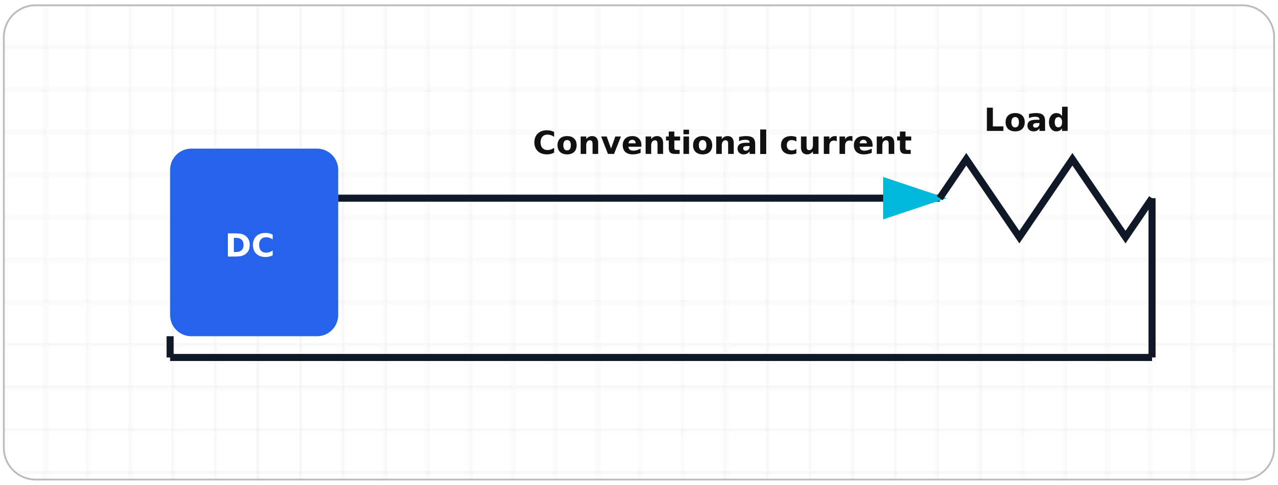
Analog and digital signals. Analog values vary continuously while digital signals use defined discrete levels.



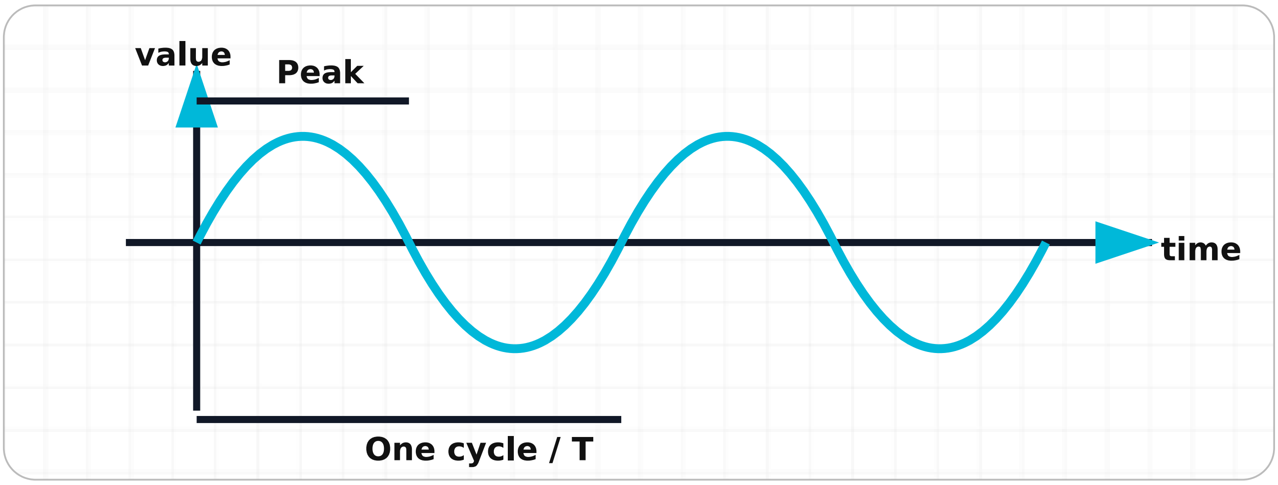
Voltmeter connection. A voltmeter is connected in parallel across the two points whose potential difference is required.



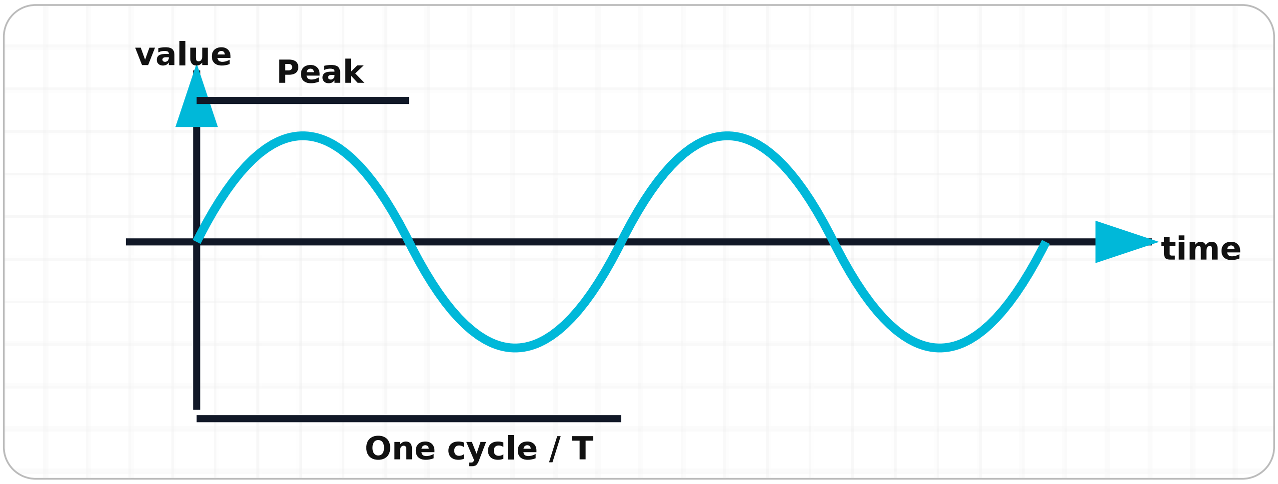
Ammeter connection. An ammeter must be inserted in series so the circuit current passes through it.



Multimeter controls. Function selector, common terminal, measurement terminal, display, and range are checked before connection.



Function-generator front panel. Frequency, waveform, amplitude, offset, and output controls define the test signal.



DSO waveform measurement. Vertical divisions measure voltage and horizontal divisions measure time; triggering stabilizes the display.

Passive Electronic Components

Resistor colour code, capacitor charging, transformer action □□□□□□ calculation-□□ practical verification-□□ □□□□□□□ □□□□□□□□.

Learning target: Explain the theory, interpret diagrams and measurements, apply formulae, perform the listed laboratory work, and answer one-, three- and seven-mark questions.

Core theory chapters (42)

1. Resistor symbol and construction

A resistor is represented by a zig-zag or rectangular symbol and is built from a resistive element with two terminals.

How it works

Its material and geometry determine resistance, power handling, stability, noise, and temperature behaviour.

Engineering practice

Identify leaded and surface-mount packages and read their markings.

Common mistake / safety: A visually undamaged resistor may be open or drifted; verify with a meter.

Malayalam explanation: Resistance current flow-□□ oppose □□□□□□□□□□. Value, tolerance, power rating □□□□□□ □□□□□□□□□ □□□□□□□□□□□□ resistor □□□□□□□□□□□□□□□□□□□.

2. Resistor operation

A resistor limits current, drops voltage, biases devices, terminates signals, and converts energy to heat.

How it works

For an ohmic resistor, current is proportional to voltage at constant temperature.

Engineering practice

Use Ohm's law and power equations together when selecting a resistor.

Common mistake / safety: A resistor does not 'consume current'; it establishes current for a given voltage.

Malayalam explanation: Resistance current flow-□□ oppose □□□□□□□□□□. Value, tolerance, power rating □□□□□□ □□□□□□□□□ □□□□□□□□□□□□□ resistor □□□□□□□□□□□□□□□□□□□.

3. Resistance parameters

Important parameters include nominal resistance, tolerance, power rating, temperature coefficient, maximum working voltage, noise, and stability.

How it works

The nominal value alone is not sufficient for engineering selection.

Engineering practice

Compare a resistor datasheet with the requirements of a voltage-divider circuit.

Common mistake / safety: Maximum voltage can limit use even when calculated power is acceptable.

Malayalam explanation: Resistance current flow-□□ oppose □□□□□□□□□□. Value, tolerance, power rating □□□□□□ □□□□□□□□□ □□□□□□□□□□□□□ resistor □□□□□□□□□□□□□□□□□□□.

4. Tolerance

Tolerance states the permitted percentage deviation from nominal value.

How it works

Rmin = R(1 − tolerance) and Rmax = R(1 + tolerance).

Engineering practice

Calculate the expected resistance range before judging a measured component.

Common mistake / safety: A measured value inside tolerance is not automatically suitable if the wrong nominal value was installed.

Malayalam explanation: □ concept □□□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□□.

5. Power rating

Power rating is the maximum continuous dissipation under specified ambient and mounting conditions.

How it works

Temperature rise and derating determine safe operating power.

Engineering practice

Select a resistor with practical margin above calculated dissipation.

Common mistake / safety: Operating continuously at the nameplate maximum reduces reliability.

Malayalam explanation: Power energy conversion rate □□□; Energy = Power × Time. W, Wh, kWh, J □□□□□ units □□□□□□ mix □□□□□□□□ conversion □□□□□□□.

6. Carbon-composition resistor

Carbon-composition resistors use a moulded carbon mixture and tolerate short pulses but have relatively high noise and poor stability.

How it works

Their resistance depends on the carbon/binder composition.

Engineering practice

Recognize older carbon-composition parts by construction and context.

Common mistake / safety: Do not replace a pulse-rated part solely by matching resistance and wattage.

Malayalam explanation: Resistance current flow-□□ oppose □□□□□□□□□□. Value, tolerance, power rating □□□□□□ □□□□□□□□□ □□□□□□□□□□□□□ resistor □□□□□□□□□□□□□□□□□□.

7. Carbon-film resistor

Carbon-film resistors use a carbon film deposited on a ceramic body, often trimmed helically.

How it works

They provide better stability and lower noise than carbon-composition types.

Engineering practice

Identify the spiral-cut construction in educational illustrations.

Common mistake / safety: Film damage can cause open circuit or value increase.

Malayalam explanation: Resistance current flow-□□ oppose □□□□□□□□□□. Value, tolerance, power rating □□□□□□ □□□□□□□□□ □□□□□□□□□□□□□ resistor □□□□□□□□□□□□□□□□□□□.

8. Metal-film resistor

Metal-film resistors use a thin metal layer and offer good accuracy, stability, and low noise.

How it works

They are common in measurement and precision signal circuits.

Engineering practice

Choose metal film for stable sensor scaling and reference dividers.

Common mistake / safety: Do not infer wattage from body colour alone.

Malayalam explanation: Resistance current flow-□□ oppose □□□□□□□□□□. Value, tolerance, power rating □□□□□□ □□□□□□□□□ □□□□□□□□□□□□□ resistor □□□□□□□□□□□□□□□□□□□.

9. Wire-wound resistor

Wire-wound resistors use resistance wire wound on an insulating former and handle higher power.

How it works

Winding inductance can limit high-frequency use unless a non-inductive construction is specified.

Engineering practice

Use for braking, load, and current-limiting applications where power is significant.

Common mistake / safety: A hot power resistor requires clearance from wires and heat-sensitive parts.

Malayalam explanation: Resistance current flow-□□ oppose □□□□□□□□□□□. Value, tolerance, power rating □□□□□□ □□□□□□□□□ □□□□□□□□□□□□□ resistor □□□□□□□□□□□□□□□□□□□.

10. Preset

A preset is a small adjustable resistor intended for infrequent calibration.

How it works

Its wiper divides the resistive track and may be single-turn or multi-turn.

Engineering practice

Mark the initial position before adjustment during servicing.

Common mistake / safety: Do not use a preset as a user control or high-current rheostat unless rated.

Malayalam explanation: □ concept □□□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

11. Potentiometer

A potentiometer has three terminals and produces an adjustable output voltage from a resistive track.

How it works

The outer terminals span the full track; the centre terminal is the wiper.

Engineering practice

Verify end-to-end resistance and wiper continuity with a DMM.

Common mistake / safety: A dirty or worn wiper causes intermittent output.

Malayalam explanation: □ concept □□□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

12. LDR

A light-dependent resistor decreases resistance as light intensity increases.

How it works

Photoconductive material changes carrier concentration under illumination.

Engineering practice

Use in simple light-sensing and automatic-light circuits.

Common mistake / safety: Response is slower and less linear than many semiconductor light sensors.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

13. VDR

A voltage-dependent resistor or varistor has resistance that falls sharply above a threshold voltage.

How it works

It clamps transient overvoltage by conducting surge current.

Engineering practice

Inspect surge protectors and power inputs for MOV/VDR devices.

Common mistake / safety: A degraded varistor can fail short or show thermal damage; replace with correct rating.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

14. Thermistor

A thermistor changes resistance strongly with temperature; NTC types decrease and PTC types increase.

How it works

The resistance-temperature relation is nonlinear.

Engineering practice

Use NTC sensors for temperature measurement and inrush limiting.

Common mistake / safety: Inrush-limiter thermistors can remain hot after power is removed.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□.

15. Four-band resistor colour code

Four bands represent first digit, second digit, multiplier, and tolerance.

How it works

Read from the end where bands are grouped closer; gold or silver commonly indicates tolerance.

Engineering practice

Decode the bands, then verify by meter.

Common mistake / safety: Brown can be confused with red under poor lighting.

Malayalam explanation: Resistance current flow-□□ oppose □□□□□□□□□□. Value, tolerance, power rating □□□□□□ □□□□□□□□□ □□□□□□□□□□□□□ resistor □□□□□□□□□□□□□□□□□□□.

16. Five-band resistor colour code

Five bands represent three significant digits, multiplier, and tolerance.

How it works

Five-band coding supports precision values.

Engineering practice

Distinguish 4-band and 5-band parts before decoding.

Common mistake / safety: Some special resistors use non-standard markings; check the datasheet when uncertain.

Malayalam explanation: Resistance current flow-□□ oppose □□□□□□□□□□. Value, tolerance, power rating □□□□□□ □□□□□□□□□ □□□□□□□□□□□□□ resistor

□□□□□□□□□□□□□□□□□□□.

17. Series resistors

Series resistors carry the same current and their voltage drops add.

How it works

Req = R1 + R2 + …

Engineering practice

Use series combinations for voltage division and obtaining non-standard values.

Common mistake / safety: Power dissipation must be checked for each resistor, not only the total.

Malayalam explanation: Series circuit-□ □□□ Current □□□□□ components-□□□□□□□ □□□□□□□□□. Resistances add □□□□□□□; □□□ open connection □□□□□□ path □□□□□.

18. Parallel resistors

Parallel resistors share the same voltage and branch currents add.

How it works

1/Req = 1/R1 + 1/R2 + …; for two resistors Req = R1R2/(R1+R2).

Engineering practice

Use parallel combinations to reduce effective resistance or share power with matched parts.

Common mistake / safety: The equivalent resistance is always less than the smallest branch resistance.

Malayalam explanation: Parallel branches-□ Voltage common □□□. Total current branch currents-□□□□ sum □□□; equivalent resistance □□□□□□□ □□□□□ branch resistance-□□□□□□□ □□□□□□□□□□□□□□□.

19. Capacitor construction

A capacitor contains two conductors separated by a dielectric.

How it works

Geometry and dielectric determine capacitance, voltage rating, losses, leakage, and stability.

Engineering practice

Identify plates, dielectric, terminals, and enclosure in a cross-section.

Common mistake / safety: Stored charge can remain after power-off.

Malayalam explanation: Capacitor charge store □□□□□□□□□□. Charging/Discharging speed-□□ RC Time Constant □□□□□□□□□□□□□□□□□□; electrolytic capacitor-□□□□ polarity□□□ voltage rating-□□□ □□□□□□□□□□□ □□□□□□□□□□.

20. Capacitor charging and discharging

A capacitor voltage changes exponentially through a resistor.

How it works

During charging, $v_c = V(1 - e^{-t/RC})$; during discharge, $v_c = V_0 e^{-t/RC}$.

Engineering practice

Measure voltage at 1τ, 3τ, and 5τ.

Common mistake / safety: Discharge capacitors safely through a suitable resistor, not by shorting.

Malayalam explanation: Capacitor charge store □□□□□□□□□□. Charging/Discharging speed-□□ RC Time Constant □□□□□□□□□□□□□□□□□□; electrolytic capacitor-□□□□ polarity□□□ voltage rating-□□□ □□□□□□□□□□□ □□□□□□□□□□.

21. Capacitance

Capacitance is charge stored per volt: $C = Q/V$.

How it works

The SI unit is farad; practical values use µF, nF, and pF.

Engineering practice

Convert prefixes carefully before combination calculations.

Common mistake / safety: A capacitor value alone does not define suitability; voltage and dielectric matter.

Malayalam explanation: Capacitor charge store □□□□□□□□□□. Charging/Discharging speed-□□ RC Time Constant □□□□□□□□□□□□□□□□□□; electrolytic capacitor-□□□□ polarity□□□ voltage rating-□□□ □□□□□□□□□□□ □□□□□□□□□.

22. Voltage rating

Voltage rating is the maximum safe voltage under specified conditions.

How it works

Applied DC plus ripple and transients must remain within the rating.

Engineering practice

Use margin and consider derating.

Common mistake / safety: Exceeding the rating can cause dielectric breakdown or explosion.

Malayalam explanation: Voltage □□□□□ points □□□□□□□□□□ potential difference □□□. Voltmeter parallel □□□ connect □□□□□□□□□□ reference point □□□□□□□□□□□□□□□□ □□□□.

23. Electrolytic capacitor

Electrolytic capacitors provide high capacitance and are usually polarized.

How it works

The oxide dielectric requires correct polarity.

Engineering practice

Identify the negative stripe and board polarity before installation.

Common mistake / safety: Reverse polarity or excessive ripple current can cause venting.

Malayalam explanation: Capacitor charge store □□□□□□□□□□. Charging/Discharging speed-□□ RC Time Constant □□□□□□□□□□□□□□□□□□; electrolytic capacitor-□□□□ polarity□□□ voltage rating-□□□ □□□□□□□□□□□ □□□□□□□□□.

24. Ceramic capacitor

Ceramic capacitors are non-polarized and cover values from pF to µF.

How it works

Different ceramic classes have different stability and voltage dependence.

Engineering practice

Use stable types for timing or frequency-sensitive circuits.

Common mistake / safety: High-capacitance ceramic values can fall significantly under DC bias.

Malayalam explanation: Capacitor charge store □□□□□□□□□□. Charging/Discharging speed-□□ RC Time Constant □□□□□□□□□□□□□□□□□□; electrolytic capacitor-□□□□ polarity□□□ voltage rating-□□□ □□□□□□□□□□□ □□□□□□□□□.

25. Film or paper capacitor

Film capacitors use plastic film dielectric and offer low loss and good pulse performance.

How it works

Metallized constructions can self-heal after small breakdowns.

Engineering practice

Use in timing, filtering, snubber, and audio applications as rated.

Common mistake / safety: Mains-connected positions require approved safety classes.

Malayalam explanation: Capacitor charge store □□□□□□□□□□. Charging/Discharging speed-□□ RC Time Constant □□□□□□□□□□□□□□□□□□; electrolytic capacitor-□□□□ polarity□□□ voltage rating-□□□ □□□□□□□□□□□ □□□□□□□□□.

26. Mica capacitor

Mica capacitors offer high stability, low loss, and good high-frequency performance.

How it works

The mica dielectric is physically and electrically stable.

Engineering practice

Recognize use in RF and precision circuits.

Common mistake / safety: Do not substitute with a general capacitor where Q and stability are critical.

Malayalam explanation: Capacitor charge store □□□□□□□□□□. Charging/Discharging speed-□□ RC Time Constant □□□□□□□□□□□□□□□□□□; electrolytic capacitor-□□□□ polarity□□□ voltage rating-□□□ □□□□□□□□□□□ □□□□□□□□□.

27. Ganged capacitor

A ganged capacitor mechanically varies two or more capacitor sections together.

How it works

It allows simultaneous tuning of multiple resonant circuits.

Engineering practice

Identify rotor, stator, and multiple sections.

Common mistake / safety: Bent plates can short and damage tuning performance.

Malayalam explanation: Capacitor charge store □□□□□□□□□□. Charging/Discharging speed-□□ RC Time Constant □□□□□□□□□□□□□□□□□□; electrolytic capacitor-□□□□ polarity□□□ voltage rating-□□□ □□□□□□□□□□□ □□□□□□□□□.

28. Padder

A padder is a small adjustable capacitor commonly used for lower-frequency alignment.

How it works

It is usually connected in series or parallel as required by the tuning design.

Engineering practice

Adjust only with an insulated alignment tool.

Common mistake / safety: Random adjustment destroys calibration.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□□.

29. Trimmer

A trimmer capacitor provides fine adjustment during alignment.

How it works

Its capacitance changes with plate overlap or dielectric spacing.

Engineering practice

Record the original position before adjustment.

Common mistake / safety: Avoid excessive force and use low-capacitance tools for RF circuits.

Malayalam explanation: □ concept □□□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

30. Direct capacitor marking

Large capacitors often print capacitance and voltage directly, such as 47 µF 25 V.

How it works

Polarity and temperature rating may also be marked.

Engineering practice

Read all markings, not only capacitance.

Common mistake / safety: A similar physical size does not guarantee identical rating.

Malayalam explanation: Capacitor charge store □□□□□□□□□□. Charging/Discharging speed-□□ RC Time Constant □□□□□□□□□□□□□□□□□; electrolytic capacitor-□□□□ polarity□□□ voltage rating-□□□ □□□□□□□□□□□ □□□□□□□□□.

31. Numeric capacitor coding

A three-digit code gives two significant digits and a multiplier in picofarads; 104 means 10 × 10^4 pF = 100 nF.

How it works

Letter codes may specify tolerance.

Engineering practice

Decode to pF first, then convert to nF or µF.

Common mistake / safety: Do not read 104 as 104 pF.

Malayalam explanation: Capacitor charge store ▯▯▯▯▯▯▯▯▯▯. Charging/Discharging speed-▯▯ RC Time Constant ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯; electrolytic capacitor-▯▯▯▯ polarity▯▯▯ voltage rating-▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯.

32. Series capacitors

Series capacitors carry equal charge and the total voltage divides.

How it works

1/Ceq = 1/C1 + 1/C2 + …

Engineering practice

Use series connection only with attention to voltage sharing and polarity.

Common mistake / safety: The equivalent capacitance is less than the smallest capacitor.

Malayalam explanation: Capacitor charge store ▯▯▯▯▯▯▯▯▯▯. Charging/Discharging speed-▯▯ RC Time Constant ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯; electrolytic capacitor-▯▯▯▯ polarity▯▯▯ voltage rating-▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯.

33. Parallel capacitors

Parallel capacitors share the same voltage and capacitances add.

How it works

Ceq = C1 + C2 + …

Engineering practice

Use parallel parts to increase capacitance or improve frequency response.

Common mistake / safety: Ripple current sharing may be unequal if ESR differs.

Malayalam explanation: Capacitor charge store ▯▯▯▯▯▯▯▯▯▯. Charging/Discharging speed-▯▯ RC Time Constant ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯; electrolytic capacitor-▯▯▯▯ polarity▯▯▯ voltage rating-▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯.

34. Inductors

An inductor stores energy in a magnetic field and opposes change in current.

How it works

$v = L(di/dt)$; energy $W = ½LI^2$.

Engineering practice

Identify inductance, current rating, core type, and DC resistance.

Common mistake / safety: Opening an inductive current path can generate high voltage.

Malayalam explanation: Inductor current change-□□ oppose □□□□□□ magnetic field-□ energy store □□□□□□□□□□. Sudden current interruption voltage spike □□□□□□□□□□.

35. Air-core inductor

An air-core inductor uses no ferromagnetic core and avoids core saturation.

How it works

It is useful at high frequency but needs more turns for a given inductance.

Engineering practice

Observe coil geometry and spacing.

Common mistake / safety: Nearby metal and coils can alter inductance.

Malayalam explanation: Inductor current change-□□ oppose □□□□□□ magnetic field-□ energy store □□□□□□□□□□. Sudden current interruption voltage spike □□□□□□□□□□.

36. Transformer construction

A transformer has primary and secondary windings coupled by a magnetic core.

How it works

Insulation system, core material, winding arrangement, and enclosure determine performance and safety.

Engineering practice

Identify primary and secondary from nameplate and resistance measurements only when de-energized.

Common mistake / safety: Never assume the low-resistance winding is safe to energize without nameplate confirmation.

Malayalam explanation: Transformer changing magnetic flux □□□□□□□□□□□ voltage transfer □□□□□□□□□□□. Steady DC transformer action □□□□□□□; turns ratio □□□□□□□□□□□□.

37. Transformer operation

Alternating current in the primary produces changing magnetic flux that induces voltage in the secondary.

How it works

For an ideal transformer $V_p/V_s = N_p/N_s$ and power is approximately conserved.

Engineering practice

Relate turns ratio to measured no-load voltage.

Common mistake / safety: A transformer requires changing flux; steady DC causes excessive current.

Malayalam explanation: Transformer changing magnetic flux □□□□□□□□□□□ voltage transfer □□□□□□□□□□□. Steady DC transformer action □□□□□□□; turns ratio □□□□□□□□□□□□.

38. Step-up transformer

A step-up transformer has more secondary turns than primary turns and increases voltage while reducing available current ideally.

How it works

$N_s > N_p$ and $V_s > V_p$.

Engineering practice

Check insulation and output hazards.

Common mistake / safety: Higher voltage remains dangerous even at lower current rating.

Malayalam explanation: Transformer changing magnetic flux □□□□□□□□□□□ voltage transfer □□□□□□□□□□□. Steady DC transformer action □□□□□□□; turns ratio □□□□□□□□□□□□.

39. Step-down transformer

A step-down transformer has fewer secondary turns and reduces voltage.

How it works

$N_s < N_p$ and $V_s < V_p$.

Engineering practice

Observe common adapters and control transformers.

Common mistake / safety: A low-voltage secondary does not remove primary-side mains hazards.

Malayalam explanation: Transformer changing magnetic flux □□□□□□□□□□□ voltage transfer □□□□□□□□□□□. Steady DC transformer action □□□□□□□; turns ratio □□□□□□□□□□□□.

40. Isolation transformer

An isolation transformer often has approximately 1:1 turns ratio and galvanically separates circuits.

How it works

It reduces direct conductive connection but does not make all contact safe.

Engineering practice

Use only approved isolation procedures in supervised labs.

Common mistake / safety: Touching both isolated secondary conductors can still cause shock.

Malayalam explanation: Transformer changing magnetic flux □□□□□□□□□□□ voltage transfer □□□□□□□□□□□. Steady DC transformer action □□□□□□□; turns ratio □□□□□□□□□□□□.

41. Voltage-divider circuit

A voltage divider uses series resistors to obtain a fraction of input voltage.

How it works

$V_{out} = V_{in}R_2/(R_1+R_2)$ for the unloaded two-resistor divider.

Engineering practice

Measure input, output, and loading effect.

Common mistake / safety: A divider is not a regulated power supply and cannot drive heavy loads.

Malayalam explanation: Voltage Divider output resistor ratio □□□□□□□□□□□□□. Load connect □□□□□□□ output □□□□□□; loading effect □□□□□□□□□□□□.

42. RC timing circuit

An RC circuit creates a predictable exponential voltage change.

How it works

The time constant τ = RC; about 63.2% charging at 1τ and 99.3% at 5τ.

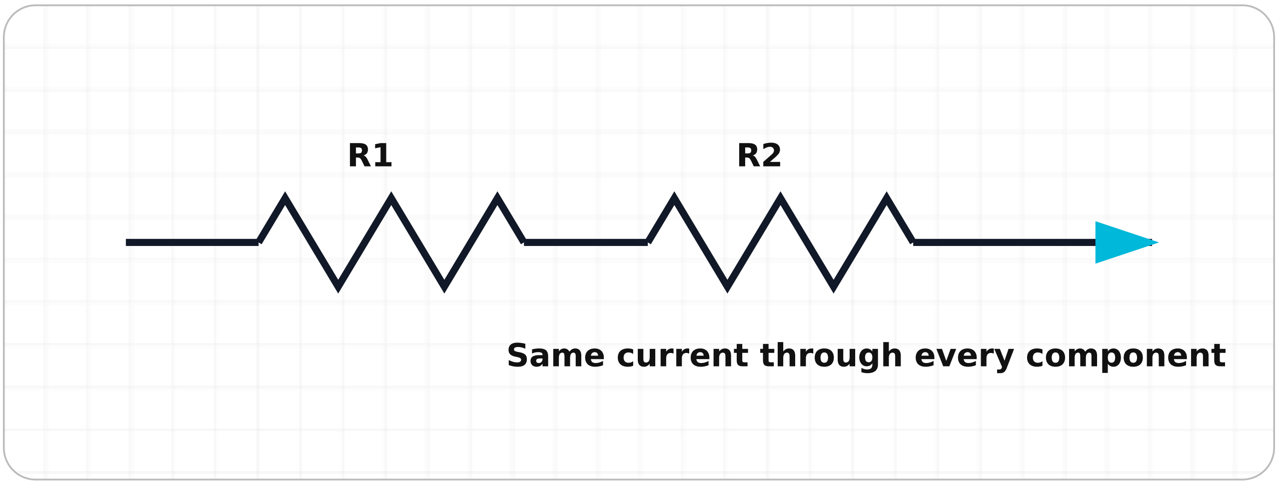
Engineering practice

Plot capacitor voltage against time.

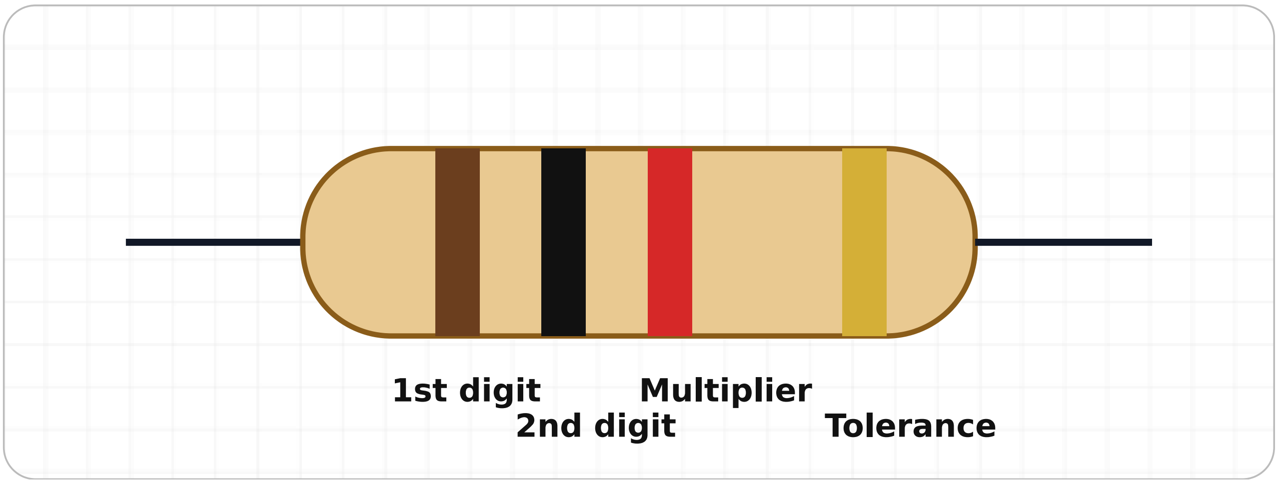
Common mistake / safety: Meter input resistance and capacitor leakage affect long time constants.

Malayalam explanation: Capacitor charge store □□□□□□□□□□. Charging/Discharging speed-□□ RC Time Constant □□□□□□□□□□□□□□□□□□; electrolytic capacitor-□□□□ polarity□□□ voltage rating-□□□ □□□□□□□□□□□ □□□□□□□□□.

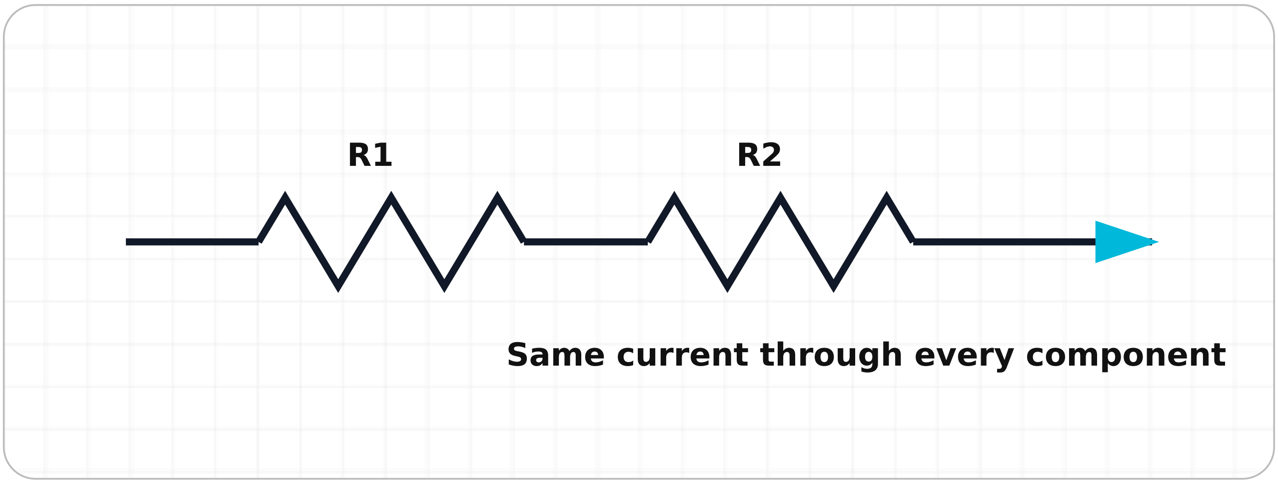
Diagram reference



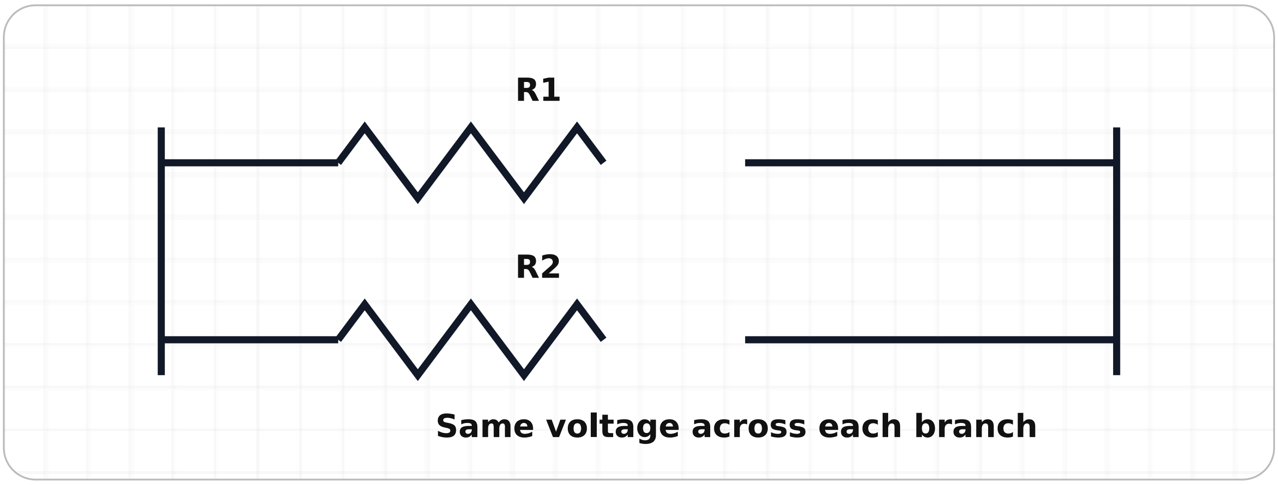
Resistor construction. A resistive element is connected to two leads and protected by an insulating body.



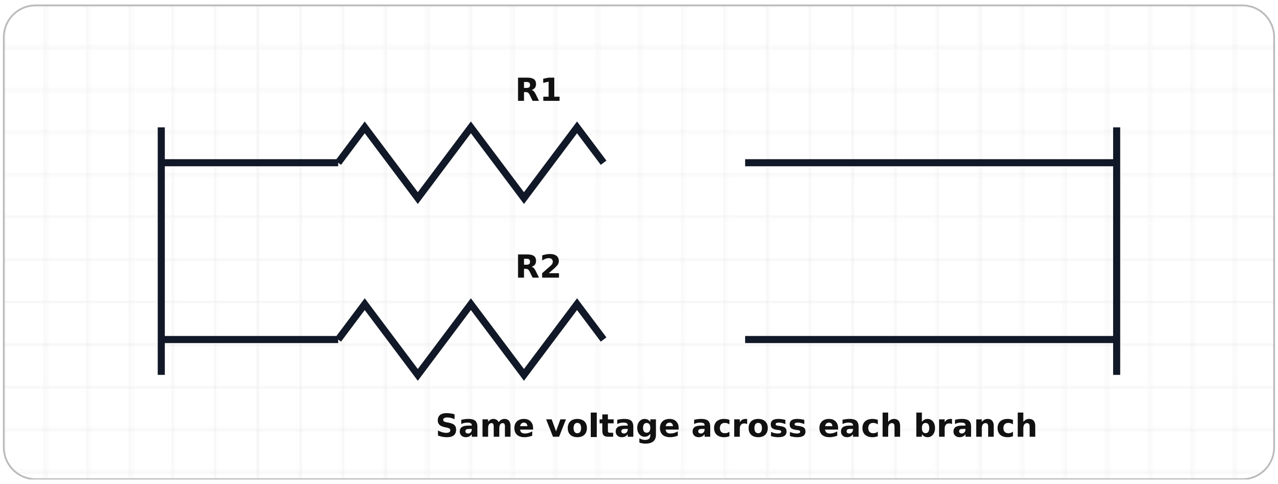
Resistor colour bands. Band position identifies significant digits, multiplier, and tolerance.



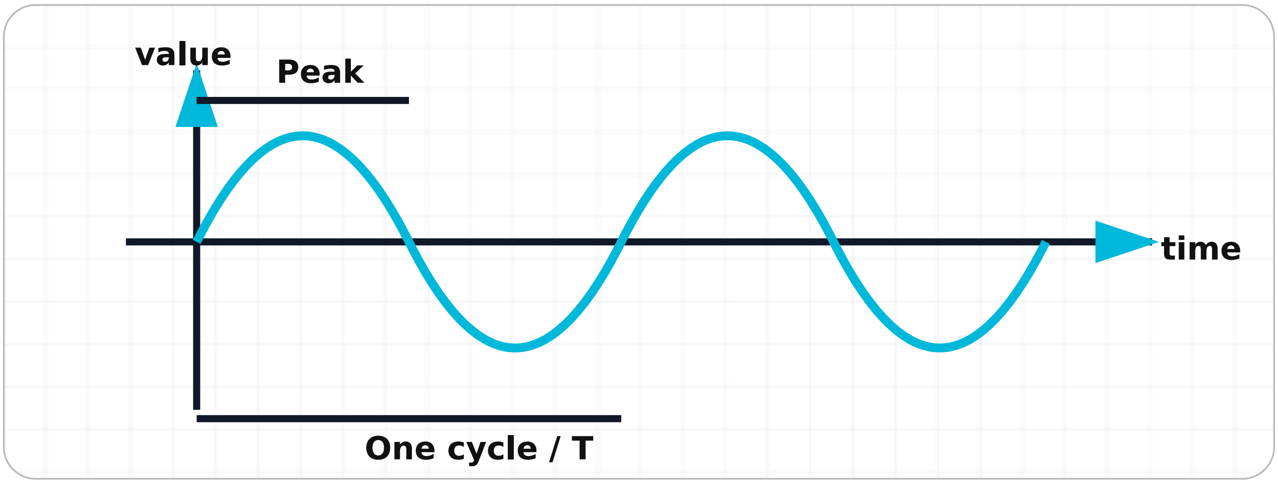
Series resistors. Series resistance is the sum because the same current passes through every resistor.



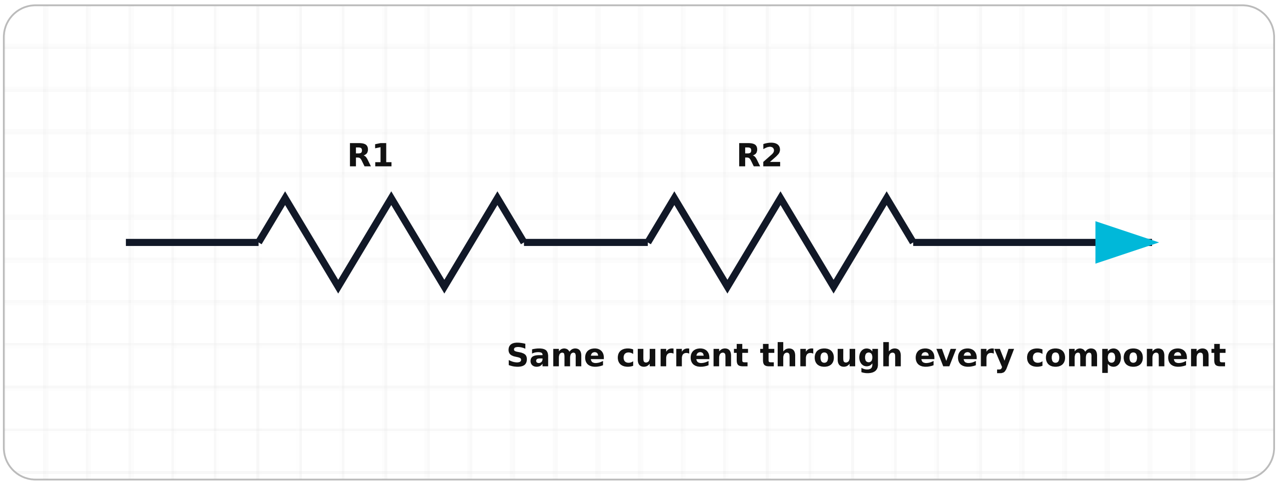
Parallel resistors. Parallel branches have the same voltage and their currents add.



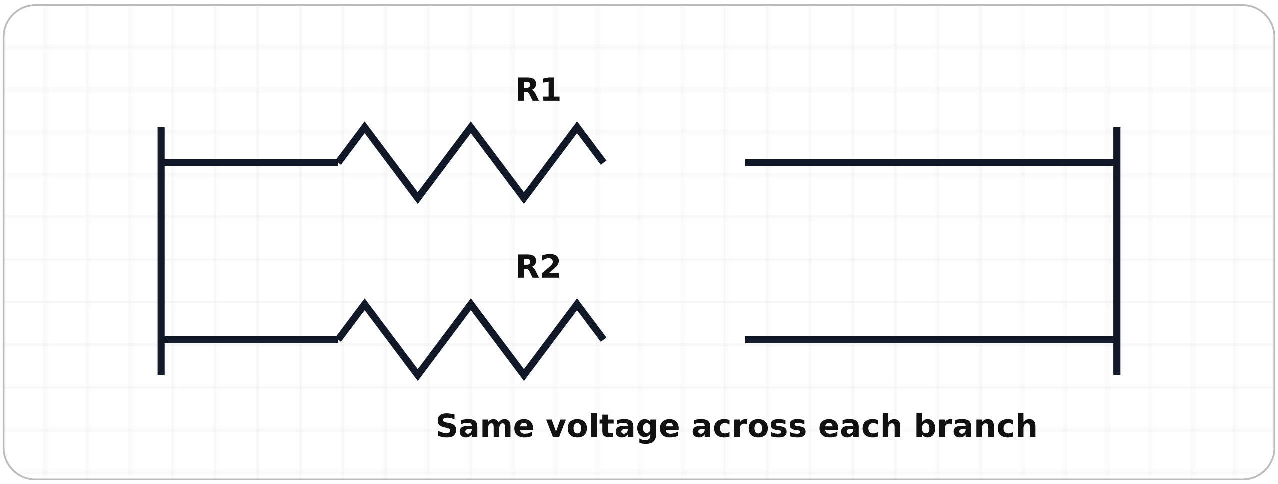
Capacitor construction. Two conductive plates separated by dielectric store charge.



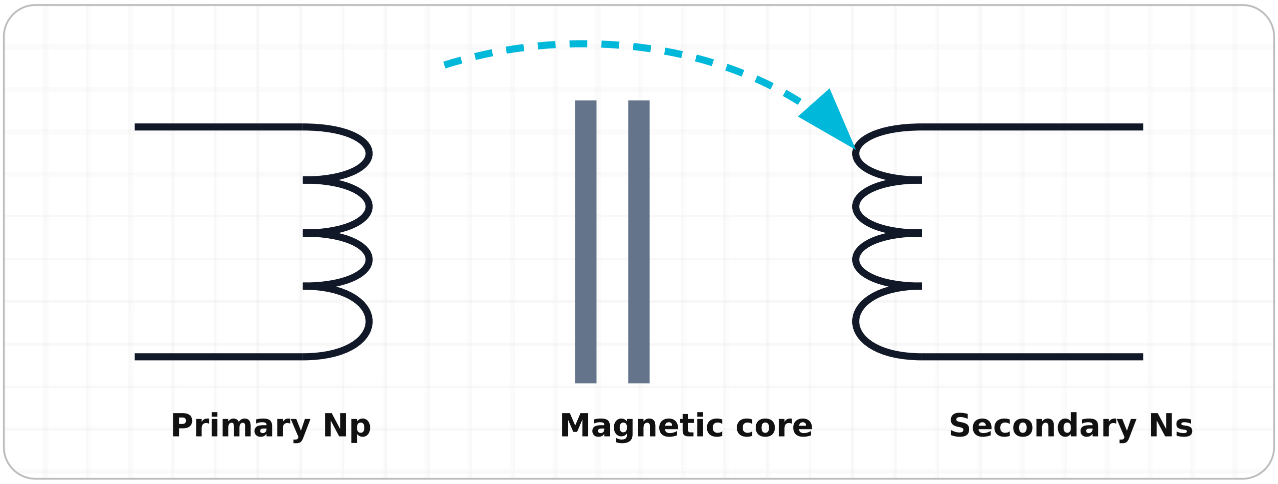
Capacitor charging. Capacitor voltage rises exponentially through the resistor toward the supply voltage.



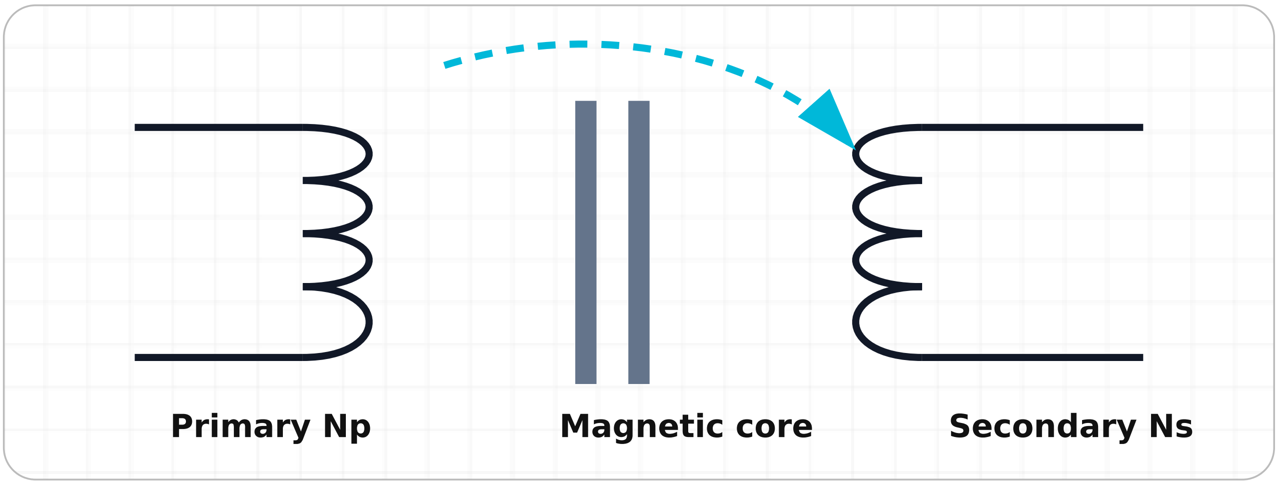
Series capacitors. Series capacitors carry equal charge and their reciprocal capacitances add.



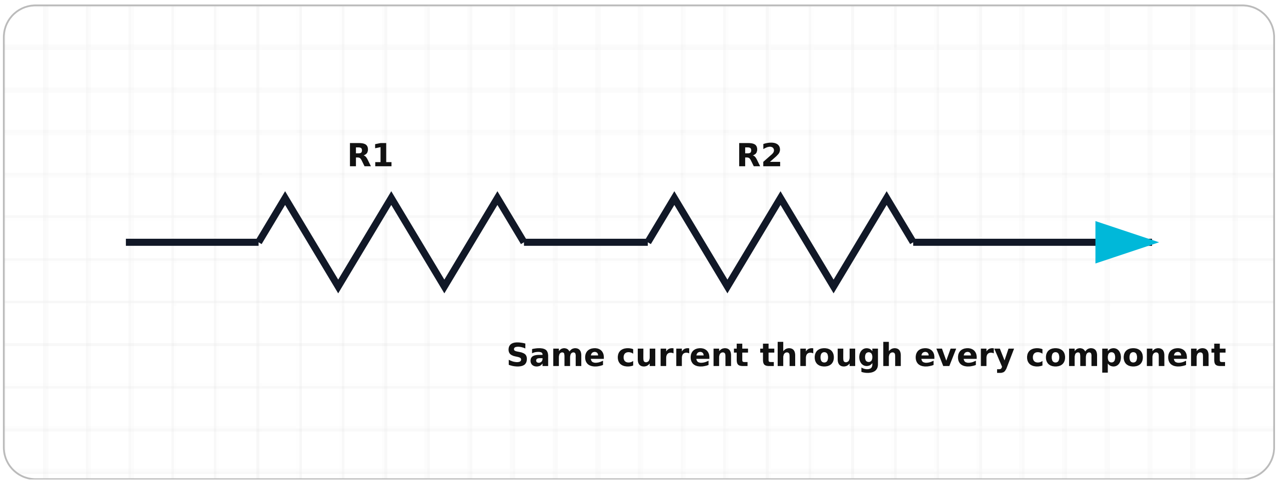
Parallel capacitors. Parallel capacitances add because the effective plate area increases.



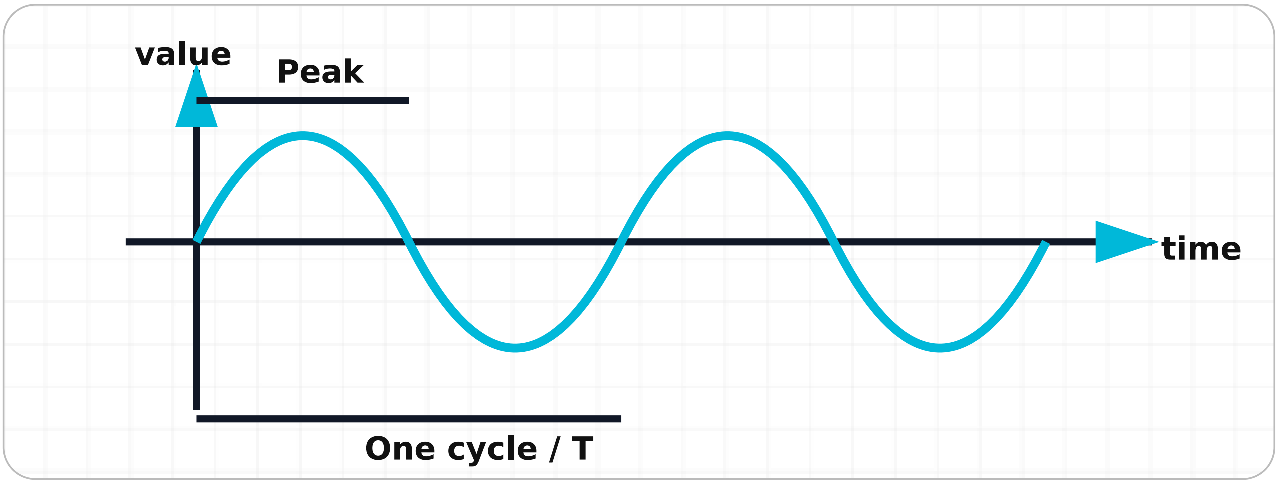
Inductor construction. A coil produces magnetic flux when current flows.



Transformer construction. Primary and secondary windings are magnetically coupled by a core.



Voltage-divider circuit. Output is taken across the lower resistor of two series resistors.



RC timing circuit. The resistor controls charge current and the capacitor voltage changes with time constant RC.

Active Electronic Components

Doping, PN junction, forward/reverse bias □□□□□□ carrier movement □□□ □□□□□□□□□□□ diode behaviour □□□□□□□ □□□□□□□□□□□.

Learning target: Explain the theory, interpret diagrams and measurements, apply formulae, perform the listed laboratory work, and answer one-, three- and seven-mark questions.

Core theory chapters (42)

1. Energy-band theory

Energy bands describe allowed electron-energy ranges in solids and explain electrical conduction.

How it works

The valence band contains bonding electrons; the conduction band supports mobile carriers; the forbidden gap separates them.

Engineering practice

Compare conductor, semiconductor, and insulator band diagrams.

Common mistake / safety: Band diagrams are energy models, not physical layers inside the material.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□ □□□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□ □□□□□□□□□□□□.

2. Conductors

Conductors have many available charge carriers and overlapping or partially filled energy bands.

How it works

Small applied electric fields produce significant current.

Engineering practice

Relate copper wiring to low resistance and current-carrying capacity.

Common mistake / safety: Good conductivity does not remove heating or voltage-drop limits.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

3. Insulators

Insulators have a large energy gap and very few free carriers under normal conditions.

How it works

They resist current but can break down under excessive electric field, temperature, contamination, or damage.

Engineering practice

Inspect insulation rating and condition.

Common mistake / safety: Insulation is not absolute; every material has limits.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

4. Semiconductors

Semiconductors have conductivity between conductors and insulators and can be controlled by doping, temperature, light, and electric fields.

How it works

Silicon is widely used because of material properties and manufacturing maturity.

Engineering practice

Relate controllable conductivity to diodes, transistors, sensors, and ICs.

Common mistake / safety: Semiconductor resistance is strongly condition-dependent.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□ □□□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□ □□□□□□□□□□□□□.

5. Intrinsic semiconductor

An intrinsic semiconductor is ideally pure, with equal electron and hole concentrations.

How it works

Thermal energy generates electron–hole pairs.

Engineering practice

Visualize covalent bonds and thermally released carriers.

Common mistake / safety: At room temperature intrinsic conductivity is limited compared with doped material.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□□ □□□□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□□ □□□□□□□□□□□□□□□.

6. Silicon and germanium

Silicon and germanium are group-IV semiconductor materials with four valence electrons.

How it works

Germanium has a smaller band gap and lower typical forward voltage but higher leakage; silicon dominates many applications.

Engineering practice

Compare approximate diode knee voltages under similar current.

Common mistake / safety: These values are approximate and depend on current and temperature.

Malayalam explanation: □ concept □□□□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□□ □□□□□□□ □□□□□□□□□□□□□□□ □□□□□□□□□□□□□□□.

7. Covalent bonding

Neighbouring semiconductor atoms share valence electrons through covalent bonds.

How it works

The crystal lattice is stable until energy frees an electron, leaving a hole.

Engineering practice

Use a lattice diagram to track one electron–hole pair.

Common mistake / safety: A hole is a useful carrier model, not a positively charged particle detached from an atom.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

8. Thermal generation of carriers

Heat can provide enough energy to break covalent bonds and create electron–hole pairs.

How it works

Carrier concentration increases strongly with temperature.

Engineering practice

Relate temperature rise to increased leakage current.

Common mistake / safety: Excess temperature can cause thermal runaway in semiconductor circuits.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□ □□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□ □□□□□□□□□□□□□.

9. Electron-hole pairs

When a valence electron gains enough energy to enter the conduction band, it leaves a vacancy called a hole.

How it works

Electrons and holes can move and later recombine.

Engineering practice

Track carrier creation, movement, and recombination separately.

Common mistake / safety: Charge is conserved: pair generation creates carriers, not net charge.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□ □□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□

□□□□□□□□□□□□□□.

10. Doping

Doping intentionally adds controlled impurity atoms to change carrier concentration.

How it works

Pentavalent donors produce N-type material; trivalent acceptors produce P-type material.

Engineering practice

Identify the purpose of doping before discussing junctions.

Common mistake / safety: Doping concentration is carefully controlled; it is not random contamination.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□□ □□□□□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□□ □□□□□□□□□□□□□□.

11. Donor impurities

Donor atoms contribute loosely bound electrons to the conduction process.

How it works

Common conceptual examples are phosphorus or arsenic in silicon.

Engineering practice

Show the fifth valence electron becoming available.

Common mistake / safety: The ionized donor atom remains fixed in the lattice.

Malayalam explanation: □ concept □□□□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□□ □□□□□□□□□□□□□□.

12. Acceptor impurities

Acceptor atoms create holes by accepting electrons into incomplete bonds.

How it works

Boron is a common conceptual example in silicon.

Engineering practice

Show the missing bond electron as a mobile hole model.

Common mistake / safety: The ionized acceptor is fixed and negatively charged.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□□□.

13. N-type semiconductor

N-type material has electrons as majority carriers and holes as minority carriers.

How it works

It remains electrically neutral overall because fixed ionized donors balance mobile charge.

Engineering practice

Use carrier symbols and fixed-ion labels in diagrams.

Common mistake / safety: N-type does not mean the entire material has a net negative charge.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□ □□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□ □□□□□□□□□□□□.

14. P-type semiconductor

P-type material has holes as majority carriers and electrons as minority carriers.

How it works

Fixed ionized acceptors balance mobile carrier charge.

Engineering practice

Distinguish mobile holes from fixed acceptor ions.

Common mistake / safety: P-type does not mean the entire material has a net positive charge.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□□ □□□□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□□ □□□□□□□□□□□□□□.

15. Majority carriers

Majority carriers are the dominant mobile carrier type created mainly by doping.

How it works

Electrons dominate in N-type; holes dominate in P-type.

Engineering practice

Use majority-carrier direction to explain diode conduction.

Common mistake / safety: Minority carriers still matter, especially in reverse current.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□□ □□□□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□□ □□□□□□□□□□□□□□.

16. Minority carriers

Minority carriers are present in smaller concentration, mainly through thermal generation.

How it works

They contribute to reverse saturation current and temperature behaviour.

Engineering practice

Relate increased temperature to increased minority-carrier concentration.

Common mistake / safety: Do not assume minority-carrier current is exactly zero.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□□ □□□□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□□ □□□□□□□□□□□□□□.

17. PN-junction formation

When P-type and N-type regions join, carriers diffuse across the boundary and recombine, leaving fixed ions.

How it works

The exposed ions create a depletion region and electric field that opposes further diffusion.

Engineering practice

Follow electrons from N to P and holes from P to N during formation.

Common mistake / safety: The depletion region is depleted of mobile carriers, not of all charge.

Malayalam explanation: PN Junction □□□□□□□□□□□□□□ Depletion Region□□□ Barrier Potential□□□ □□□□□□□□. Applied Bias □□□□□□□□□□ barrier width □□□□□□□□.

18. Diffusion

Diffusion is carrier movement from high concentration to low concentration.

How it works

Carrier concentration gradients drive diffusion current.

Engineering practice

Use concentration arrows across an unbiased junction.

Common mistake / safety: Diffusion is distinct from drift caused by an electric field.

Malayalam explanation: □ concept □□□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

19. Recombination

Recombination occurs when a conduction electron fills a hole, releasing energy.

How it works

It reduces mobile carrier population.

Engineering practice

Mark recombination near the junction during formation.

Common mistake / safety: Recombination does not annihilate charge; electrons return to lower-energy states.

Malayalam explanation: ▯ concept ▯▯▯▯▯▯▯▯▯▯▯▯ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety ▯▯▯▯▯▯ ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯.

20. Depletion region

The depletion region contains fixed ionized donors and acceptors with few mobile carriers.

How it works

Its electric field forms a potential barrier.

Engineering practice

Compare width under forward and reverse bias.

Common mistake / safety: The region can change width with applied voltage.

Malayalam explanation: PN Junction ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Depletion Region▯▯▯ Barrier Potential▯▯▯ ▯▯▯▯▯▯▯▯. Applied Bias ▯▯▯▯▯▯▯▯▯▯ barrier width ▯▯▯▯▯▯▯▯.

21. Barrier potential

Barrier potential is the built-in potential across the depletion region that opposes majority-carrier diffusion.

How it works

Typical classroom approximations are about 0.7 V for silicon and 0.3 V for germanium at useful currents.

Engineering practice

Relate applied forward voltage to barrier reduction.

Common mistake / safety: The barrier is not a fixed ideal battery and varies with temperature and doping.

Malayalam explanation: Voltage ▯▯▯▯▯ points ▯▯▯▯▯▯▯▯▯▯ potential difference ▯▯▯. Voltmeter parallel ▯▯▯ connect ▯▯▯▯▯▯▯▯▯▯ reference point ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯.

22. Drift current

Drift current results from carriers moving under an electric field.

How it works

In a PN junction it is associated strongly with minority carriers across the depletion region.

Engineering practice

Show carrier movement due to the built-in field.

Common mistake / safety: Drift direction depends on carrier charge and field direction.

Malayalam explanation: Current □□□□□□ charge flow rate □□□. Ammeter □□□□□□□□ circuit-□ series □□□ connect □□□□□□□; source-□□ across □□□ connect □□□□□□□□□.

23. Diffusion current

Diffusion current results from concentration-gradient-driven carrier motion.

How it works

At equilibrium, diffusion and drift currents balance so net current is zero.

Engineering practice

Use equilibrium to explain why an unbiased diode has no external current.

Common mistake / safety: Internal carrier motion can exist even when net terminal current is zero.

Malayalam explanation: Current □□□□□□ charge flow rate □□□. Ammeter □□□□□□□□ circuit-□ series □□□ connect □□□□□□□; source-□□ across □□□ connect □□□□□□□□□.

24. Diode symbol and construction

A PN-junction diode has an anode connected to P-type material and a cathode connected to N-type material.

How it works

The symbol indicates the two terminals and conventional current direction in forward conduction.

Engineering practice

Identify cathode band on common packages.

Common mistake / safety: Package markings vary; confirm uncertain devices with a meter and datasheet.

Malayalam explanation: Diode □□□□□□□□□□ one-direction conduction □□□□□□□ semiconductor device □□□. Anode, Cathode, polarity, forward drop, current limit □□□□□□ □□□□□□□□□□□.

25. Anode and cathode

The anode is the P-side terminal and the cathode is the N-side terminal of a basic PN diode.

How it works

In forward bias the anode is more positive than the cathode.

Engineering practice

Use DMM diode mode to identify the conducting direction.

Common mistake / safety: Do not infer polarity from lead length on every diode package.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□□□.

26. Silicon and germanium diodes

Silicon diodes typically show higher forward drop and lower leakage than germanium diodes.

How it works

Material properties affect knee behaviour, reverse leakage, temperature sensitivity, and application.

Engineering practice

Compare measured diode-mode readings.

Common mistake / safety: DMM readings depend on test current and are not universal constants.

Malayalam explanation: Diode □□□□□□□□□□ one-direction conduction □□□□□□□ semiconductor device □□□. Anode, Cathode, polarity, forward drop, current limit □□□□□□ □□□□□□□□□□□□□.

27. Forward bias

Forward bias connects P to positive and N to negative, reducing the barrier and narrowing the depletion region.

How it works

Current rises rapidly after the applied voltage reaches the practical conduction region.

Engineering practice

Use a series resistor to limit current.

Common mistake / safety: Never forward-bias a diode directly from a low-impedance supply without current limiting.

Malayalam explanation: Forward Bias junction barrier □□□□□□□ current flow □□□□□□□□□□□□□□. Diode current safe value-□ limit □□□□□□□.

28. Reverse bias

Reverse bias connects P to negative and N to positive, increasing barrier and widening the depletion region.

How it works

Only a small reverse saturation current flows until breakdown.

Engineering practice

Respect the peak inverse voltage rating.

Common mistake / safety: Breakdown can destroy an ordinary diode unless current is safely limited and the device is designed for it.

Malayalam explanation: Reverse Bias depletion region □□□□□□□□□□□□□□□□ majority-carrier current □□□□□□□□ □□□□□□□□□□. Breakdown rating exceed □□□□□□□□□.

29. PN diode V-I characteristics

The V-I curve shows low forward current below the knee, rapid current rise in forward conduction, and small reverse current before breakdown.

How it works

The curve is nonlinear and temperature-dependent.

Engineering practice

Plot measured current against diode voltage using safe steps.

Common mistake / safety: Do not connect data points in a way that suggests unmeasured accuracy.

Malayalam explanation: PN Junction ▯▯▯▯▯▯▯▯▯▯▯▯▯ Depletion Region▯▯▯ Barrier Potential▯▯▯ ▯▯▯▯▯▯▯▯▯. Applied Bias ▯▯▯▯▯▯▯▯▯▯ barrier width ▯▯▯▯▯▯▯▯▯.

30. Knee voltage

Knee voltage is the approximate forward-voltage region where current begins increasing rapidly.

How it works

It is a useful teaching approximation rather than a sharp physical switch point.

Engineering practice

Estimate from a plotted curve.

Common mistake / safety: Treating a diode as exactly off below 0.7 V and exactly on above it is an oversimplification.

Malayalam explanation: Voltage ▯▯▯▯▯ points ▯▯▯▯▯▯▯▯▯▯ potential difference ▯▯▯. Voltmeter parallel ▯▯▯ connect ▯▯▯▯▯▯▯▯▯▯ reference point ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯.

31. Reverse saturation current

Reverse saturation current is a small minority-carrier current under reverse bias before breakdown.

How it works

It increases strongly with temperature.

Engineering practice

Use proper instruments and ranges when measuring small current.

Common mistake / safety: Leakage can be too small for a basic lab DMM to resolve accurately.

Malayalam explanation: Current □□□□□□ charge flow rate □□□. Ammeter □□□□□□□□ circuit-□ series □□□ connect □□□□□□□; source-□□ across □□□ connect □□□□□□□□□.

32. Breakdown voltage

Breakdown voltage is the reverse voltage at which current increases sharply.

How it works

Mechanisms include Zener and avalanche effects depending on junction design and voltage.

Engineering practice

Use current limiting and approved lab circuits.

Common mistake / safety: Exceeding breakdown without limiting current can permanently damage the device.

Malayalam explanation: Zener diode controlled Reverse Breakdown region-□ voltage regulation-□□□□□□ □□□□□□□□□□□□□□. Series current-limiting resistor □□□□□□□□□□□.

33. Diode ratings

Key ratings include forward current, repetitive peak reverse voltage, power dissipation, surge current, junction temperature, and recovery time.

How it works

Ratings interact and require derating.

Engineering practice

Check datasheet limits before substitution.

Common mistake / safety: A diode with sufficient current rating may still have inadequate reverse-voltage or speed rating.

Malayalam explanation: Diode □□□□□□□□□□□ one-direction conduction □□□□□□□ semiconductor device □□□. Anode, Cathode, polarity, forward drop, current limit □□□□□□ □□□□□□□□□□□□.

34. Static resistance

Static or DC resistance at an operating point is V/I.

How it works

It is the ratio from origin to the selected point on the curve.

Engineering practice

Calculate using measured diode voltage and current at the same point.

Common mistake / safety: Static resistance is not the same as the slope resistance.

Malayalam explanation: Resistance current flow-□□ oppose □□□□□□□□□□. Value, tolerance, power rating □□□□□□ □□□□□□□□□ □□□□□□□□□□□□□ resistor □□□□□□□□□□□□□□□□□□□.

35. Dynamic resistance

Dynamic or AC resistance is the local slope ΔV/ΔI around an operating point.

How it works

It describes small-signal behaviour.

Engineering practice

Choose two nearby points on the curve and use consistent units.

Common mistake / safety: Using widely separated points gives an average, not true local dynamic resistance.

Malayalam explanation: Resistance current flow-□□ oppose □□□□□□□□□□. Value, tolerance, power rating □□□□□□ □□□□□□□□□ □□□□□□□□□□□□□ resistor □□□□□□□□□□□□□□□□□□□.

36. Zener diode

A Zener diode is designed to operate safely in reverse breakdown with controlled current.

How it works

It provides voltage reference or regulation in suitable circuits.

Engineering practice

Always use a series resistor to limit current.

Common mistake / safety: The Zener voltage varies with current, tolerance, and temperature.

Malayalam explanation: Zener diode controlled Reverse Breakdown region-□ voltage regulation-□□□□□□ □□□□□□□□□□□□□□. Series current-limiting resistor □□□□□□□□□□□.

37. Zener V-I characteristics

The Zener curve includes ordinary forward conduction and a reverse breakdown region near the rated Zener voltage.

How it works

Useful regulation occurs between minimum and maximum specified current.

Engineering practice

Plot reverse voltage carefully using a current-limited setup.

Common mistake / safety: Do not exceed Zener power PZ = VZIZ.

Malayalam explanation: Zener diode controlled Reverse Breakdown region-□ voltage regulation-□□□□□□ □□□□□□□□□□□□□□. Series current-limiting resistor □□□□□□□□□□□.

38. LED

A light-emitting diode emits photons when forward-biased carriers recombine.

How it works

Colour depends mainly on semiconductor band gap.

Engineering practice

Use a series resistor selected from supply voltage, forward voltage, and desired current.

Common mistake / safety: Never test an LED directly across a battery or power supply without current limiting.

Malayalam explanation: LED forward current □□□□□□□□□□□□ light emit □□□□□□□□□□. Polarity, forward voltage, current-limiting resistor □□□□□□ □□□□□□□□□□□□.

39. LED polarity

Common through-hole LEDs often have a longer anode lead, shorter cathode lead, flat package edge on cathode side, and larger internal cathode electrode.

How it works

DMM diode mode can confirm polarity and may produce a faint glow.

Engineering practice

Verify package-specific markings.

Common mistake / safety: Lead length may have been trimmed, so it is not always reliable.

Malayalam explanation: LED forward current □□□□□□□□□□□ light emit □□□□□□□□□□. Polarity, forward voltage, current-limiting resistor □□□□□□ □□□□□□□□□□□□.

40. LED current-limiting resistor

The resistor sets LED current: $R = (VS - VF)/I$.

How it works

Resistor power is approximately I^2R.

Engineering practice

Choose the next standard value that keeps current within rating.

Common mistake / safety: Using nominal VF without considering supply tolerance can overdrive the LED.

Malayalam explanation: LED forward current □□□□□□□□□□□ light emit □□□□□□□□□□. Polarity, forward voltage, current-limiting resistor □□□□□□ □□□□□□□□□□□□.

41. Photodiode

A photodiode converts light into current and is commonly operated in reverse bias or zero-bias mode.

How it works

Light increases photocurrent approximately with incident optical power over a useful range.

Engineering practice

Observe response with a safe low-voltage circuit.

Common mistake / safety: Polarity, wavelength sensitivity, dark current, and amplifier design affect results.

Malayalam explanation: Photodiode incident light-□□ electrical current □□□ convert □□□□□□□□□□□. □□□□□□□ reverse-bias operation-□ sensitivity □□□□□□□□□□□□□.

42. Photodiode applications

Photodiodes are used in optical communication, encoders, smoke detectors, light meters, remote receivers, and safety sensors.

How it works

A transimpedance amplifier commonly converts photocurrent to voltage.

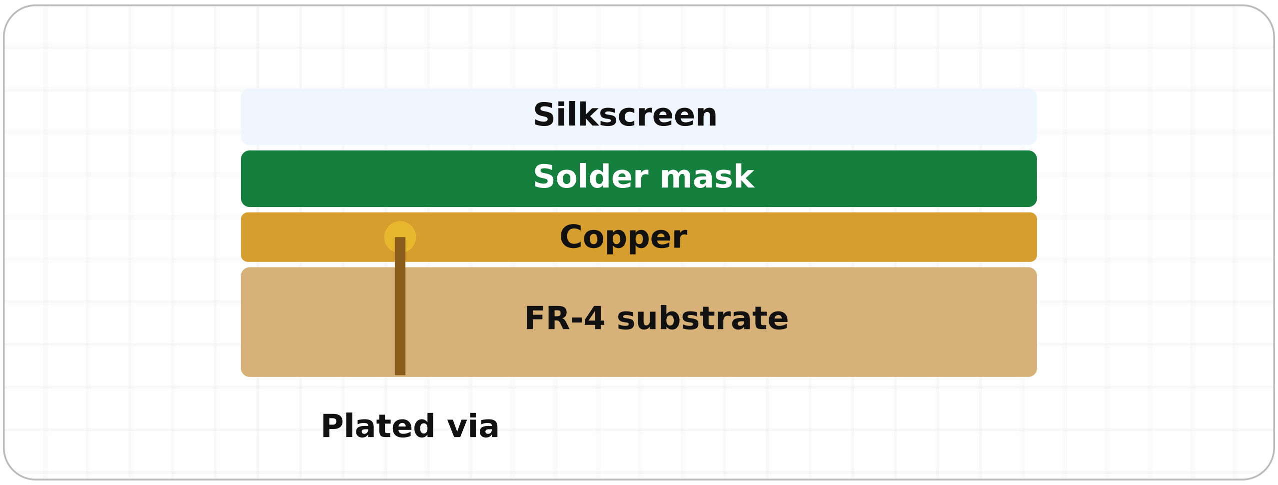
Engineering practice

Relate an optical escalator entry sensor to emitter, receiver, alignment, and contamination.

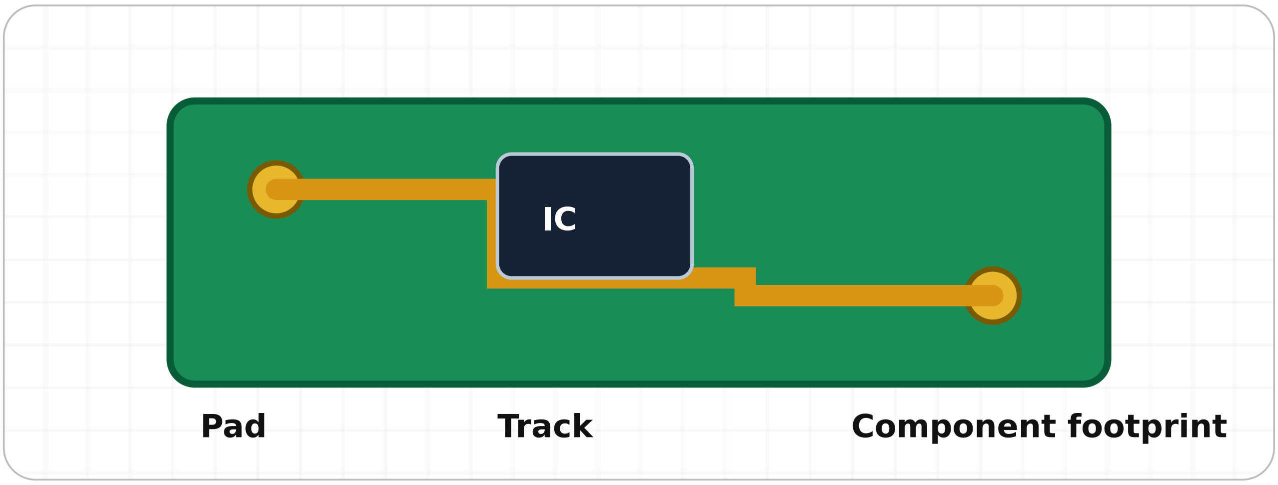
Common mistake / safety: Do not stare into unknown infrared emitters or lasers.

Malayalam explanation: Photodiode incident light-□□ electrical current □□□ convert □□□□□□□□□□□. □□□□□□□ reverse-bias operation-□ sensitivity □□□□□□□□□□□□□.

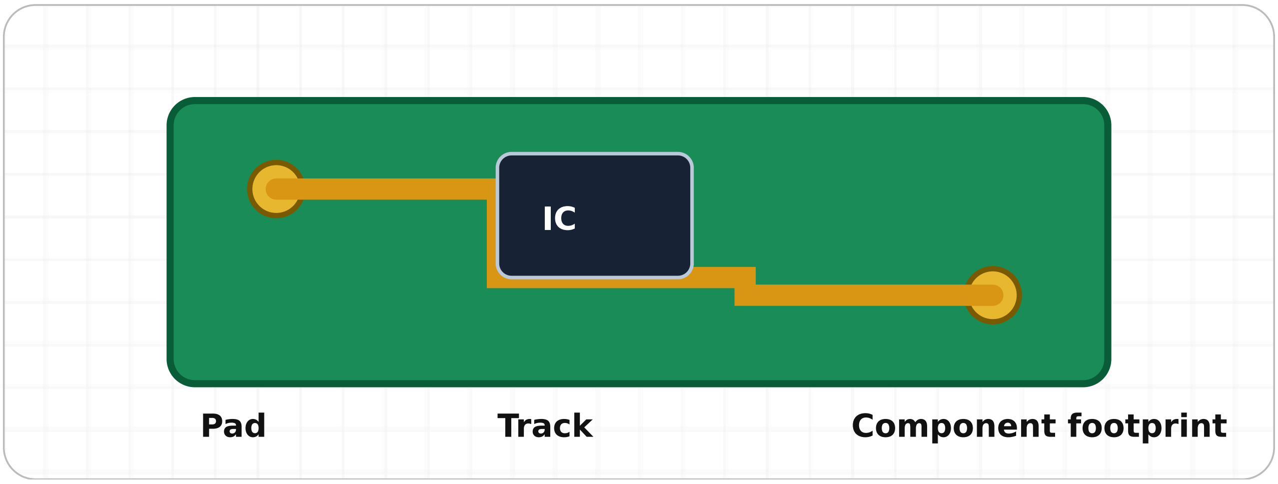
Diagram reference



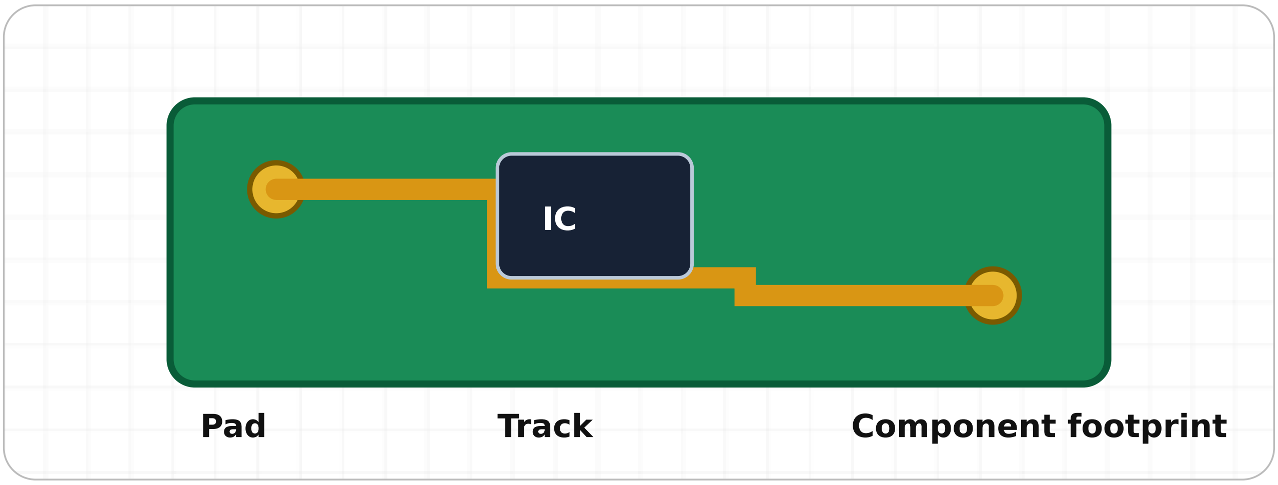
Energy-band diagrams. Relative band gap distinguishes conductor, semiconductor, and insulator behaviour.



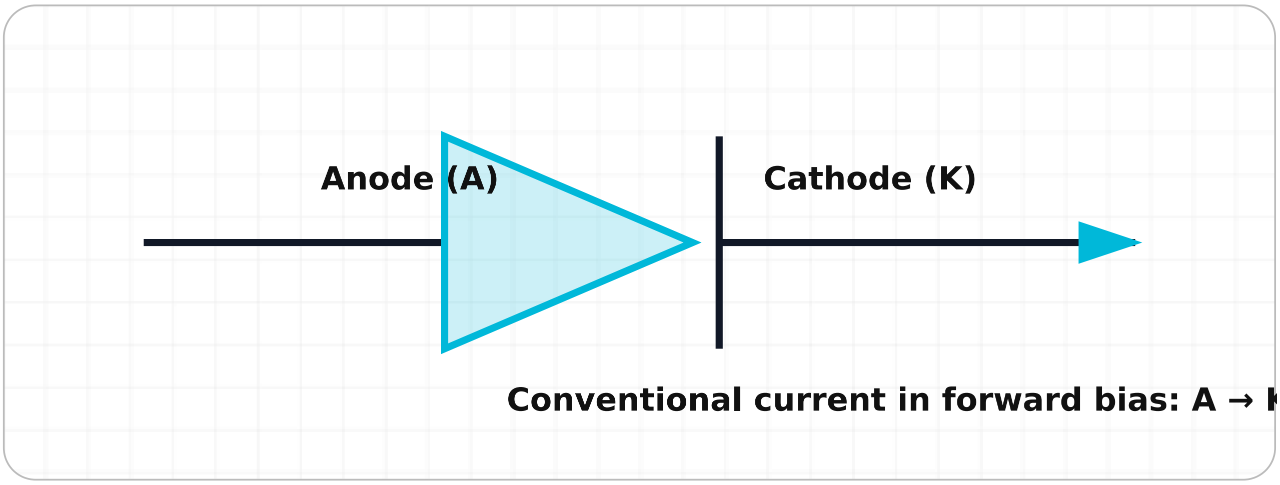
Intrinsic semiconductor. Thermal energy creates an electron-hole pair in pure semiconductor material.



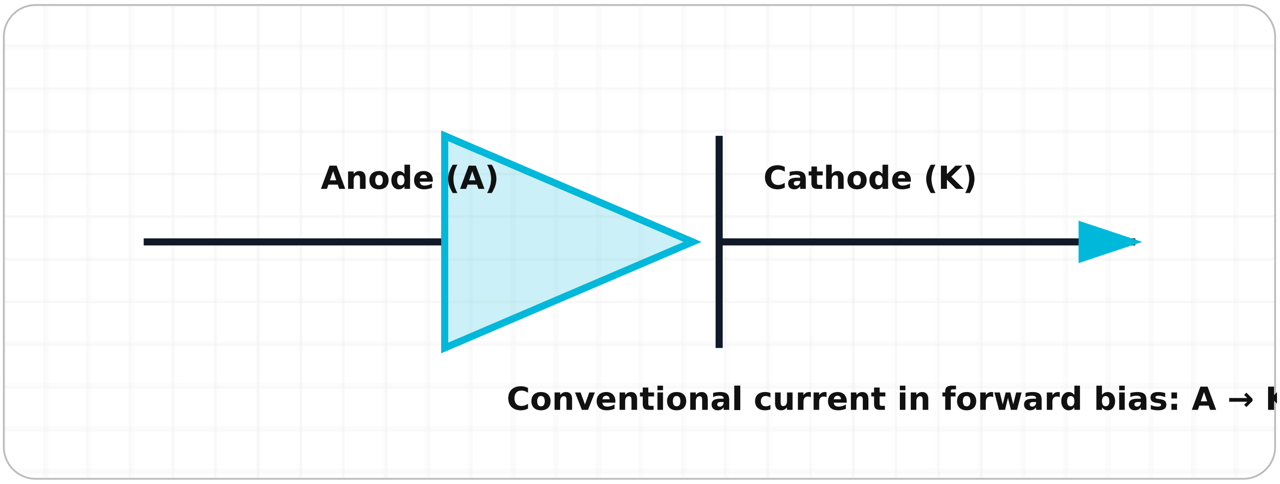
N-type semiconductor. Donor atoms create electrons as majority carriers while the material remains neutral overall.



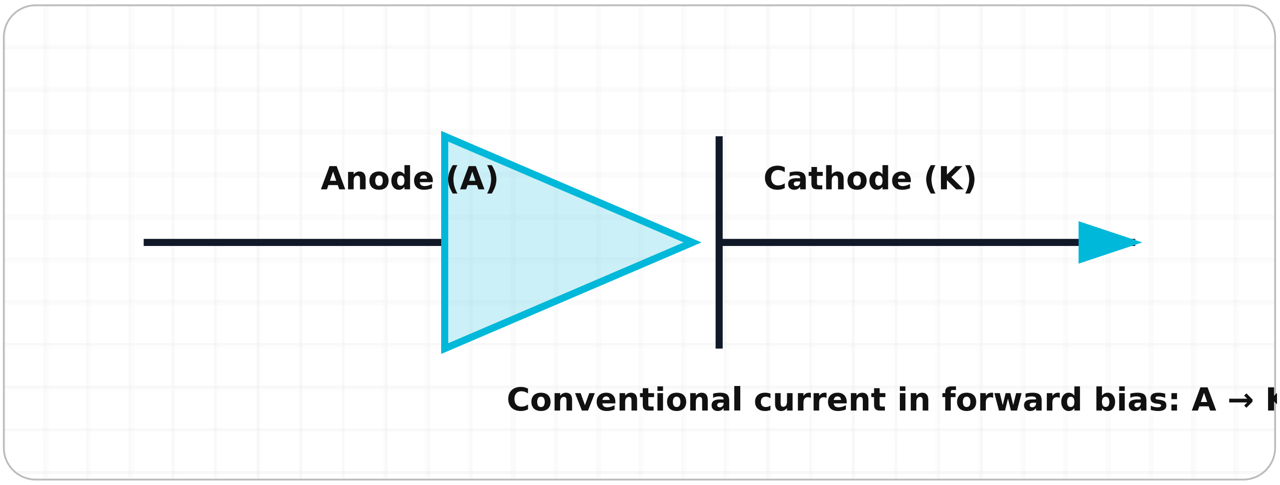
P-type semiconductor. Acceptor atoms create holes as majority carriers while the material remains neutral overall.



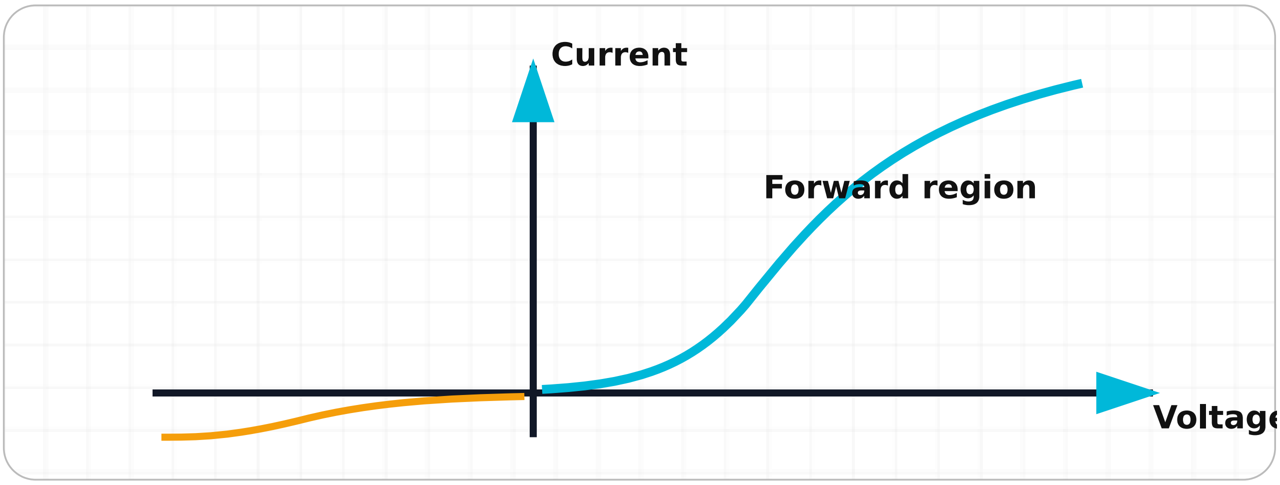
PN-junction formation. Carrier diffusion and recombination expose fixed ions and create a depletion region.



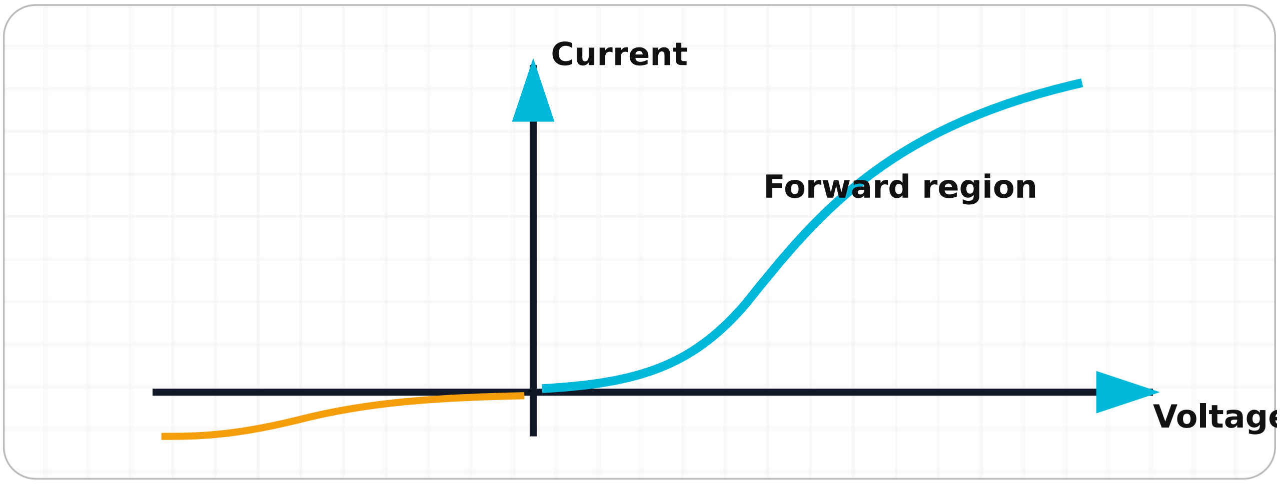
Forward-bias circuit. Positive supply at the anode reduces the barrier; a series resistor limits current.



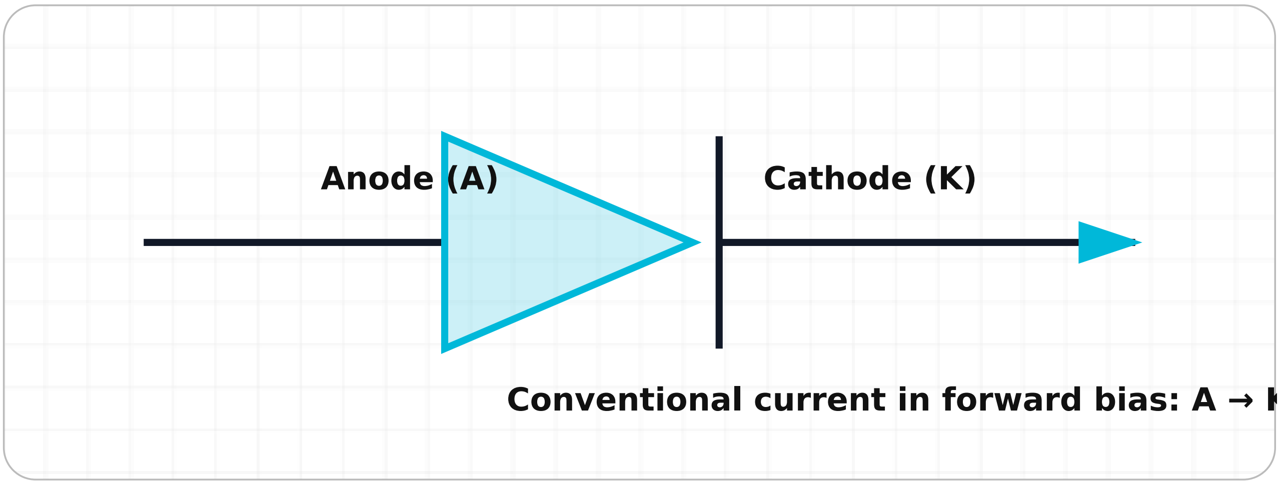
Reverse-bias circuit. Positive supply at the cathode widens the depletion region and only small leakage flows before breakdown.



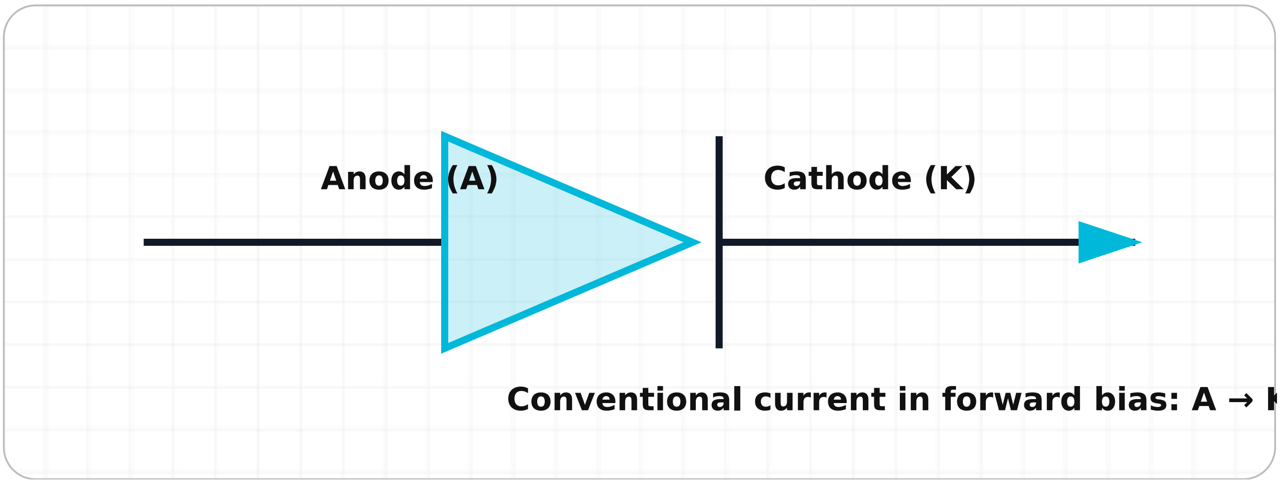
Diode V-I graph. Forward current rises rapidly beyond the knee while reverse current stays small before breakdown.



Zener V-I graph. The reverse characteristic enters controlled breakdown near the Zener voltage when current is limited.



LED connection. An LED is forward biased through a current-limiting resistor.



Photodiode connection. A photodiode is commonly reverse biased so incident light controls photocurrent.

MODULE 4 · CO4 · THEORY 10 H · PRACTICAL 12 H

PCB and Soldering

PCB fabrication-□□□ soldering-□□□ workmanship, ESD, electrical safety □□□□□□ result-□□□□□□□ □□□□□□□□□ □□□□□□□□□□.

Learning target: Explain the theory, interpret diagrams and measurements, apply formulae, perform the listed laboratory work, and answer one-, three- and seven-mark questions.

Core theory chapters (52)

1. Purpose of a PCB

A printed circuit board mechanically supports components and electrically interconnects them with patterned copper conductors.

How it works

PCBs improve repeatability, compactness, reliability, assembly speed, and service documentation.

Engineering practice

Trace power, ground, and signal paths on a simple board.

Common mistake / safety: A PCB drawing must distinguish copper side, component side, and viewing orientation.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□□□.

2. PCB structure

A PCB combines insulating substrate, copper conductors, holes or vias, protective coatings, markings, and component footprints.

How it works

Layer count and construction depend on electrical, mechanical, thermal, and cost requirements.

Engineering practice

Identify visible structural features before testing.

Common mistake / safety: Do not scrape or drill a board without considering hidden tracks and multilayers.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□□□.

3. Substrate

The substrate provides mechanical support and electrical insulation.

How it works

Common materials include FR-4 glass-reinforced epoxy; properties include dielectric strength, thermal rating, moisture resistance, and flame performance.

Engineering practice

Check board material and thickness where documented.

Common mistake / safety: Charred substrate can become conductive and is not reliably repaired by cleaning alone.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□.

4. Copper layer

Copper foil forms tracks, pads, planes, and shielding features.

How it works

Copper thickness and track geometry determine resistance, current capacity, and heat rise.

Engineering practice

Identify heavy-current tracks and compare width with signal tracks.

Common mistake / safety: A narrow track can act as an unintended fuse or hot spot.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□.

5. Solder mask

Solder mask is an insulating protective coating over copper except at intended solderable areas.

How it works

It reduces solder bridges, contamination, and corrosion.

Engineering practice

Inspect for lifted or burnt mask around repaired joints.

Common mistake / safety: Solder mask is not primary electrical insulation for hazardous spacing.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□□.

6. Silkscreen

Silkscreen identifies references, polarity, orientation, test points, warnings, and assembly information.

How it works

Clear legends reduce assembly and service errors.

Engineering practice

Verify component designators against the circuit diagram and BOM.

Common mistake / safety: Silkscreen polarity can be wrong on poor-quality boards; confirm with design data.

Malayalam explanation: □ concept □□□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□□ □□□□□□□□□□□□□□.

7. Tracks

Tracks are copper conductors connecting circuit nodes.

How it works

Width, thickness, length, temperature, and allowable voltage drop determine current capability.

Engineering practice

Use continuity mode only on de-energized boards.

Common mistake / safety: A continuity beep does not prove low enough resistance for a power path.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□.

8. Pads

Pads provide solderable connection areas for component leads, terminals, vias, or test points.

How it works

Pad diameter, hole size, annular ring, thermal relief, and mechanical stress determine reliability.

Engineering practice

Inspect for cracks and lifted pads under magnification.

Common mistake / safety: Repeated heating can detach a pad from the substrate.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□.

9. Vias

Vias are plated holes that connect copper between layers.

How it works

Through, blind, and buried vias serve different layer connections.

Engineering practice

Check via continuity when a track appears intact but a node is open.

Common mistake / safety: Drilling a multilayer via destroys hidden connections.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□.

10. Single-sided PCB

A single-sided PCB has copper on one side and components commonly mounted on the opposite side.

How it works

It is simple and economical but may require jumpers for routing.

Engineering practice

Identify component and solder sides.

Common mistake / safety: Viewing the underside mirrors left-right orientation relative to the component side.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□.

11. Double-sided PCB

A double-sided PCB has copper on both sides with plated holes or vias connecting them.

How it works

It supports denser routing and improved grounding.

Engineering practice

Inspect solder joints on both sides where accessible.

Common mistake / safety: Removing a plated-through component can damage the barrel connection.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□.

12. Multilayer PCB

A multilayer PCB contains internal copper layers laminated within the board.

How it works

Internal layers commonly form power planes, ground planes, or dense signal routing.

Engineering practice

Use design files or X-ray methods for full visibility.

Common mistake / safety: Do not assume an apparently unused hole or area has no internal connection.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□.

13. PCB materials

PCB material selection affects electrical loss, temperature capability, mechanical strength, fire performance, and cost.

How it works

FR-4 is common; specialised materials serve high-frequency, high-temperature, flexible, or metal-core applications.

Engineering practice

Match material to operating environment.

Common mistake / safety: Material colour alone is not a reliable specification.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□.

14. FR-4

FR-4 is a flame-retardant glass-reinforced epoxy laminate widely used for rigid PCBs.

How it works

It offers good general mechanical and electrical performance.

Engineering practice

Recognize woven glass texture at cut edges where safe.

Common mistake / safety: FR-4 grade and temperature rating vary; use documentation.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□□□ □□□□□□□□□□□□□.

15. Copper-clad laminate

Copper-clad laminate is substrate bonded with copper foil before patterning.

How it works

PCB fabrication selectively removes or builds copper to create the circuit.

Engineering practice

Handle edges carefully and prevent oxidation before processing.

Common mistake / safety: Etchants and residues require controlled handling and disposal.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

16. Component placement

Placement determines routing quality, signal integrity, thermal performance, assembly, and service access.

How it works

Place connectors by mechanical constraints, critical parts close together, and heat sources with clearance.

Engineering practice

Review orientation, access, and wire-entry direction.

Common mistake / safety: A circuit can be electrically correct but mechanically unserviceable.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

17. Track width

Track width is selected from current, copper thickness, temperature rise, length, and manufacturing limits.

How it works

Power tracks are normally wider than signal tracks.

Engineering practice

Use a documented design rule rather than guessing.

Common mistake / safety: Neck-down sections and connector pads can be the weakest point.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□.

18. Pad size

Pad size must support soldering, annular ring, mechanical strength, and manufacturing tolerance.

How it works

Hole and lead dimensions determine through-hole pad geometry.

Engineering practice

Check lead fit before assembly.

Common mistake / safety: Oversized holes reduce joint quality and pad strength.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□.

19. Electrical clearance

Clearance is the shortest distance through air between conductive parts.

How it works

Required clearance increases with voltage, pollution, transient category, and standards.

Engineering practice

Measure actual conductor-to-conductor distance, not only nominal grid spacing.

Common mistake / safety: Solder protrusions and component leads can reduce clearance.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□□□.

20. Creepage

Creepage is the shortest distance along an insulating surface between conductive parts.

How it works

It depends strongly on working voltage, pollution degree, and material group.

Engineering practice

Inspect contamination paths and slots.

Common mistake / safety: Clearance and creepage are different and both must be satisfied.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□□□.

21. PCB routing

Routing connects placed components while controlling current paths, noise, crosstalk, manufacturability, and test access.

How it works

Short direct routes help many signals, but return paths are equally important.

Engineering practice

Trace complete loops, not only outgoing tracks.

Common mistake / safety: Autorouting does not remove the need for engineering review.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□.

22. Ground-return paths

Current returns to its source through a complete loop; ground is not an infinite sink.

How it works

High-frequency return current follows the path of lowest impedance, often near the signal path.

Engineering practice

Identify signal and return together.

Common mistake / safety: Splitting ground without understanding return paths can worsen noise.

Malayalam explanation: □ concept □□□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

23. Power tracks

Power tracks distribute energy and must meet current, voltage-drop, heat, protection, and fault requirements.

How it works

Bulk and local decoupling support transient load demand.

Engineering practice

Check connector, fuse, track, and return-path ratings as one chain.

Common mistake / safety: A wide positive track with a narrow return track is still a weak circuit.

Malayalam explanation: Power energy conversion rate □□□; Energy = Power × Time. W, Wh, kWh, J □□□□□ units □□□□□□ mix □□□□□□□□ conversion □□□□□□□.

24. Connector placement

Connectors must satisfy wiring access, keying, strain relief, creepage, current, and service orientation.

How it works

Silkscreen labels and pin-1 indication reduce wiring errors.

Engineering practice

Compare board connector orientation with the harness drawing.

Common mistake / safety: Board-mounted connectors should not carry unmanaged cable stress.

Malayalam explanation: □ concept □□□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

25. Layout preparation

Layout preparation converts schematic connectivity and mechanical constraints into footprints, placement, routing, and fabrication data.

How it works

Design-rule checks detect many but not all errors.

Engineering practice

Verify netlist, footprints, polarity, holes, board outline, and test points.

Common mistake / safety: A correct schematic does not guarantee correct footprint pin numbering.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□.

26. Artwork transfer

Artwork transfer places the required copper pattern or resist image onto copper-clad material.

How it works

Manual educational methods may use toner transfer or photoresist under controlled conditions.

Engineering practice

Inspect for breaks, pinholes, and unwanted bridges before etching.

Common mistake / safety: Reversed artwork orientation is a common fabrication error.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

27. Etching

Etching removes unwanted exposed copper while protected copper remains.

How it works

Time, temperature, agitation, concentration, and resist quality affect results.

Engineering practice

Use PPE, ventilation, labelled containers, and approved disposal.

Common mistake / safety: Never pour etchant into ordinary drains or mix unknown chemicals.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

28. Cleaning

After etching, resist and residues are removed and copper is cleaned for inspection and soldering.

How it works

Clean surfaces improve solderability and reveal defects.

Engineering practice

Use approved cleaners and dry the board fully.

Common mistake / safety: Abrasive cleaning can remove thin tracks or damage pads.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

29. Drilling

Drilling creates accurate component and mounting holes.

How it works

Correct bit size, speed, support, eye protection, and dust control are required.

Engineering practice

Drill perpendicular and inspect annular rings.

Common mistake / safety: Fibreglass dust is hazardous; avoid inhalation and uncontrolled dry brushing.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□□ □□□□□□□□□□□□□.

30. Component insertion

Components are inserted according to reference, value, orientation, polarity, and lead-forming rules.

How it works

Proper lead spacing avoids stress on the component body and pads.

Engineering practice

Use the BOM and assembly drawing.

Common mistake / safety: Do not bend leads sharply at the component body.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□□ □□□□□□□□□□□□□.

31. Component mounting

Mounting controls component height, heat clearance, mechanical support, and serviceability.

How it works

Heavy or hot components may require brackets, spacing, adhesive, or fasteners.

Engineering practice

Check connector seating and polarity before soldering.

Common mistake / safety: Solder should not be used as the only mechanical support for heavy parts.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

32. PCB inspection

Inspection checks workmanship, correct parts, polarity, solder quality, contamination, damage, and spacing.

How it works

Use visual inspection, continuity checks, and functional tests in a controlled sequence.

Engineering practice

Inspect before powering to catch low-cost errors.

Common mistake / safety: Powering first can turn a simple assembly error into major damage.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□.

33. Soldering iron

A soldering iron supplies controlled heat to the joint.

How it works

Temperature control, tip size, power, grounding, and maintenance affect results.

Engineering practice

Select a tip that transfers heat efficiently without contacting adjacent pads.

Common mistake / safety: Excessive temperature accelerates flux burning, tip wear, and board damage.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

34. Solder wire

Solder wire is an alloy, often with a flux core, that wets clean metal surfaces.

How it works

Alloy composition and diameter suit different processes; lead-free solder often needs higher temperature.

Engineering practice

Use only the approved alloy for the product and process.

Common mistake / safety: Lead-containing solder requires strict hygiene and process control.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

35. Flux

Flux removes oxides and promotes wetting during soldering.

How it works

Activity level and residue requirements determine cleaning needs.

Engineering practice

Apply the minimum suitable amount and follow the process specification.

Common mistake / safety: Flux fumes require extraction; 'no-clean' does not mean harmless.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

36. Iron stand

An iron stand safely supports the hot tool and usually includes tip-cleaning media.

How it works

It prevents burns, cable damage, and bench fires.

Engineering practice

Return the iron to the stand immediately after each joint.

Common mistake / safety: Never place a hot iron directly on the bench.

Malayalam explanation: □ concept □□□□□□□□□□□ Definition, Unit, Symbol, Working Principle, Circuit/Diagram, Application, Safety □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□.

37. Tip cleaner

Brass wool or a damp sponge removes residue from the soldering tip.

How it works

Cleaning followed by light tinning preserves heat transfer and reduces oxidation.

Engineering practice

Clean only as needed; maintain a bright tinned surface.

Common mistake / safety: Aggressive scraping removes protective plating.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

38. Desoldering pump

A desoldering pump uses suction to remove molten solder.

How it works

It is effective for through-hole joints when synchronized with heating.

Engineering practice

Reflow the joint, heat fully, then trigger suction without levering the pad.

Common mistake / safety: Repeated suction and heating can damage pads and plated holes.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

39. Desoldering wick

Copper braid with flux absorbs molten solder by capillary action.

How it works

Correct braid width, fresh flux, and controlled pressure improve removal.

Engineering practice

Place braid on the joint and the iron on the braid.

Common mistake / safety: Pulling before solder solidifies can lift pads.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

40. Tip tinning

Tip tinning coats the iron tip with a thin layer of solder.

How it works

It protects against oxidation and improves thermal contact.

Engineering practice

Tin the tip after cleaning and before storage.

Common mistake / safety: A black dry tip transfers heat poorly and should be restored according to manufacturer guidance.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

41. Component soldering

A reliable joint is made by heating the pad and lead together, applying solder to the joint, removing solder, then removing heat.

How it works

Wetting and intermetallic formation require clean surfaces and correct time-temperature exposure.

Engineering practice

Use a stable sequence and inspect after cooling.

Common mistake / safety: Do not melt solder on the tip and wipe it onto a cold joint.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

42. Desoldering

Desoldering removes components while preserving pads, barrels, tracks, and the component if required.

How it works

Add fresh solder or flux, control heat, remove solder, and release all leads before lifting.

Engineering practice

Verify each terminal is free before applying force.

Common mistake / safety: Mechanical force while solder remains solid causes lifted pads and broken vias.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

43. Good solder joint

A good through-hole joint wets both lead and pad, has a smooth concave fillet, adequate coverage, and no bridge or movement.

How it works

Appearance criteria vary with alloy, but wetting and geometry remain essential.

Engineering practice

Inspect under good lighting and magnification.

Common mistake / safety: Shiny appearance alone does not guarantee a good lead-free joint.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

44. Dry joint

A dry joint has poor wetting, often caused by oxidation, contamination, insufficient heat, or movement.

How it works

It can produce high resistance or intermittent operation.

Engineering practice

Rework by cleaning, fluxing, and reheating correctly.

Common mistake / safety: Adding more solder without correcting the cause may hide the defect.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

45. Cold joint

A cold joint forms when the metals do not reach suitable soldering temperature or the joint moves during solidification.

How it works

It may look grainy, irregular, or poorly wetted.

Engineering practice

Stabilize the work and heat both surfaces.

Common mistake / safety: Do not use excessive dwell time to compensate for an undersized or dirty tip.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

46. Solder bridge

A solder bridge unintentionally connects adjacent conductive points.

How it works

It can cause shorts, wrong logic states, or component damage.

Engineering practice

Remove with clean tip, wick, or controlled rework and re-inspect.

Common mistake / safety: Power must remain off until the bridge and any hidden splashes are cleared.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

47. Insufficient solder

Insufficient solder leaves incomplete wetting or inadequate mechanical/electrical contact.

How it works

The pad or hole may not be sufficiently filled.

Engineering practice

Add solder only after reheating pad and lead together.

Common mistake / safety: More solder is not a substitute for proper wetting.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

48. Excess solder

Excess solder obscures joint geometry, can hide poor wetting, and increases bridge risk.

How it works

The objective is adequate—not maximum—solder.

Engineering practice

Remove excess and inspect the actual fillet.

Common mistake / safety: Large blobs make troubleshooting harder.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

49. Lifted pad

A lifted pad separates from the substrate due to excessive heat, force, repeated rework, or poor adhesion.

How it works

Repair requires restoring electrical and mechanical integrity according to approved procedure.

Engineering practice

Stop heating as soon as damage is observed.

Common mistake / safety: Do not glue a pad and assume conductivity is restored.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□.

50. Damaged track

A track may crack, burn, corrode, or detach.

How it works

Repair requires finding the full damaged length, cause, current requirement, and secure replacement path.

Engineering practice

Check continuity and voltage drop after repair.

Common mistake / safety: A wire link must be correctly sized, routed, insulated, and mechanically secured.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□□.

51. ESD precautions

Electrostatic discharge can damage sensitive devices without visible evidence.

How it works

Controls include grounded work surfaces, wrist straps, antistatic packaging, humidity management, and handling by edges.

Engineering practice

Verify the ESD workstation before opening protected components.

Common mistake / safety: A wrist strap must use approved resistance and must not be used near exposed hazardous voltage.

Malayalam explanation: ESD □□□□□□ □□□□□□□□□□□□ □□□□ semiconductor devices damage □□□□□□□. Grounded mat, approved wrist strap, antistatic packaging □□□□□□ □□□□□□□□□□□.

52. Soldering safety

Soldering hazards include burns, fumes, sharp leads, splashes, chemicals, electricity, and fire.

How it works

Use extraction, eye protection, stand, housekeeping, hand washing, and controlled materials.

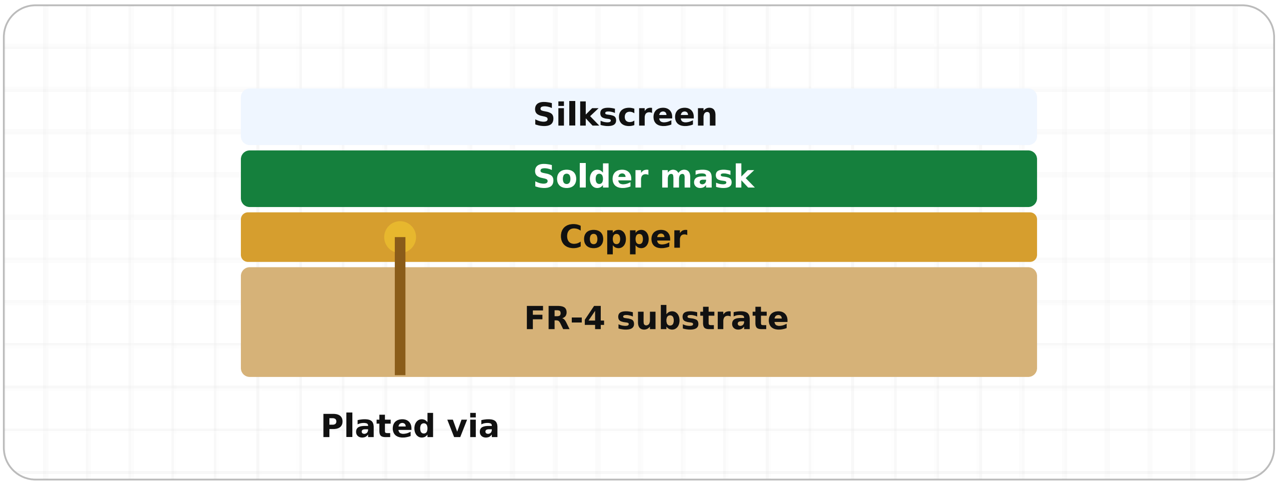
Engineering practice

Keep cables clear and know first-aid and emergency procedures.

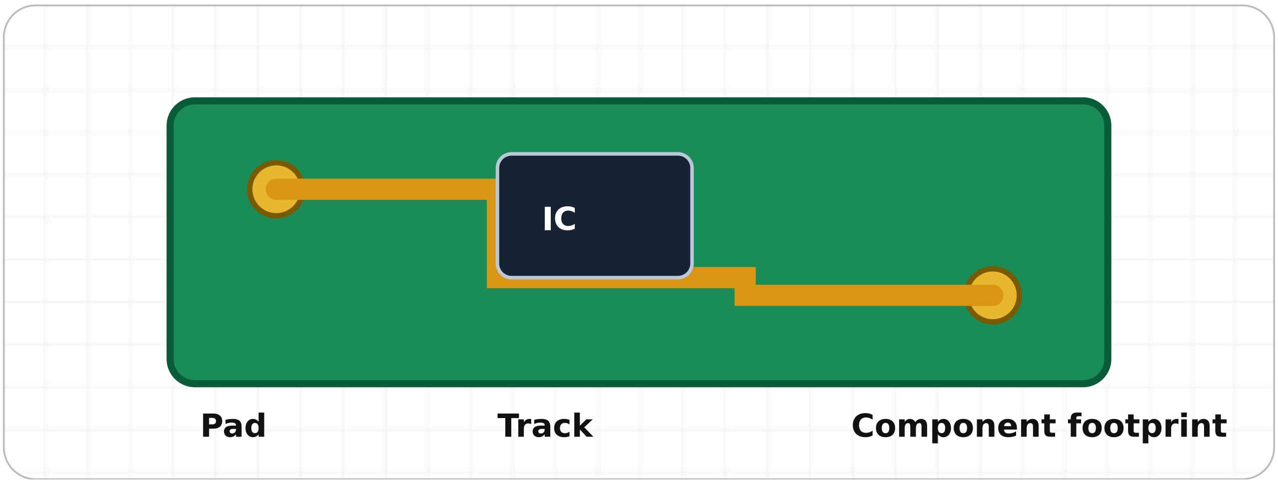
Common mistake / safety: Never eat or drink at the soldering bench; wash hands after handling solder and flux.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

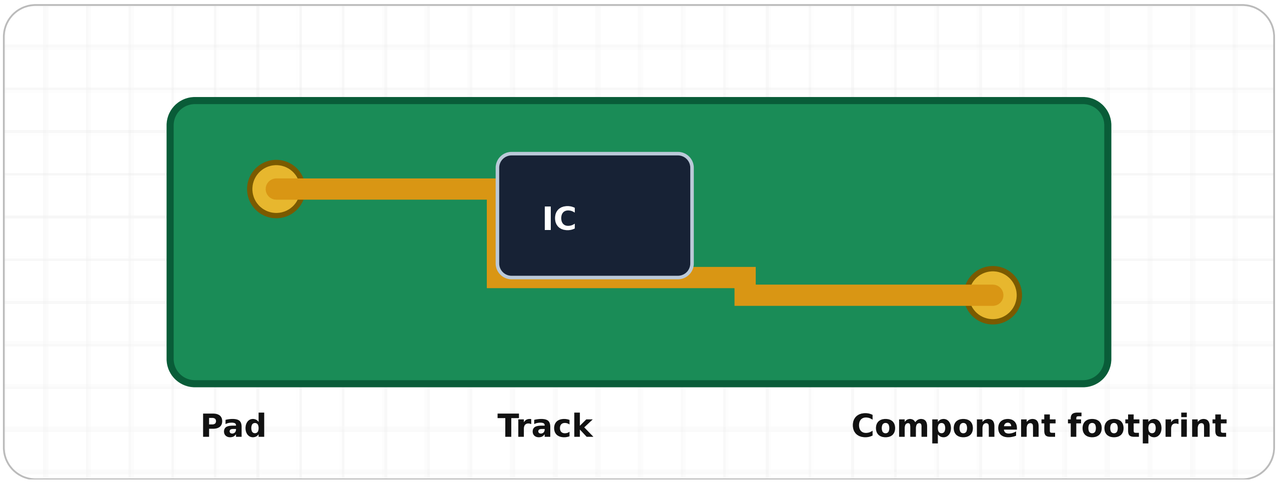
Diagram reference



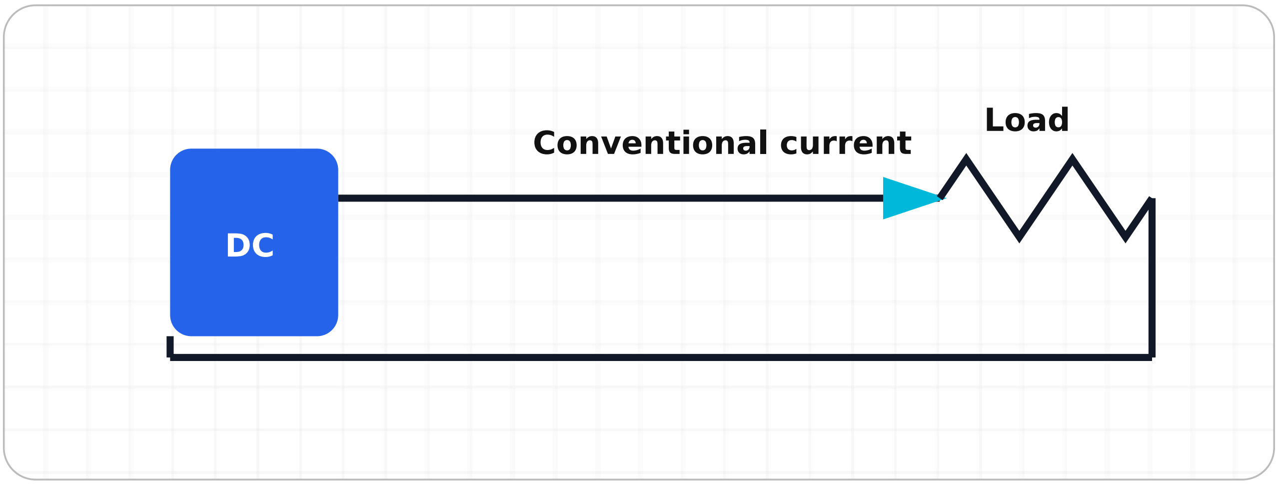
PCB layers. Silkscreen, solder mask, copper, substrate, and vias form the board structure.



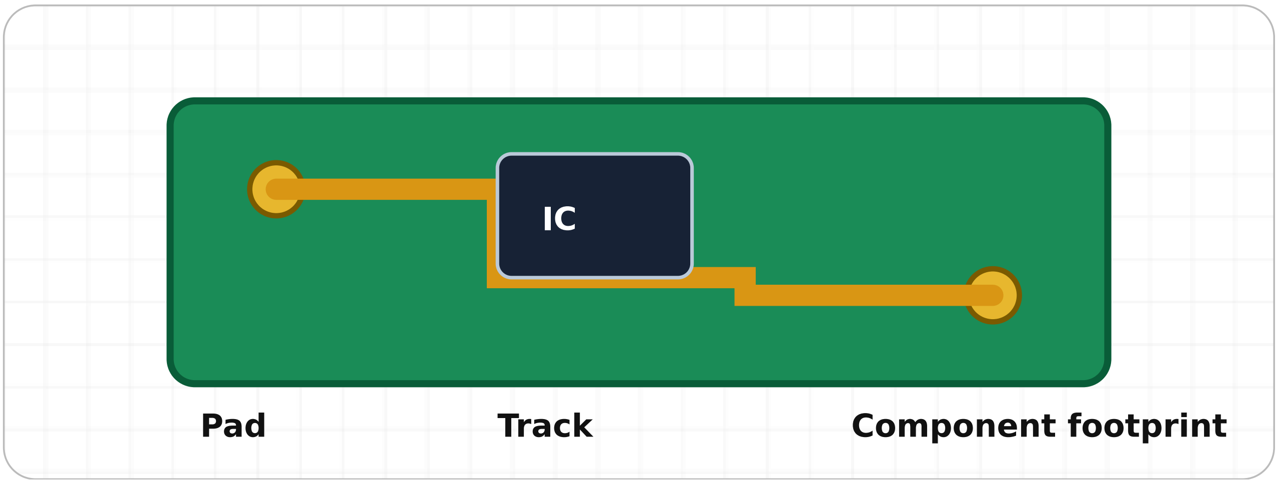
PCB fabrication stages. Artwork, transfer, etching, cleaning, drilling, mounting, soldering, and inspection form a controlled sequence.



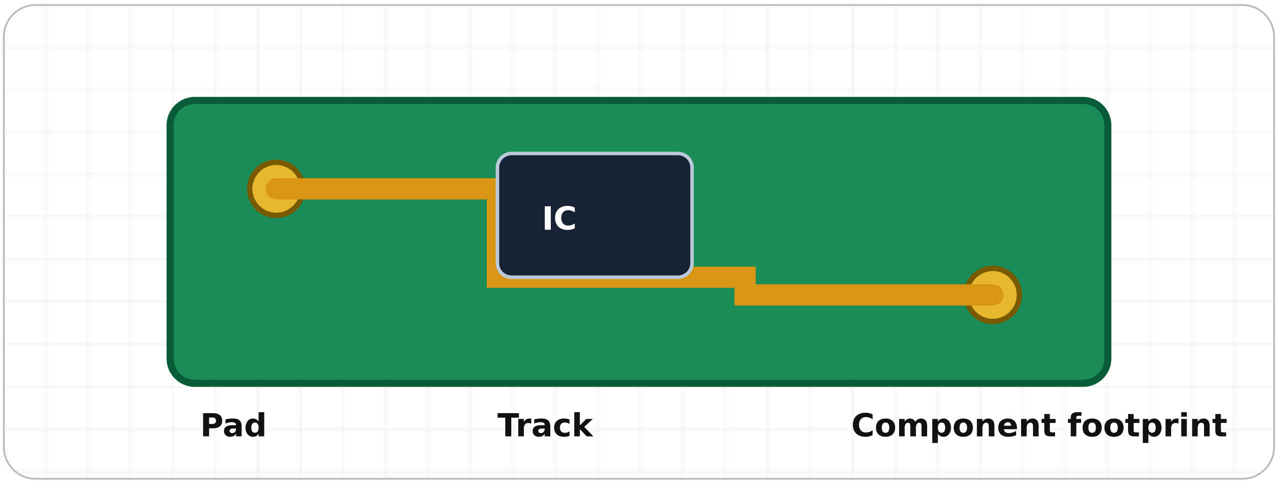
PCB tracks and pads. Tracks carry current and pads provide solderable component connections.



Soldering tools. Iron, stand, tip cleaner, solder, flux, pump, wick, cutter, and extraction support safe work.



Good solder joint. A reliable through-hole joint wets the lead and pad with a smooth concave fillet.



Defective solder joints. Dry joints, bridges, insufficient solder, excess solder, and lifted pads have distinct causes and remedies.

Practical Experiments and Record Format

Use only approved low-voltage training arrangements unless a supervised procedure explicitly requires otherwise. Mains work is not a beginner experiment.

Electronics laboratory familiarisation and daily-life system identification

Date: ________________	Batch: _________	Roll No.: _________

Aim

Familiarise learners with laboratory zones, emergency controls, instruments, safe work practices, and the functional blocks of common electronic systems.

Objectives

Identify the relevant equipment and components, make correct connections, take systematic observations, interpret the result, and work safely.

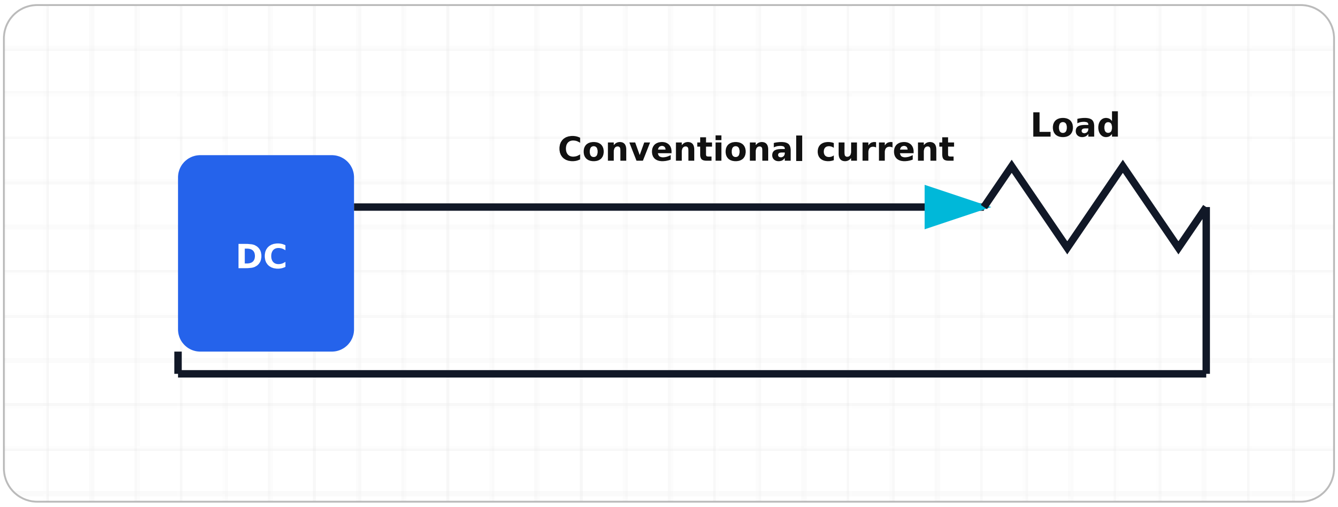
Apparatus

Laboratory safety chart, DMM, DC supply, function generator, DSO, sample consumer/industrial electronic systems or photographs.

Components

Representative resistor, capacitor, diode, LED, PCB, connector and fuse samples.

Circuit / connection diagram



Electronics laboratory familiarisation and daily-life system identification connection. Generic safe low-voltage arrangement. Follow the specific approved circuit and instructor direction for the experiment.

Theory

A safe electronics laboratory separates hazardous supply work from extra-low-voltage experiments. Every system can be described using source, input, processing, output and feedback blocks.

Procedure

1. Locate the main isolator, emergency switch, fire extinguisher, first-aid point and exits.
2. Inspect benches, earthing points, ESD facilities and instrument ratings.
3. Identify each instrument and state its purpose, terminals and one major precaution.
4. Select five daily-life electronic systems and prepare functional block diagrams.
5. Record observations without energising unidentified equipment.

Observation table

Item/system	Input/source	Processing/control	Output	Safety feature

Calculations / graph

Show formulas, substitutions, units, percentage difference and graph axes where applicable.

Result

Laboratory facilities and common electronic systems were identified and documented.

Precautions: Do not operate an instrument before checking its category, range, leads, earth reference and instructor approval.

Common errors: Confusing a protective earth terminal with a signal common; listing product names without identifying functional blocks.

Troubleshooting

If an instrument label is unclear, isolate it and consult the manual or instructor instead of guessing.

Viva questions

Why is the DSO earth clip safety-critical?

It may be connected to protective earth and can short a live non-isolated node.

What is the first action before an experiment?

Understand the circuit, expected values, hazards and instrument limits.

Record-writing format

Write: date → experiment number/title → aim → apparatus/components → neat labelled circuit → brief theory/formula → numbered procedure → observation table → sample calculation → graph → result → precautions.

Student signature: __________________ Faculty signature: __________________

EXPERIMENT 2 · MODULE 1

Identification of AC and DC sources and phase/neutral demonstration

Date: __________________ Batch: __________ Roll No.: __________

Aim

Classify electrical sources and safely demonstrate phase and neutral identification on an approved trainer.

Objectives

Identify the relevant equipment and components, make correct connections, take systematic observations, interpret the result, and work safely.

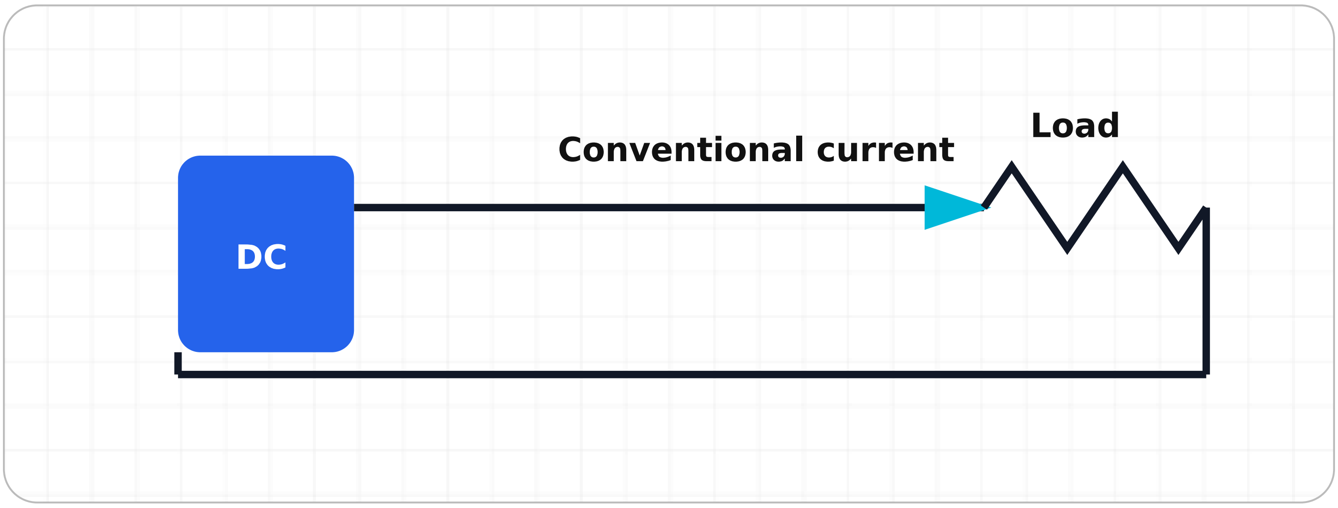
Apparatus

Approved single-phase trainer with RCD/MCB, line tester or approved voltage detector, DMM rated for the installation category.

Components

Low-voltage AC transformer output, battery, regulated DC supply.

Circuit / connection diagram



Identification of AC and DC sources and phase/neutral demonstration connection. Generic safe low-voltage arrangement. Follow the specific approved circuit and instructor direction for the experiment.

Theory

DC has fixed polarity; AC changes polarity periodically. Phase is live relative to neutral and earth. A line tester is an indication device, not a complete proof of de-energisation.

Procedure

1. Inspect instrument and lead ratings; confirm instructor supervision.
2. Classify battery, DC supply and isolated transformer output by nameplate.
3. Measure low-voltage DC with DC mode and low-voltage AC with AC mode.
4. On the approved mains trainer only, demonstrate phase indication using the prescribed detector.
5. Verify conclusions with the approved two-pole/DMM method as instructed and record results.

Observation table

Source	Rated value	Meter mode	Measured value	AC/DC	Polarity or phase indication

Calculations / graph

Show formulas, substitutions, units, percentage difference and graph axes where applicable.

Result

The available sources were classified and measured; phase and neutral were demonstrated on an approved training setup.

> **Precautions:** Never use a bare improvised circuit, never touch conductors, and never rely on a neon line tester alone to prove isolation.

> **Common errors:** Wrong meter mode, current lead left in current jack, holding probe beyond finger guard.

Troubleshooting

Unexpected readings require stopping, isolating and checking reference, range, source and protective devices.

Viva questions

Can a line tester prove a conductor is dead?

No. Use an approved verification method and safe isolation procedure.

Why measure AC and DC on different modes?

The meter processes the waveform differently and wrong mode gives misleading results.

Record-writing format

Write: date → experiment number/title → aim → apparatus/components → neat labelled circuit → brief theory/formula → numbered procedure → observation table → sample calculation → graph → result → precautions.

Student signature: ________________ Faculty signature: ________________

EXPERIMENT 3 · MODULE 1

Identification and basic operation of electronic instruments

Date: ________________ Batch: _________ Roll No.: _________

Aim

Identify controls and demonstrate safe basic operation of DMM, DC supply, function generator and DSO.

Objectives

Identify the relevant equipment and components, make correct connections, take systematic observations, interpret the result, and work safely.

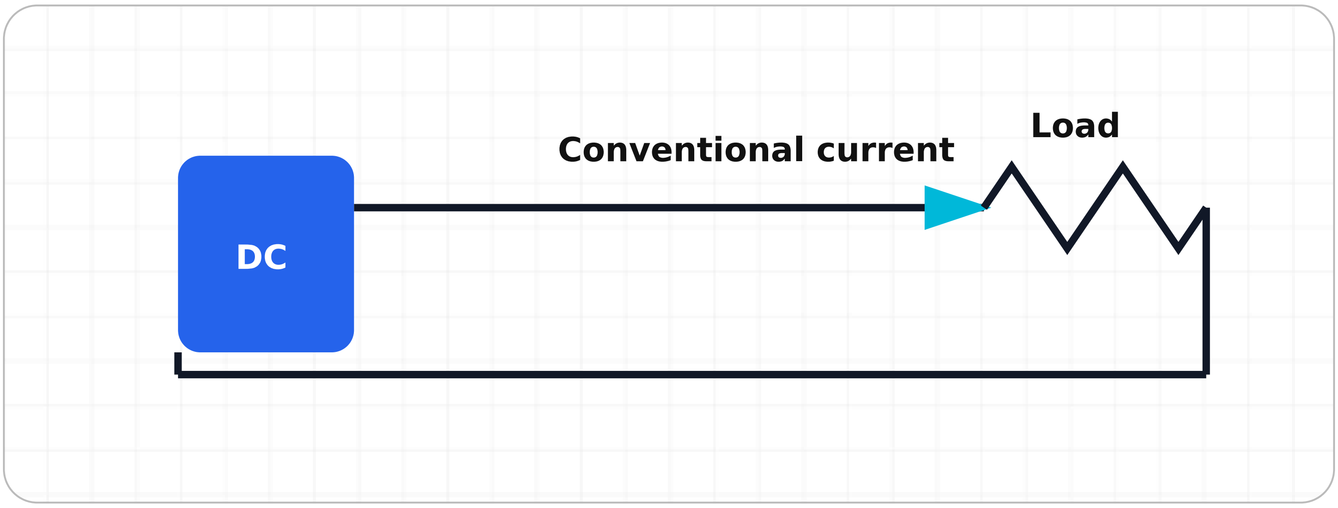
Apparatus

DMM, regulated DC supply, function generator, DSO and x10 probe.

Components

1 kΩ resistor and LED test circuit for low-voltage demonstration.

Circuit / connection diagram



Identification and basic operation of electronic instruments connection. Generic safe low-voltage arrangement. Follow the specific approved circuit and instructor direction for the experiment.

Theory

Correct function, range, terminal, probe ratio, coupling and trigger settings are essential to measurement validity.

Procedure

1. Inspect all leads and power cords.
2. Set the DC supply voltage and current limit with output disabled.
3. Generate a 1 kHz, 2 Vpp sine wave with zero offset.
4. Connect the function generator to the DSO using a common reference.
5. Set probe ratio, vertical scale, time base and trigger; verify amplitude and frequency.
6. Use DMM voltage mode to check a low-voltage DC output.

Observation table

Instrument	Control/terminal	Setting	Observed output	Precaution

Calculations / graph

Show formulas, substitutions, units, percentage difference and graph axes where applicable.

Result

The four instruments were identified and operated using safe low-voltage settings.

Precautions: Never connect the DSO ground to an unknown live circuit; verify output limits before connecting a load.

Common errors: Probe set to x10 while DSO set to x1, incorrect generator load assumption, current terminal misuse.

Troubleshooting

For an unstable waveform, check trigger source, level, coupling, ground and time base.

Viva questions

Why set current limit before enabling a supply?

To restrict fault current and protect the circuit.

What does x10 probe setting affect?

Displayed voltage scaling, loading and bandwidth.

Record-writing format

Write: date → experiment number/title → aim → apparatus/components → neat labelled circuit → brief theory/formula → numbered procedure → observation table → sample calculation → graph → result → precautions.

Student signature: ________________ Faculty signature: ________________

EXPERIMENT 4 · MODULE 1

Measurement of sinusoidal AC signal parameters

Date: ________________ Batch: _________ Roll No.: _________

Aim

Generate a sine wave and measure peak, peak-to-peak, time period and frequency using a DSO.

Objectives

Identify the relevant equipment and components, make correct connections, take systematic observations, interpret the result, and work safely.

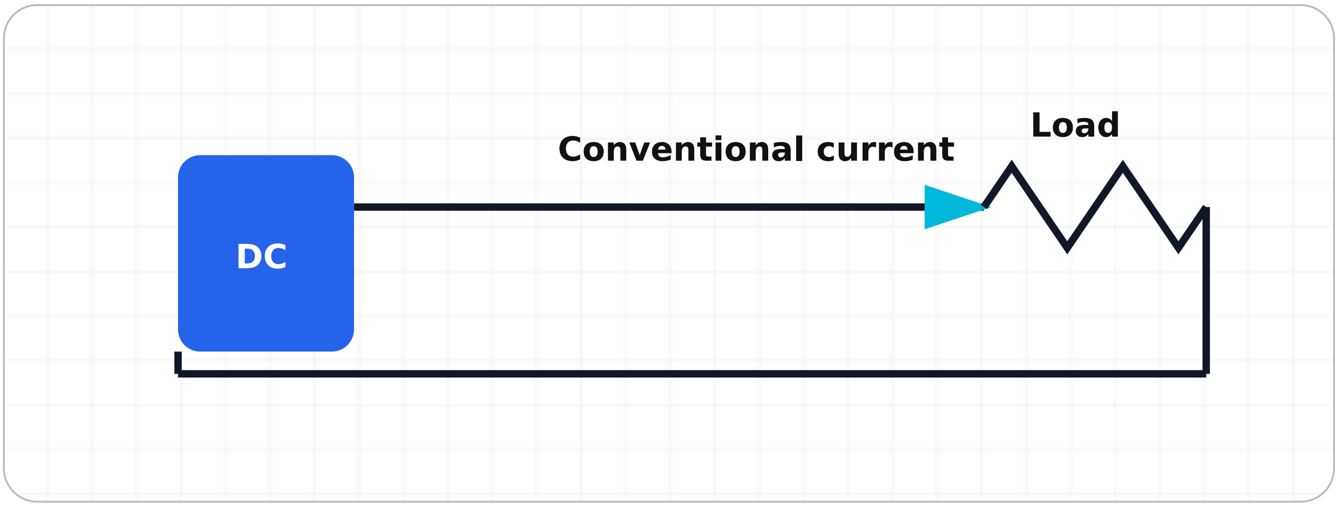
Apparatus

Function generator, DSO, compensated probe or BNC lead.

Components

No external component required for direct low-voltage signal measurement.

Circuit / connection diagram



Measurement of sinusoidal AC signal parameters connection. Generic safe low-voltage arrangement. Follow the specific approved circuit and instructor direction for the experiment.

Theory

Vpp equals the number of vertical divisions times volts/div. T equals horizontal divisions times time/div. f = 1/T.

Procedure

1. Set generator to sine wave, approximately 1 kHz and 4 Vpp with zero offset.
2. Connect generator output to DSO input with common reference.
3. Adjust vertical and horizontal scales for at least two clear cycles.
4. Stabilise with edge trigger.
5. Measure Vpp and T manually from divisions and with DSO cursors/automatic functions.
6. Calculate f and percentage difference from generator display.

Observation table

Trial	V/div	Vertical divisions	Vpp	Time/div	Horizontal divisions	T	Calculated f	Generator f	% error

Calculations / graph

Show formulas, substitutions, units, percentage difference and graph axes where applicable.

Result

Amplitude, time period and frequency of the sine wave were measured and compared.

Precautions: Use low voltage; confirm common ground; do not exceed DSO input rating.

Common errors: Using half-cycle as T, failing to convert ms to s, reading amplitude as Vpp.

Troubleshooting

If clipped, reduce amplitude or offset; if noisy, improve grounding and scale.

Viva questions

How are frequency and time period related?

They are reciprocals: f = 1/T.

What does triggering do?

It synchronises waveform acquisition to a repeatable event for a stable display.

Record-writing format

Write: date → experiment number/title → aim → apparatus/components → neat labelled circuit → brief theory/formula → numbered procedure → observation table → sample calculation → graph → result → precautions.

Student signature: ________________ Faculty signature: ________________

EXPERIMENT 5 · MODULE 2

Identification and testing of passive components

Date: ________________ Batch: _________ Roll No.: _________

Aim

Identify resistor, potentiometer, LDR, VDR, thermistor, capacitor, inductor and transformer samples and perform suitable de-energised checks.

Objectives

Identify the relevant equipment and components, make correct connections, take systematic observations, interpret the result, and work safely.

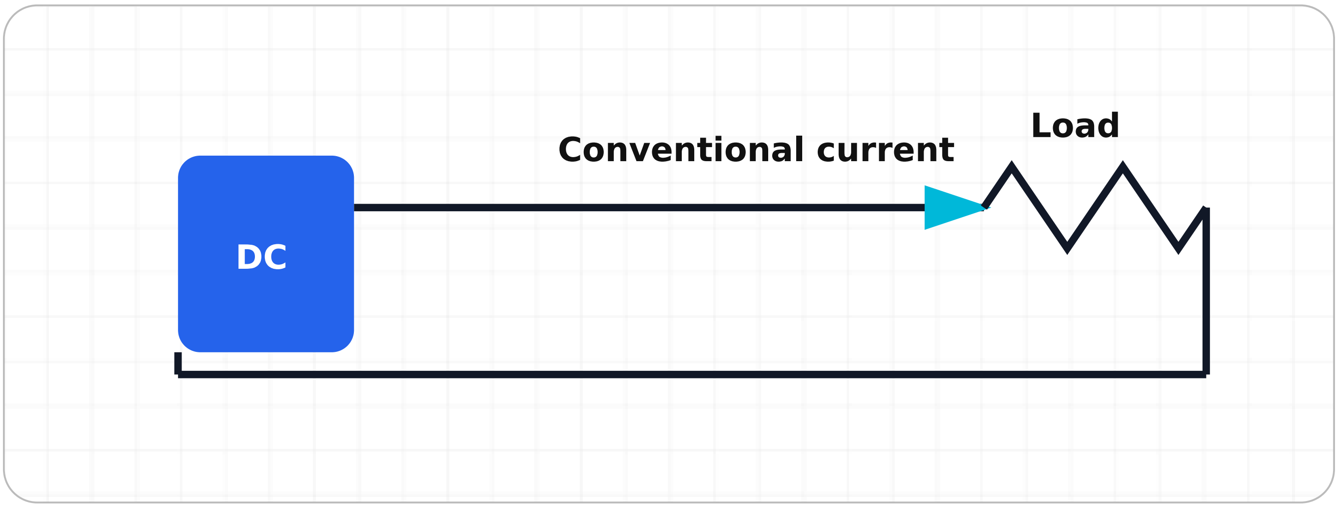
Apparatus

DMM with resistance and capacitance functions, component tray, datasheets.

Components

Assorted passive components.

Circuit / connection diagram



Identification and testing of passive components connection. Generic safe low-voltage arrangement. Follow the specific approved circuit and instructor direction for the experiment.

Theory

Component identity depends on symbol, construction, markings, terminals, polarity, ratings and measured behaviour.

Procedure

1. Sort components by physical construction without energising.
2. Read and record markings and polarity.
3. Measure resistance of fixed/variable resistors and sensor changes under safe stimulus.
4. Measure capacitance after safe discharge.
5. Check inductor/transformer winding continuity without applying mains.
6. Compare measured values with markings and tolerance.

Observation table

Component	Marking	Polarity/terminals	Rated value	Measured value	Condition

Calculations / graph

Show formulas, substitutions, units, percentage difference and graph axes where applicable.

Result

Passive components were identified and their basic condition verified.

Precautions: Discharge capacitors; do not apply mains to unknown transformers; avoid heating thermistors excessively.

Common errors: Measuring capacitance in-circuit, confusing VDR with capacitor, using continuity alone as a full test.

Troubleshooting

If reading is unstable, clean contacts, isolate the component and select a suitable range.

Viva questions

Why must a capacitor be discharged?

Stored energy can injure the user or damage the meter.

How is a potentiometer checked?

Measure end-to-end resistance and smooth wiper resistance to each end.

Record-writing format

Write: date → experiment number/title → aim → apparatus/components → neat labelled circuit → brief theory/formula → numbered procedure → observation table → sample calculation → graph → result → precautions.

Student signature: ________________ Faculty signature: ________________

EXPERIMENT 6 · MODULE 2

Resistor colour-code decoding and value verification

Date: ________________ Batch: _________ Roll No.: _________

Aim

Decode 4-band and 5-band resistor values and verify them with a DMM.

Objectives

Identify the relevant equipment and components, make correct connections, take systematic observations, interpret the result, and work safely.

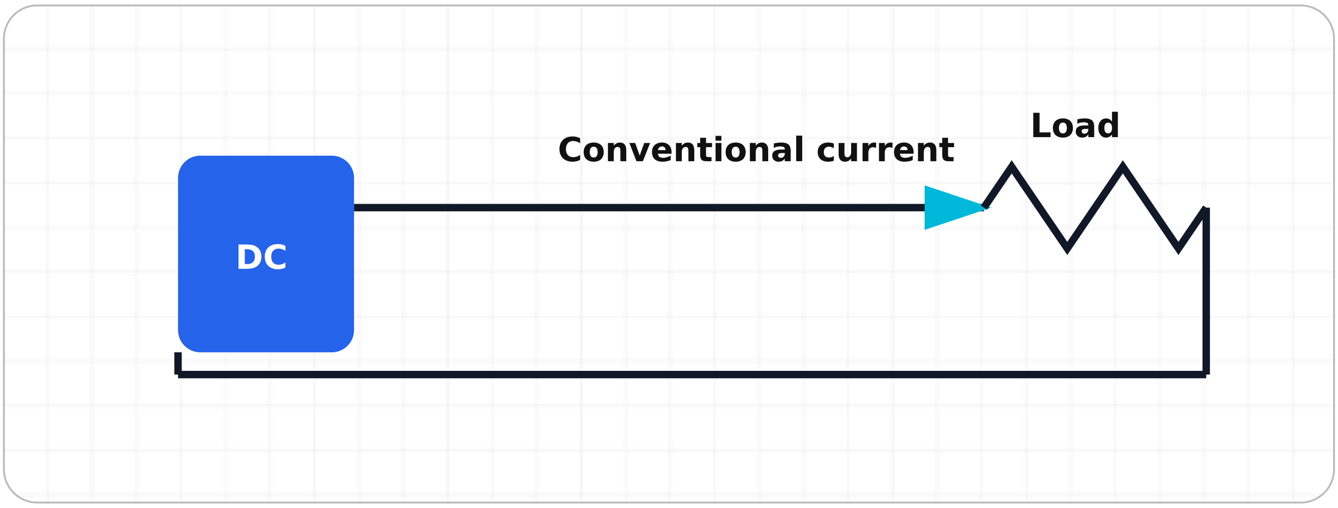
Apparatus

DMM and resistor colour chart.

Components

At least ten resistors including 4-band and 5-band types.

Circuit / connection diagram



Resistor colour-code decoding and value verification connection. Generic safe low-voltage arrangement. Follow the specific approved circuit and instructor direction for the experiment.

Theory

Colour bands encode significant digits, multiplier and tolerance. A valid measurement should lie within the calculated tolerance range.

Procedure

1. Determine reading direction from band spacing and tolerance band.
2. Decode nominal value and tolerance.
3. Calculate minimum and maximum acceptable resistance.
4. Set DMM to resistance mode and ensure circuit is de-energised.
5. Measure the isolated resistor and compare with the tolerance range.
6. Record pass/fail and possible causes for out-of-range values.

Observation table

Resistor ID	Bands	Nominal value	Tolerance	Minimum	Maximum	Measured	Status

Calculations / graph

Show formulas, substitutions, units, percentage difference and graph axes where applicable.

Result

Resistor values were decoded and verified against tolerance.

Precautions: Do not hold both leads with fingers during high-resistance measurement; never measure resistance in an energised circuit.

Common errors: Wrong reading direction, multiplier error, prefix conversion error.

Troubleshooting

For a suspicious reading, lift one lead from the circuit and repeat.

Viva questions

What does the tolerance band mean?

The permitted percentage deviation from nominal resistance.

Why is equivalent in-circuit resistance often lower?

Parallel paths provide additional current paths.

Record-writing format

Write: date → experiment number/title → aim → apparatus/components → neat labelled circuit → brief theory/formula → numbered procedure → observation table → sample calculation → graph → result → precautions.

Student signature: ____________________ Faculty signature: ____________________

EXPERIMENT 7 · MODULE 2

Series and parallel resistor circuits

Date: ____________________ Batch: ___________ Roll No.: ___________

Aim

Construct series and parallel resistor circuits, calculate equivalent resistance and verify by measurement.

Objectives

Identify the relevant equipment and components, make correct connections, take systematic observations, interpret the result, and work safely.

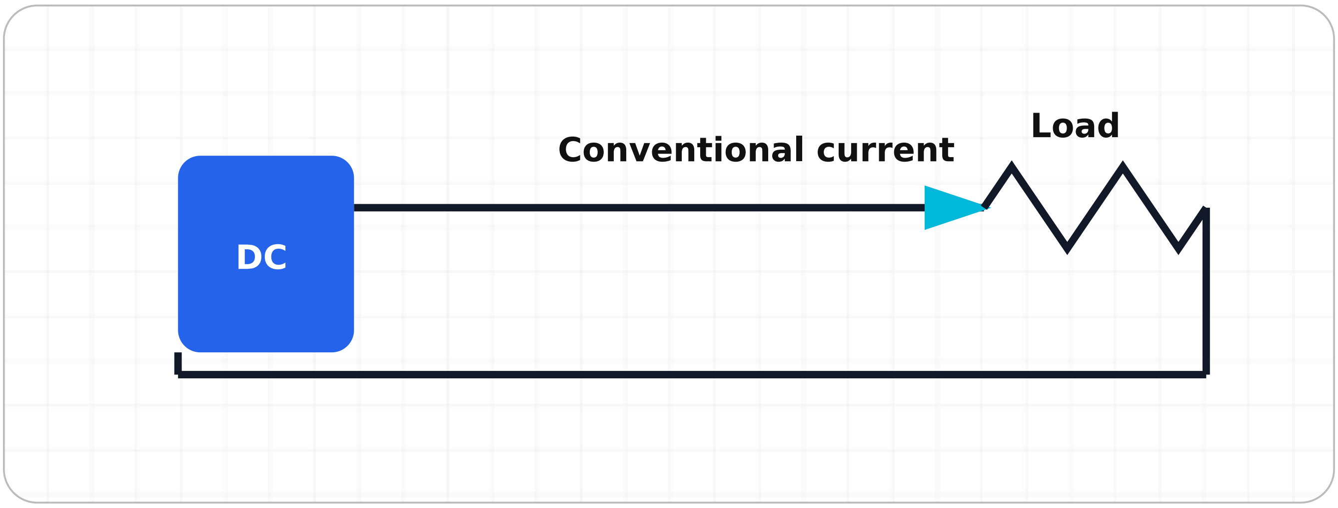
Apparatus

Breadboard, DMM, low-voltage DC supply, leads.

Components

Three resistors of known values.

Circuit / connection diagram



Series and parallel resistor circuits connection. Generic safe low-voltage arrangement. Follow the specific approved circuit and instructor direction for the experiment.

Theory

Series resistances add; parallel reciprocal conductances add. Current is common in series and voltage is common in parallel.

Procedure

1. Measure each resistor individually.
2. Calculate expected series and parallel equivalents.
3. Construct the series network with supply off and measure equivalent resistance.
4. Construct the parallel network and measure equivalent resistance.
5. Apply a safe DC voltage and measure series voltage drops or parallel branch currents as directed.
6. Compare calculated and measured values.

Observation table

Configuration	R1	R2	R3	Calculated Req	Measured Req	% difference

Calculations / graph

Show formulas, substitutions, units, percentage difference and graph axes where applicable.

Result

Series and parallel resistance laws were verified within component and measurement tolerances.

Precautions: Remove supply before resistance measurement; prevent lead shorts.

Common errors: Measuring powered resistance, breadboard row misunderstanding, wrong reciprocal calculation.

Troubleshooting

If Req is unexpected, check each node and measure individual resistors again.

Viva questions

How does parallel Req compare with the smallest resistor?

It is always smaller.

What is common in a series circuit?

The current.

Record-writing format

Write: date → experiment number/title → aim → apparatus/components → neat labelled circuit → brief theory/formula → numbered procedure → observation table → sample calculation → graph → result → precautions.

Student signature: ________________ Faculty signature: ________________

EXPERIMENT 8 · MODULE 2

Capacitor value decoding and series/parallel verification

Date: ________________ Batch: _________ Roll No.: _________

Aim

Decode capacitor markings and verify individual and combined capacitance.

Objectives

Identify the relevant equipment and components, make correct connections, take systematic observations, interpret the result, and work safely.

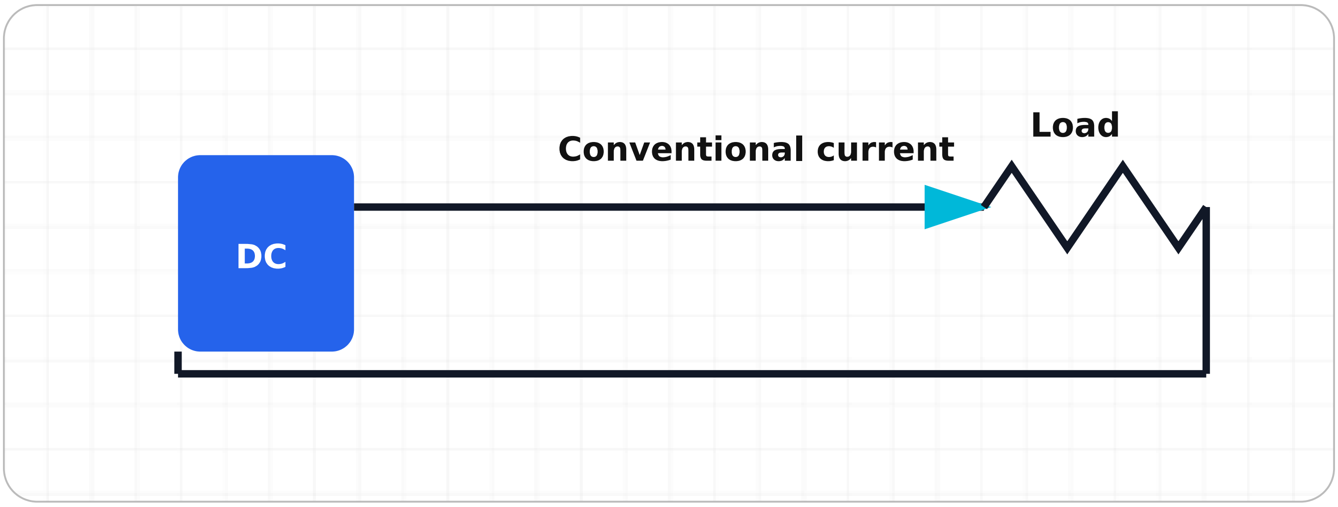
Apparatus

Capacitance meter or DMM, discharge resistor, breadboard.

Components

Three non-polarised capacitors and optional electrolytic samples for identification.

Circuit / connection diagram



Capacitor value decoding and series/parallel verification connection. Generic safe low-voltage arrangement. Follow the specific approved circuit and instructor direction for the experiment.

Theory

Three-digit codes are interpreted in pF. Parallel capacitances add; reciprocal capacitances add in series.

Procedure

1. Identify capacitor type, marking, tolerance and voltage rating.
2. Decode each value and convert units.
3. Discharge and measure each capacitor.
4. Calculate series and parallel equivalents.
5. Build each combination with supply disconnected and measure.
6. Compare results considering tolerance and meter accuracy.

Observation table

Configuration	C1	C2	C3	Calculated Ceq	Measured Ceq	% difference

Calculations / graph

Show formulas, substitutions, units, percentage difference and graph axes where applicable.

Result

Capacitor codes and combination laws were verified.

Precautions: Observe polarity for electrolytics; discharge before handling; do not exceed voltage rating.

Common errors: Reading 104 as 104 pF, reversing electrolytic polarity, ignoring meter lead capacitance for small values.

Troubleshooting

For low values, use relative/zero function and short leads.

Viva questions

What does code 104 mean?

100,000 pF = 100 nF = 0.1 μF.

How does series Ceq compare with the smallest capacitor?
It is smaller.

Record-writing format

Write: date → experiment number/title → aim → apparatus/components → neat labelled circuit → brief theory/formula → numbered procedure → observation table → sample calculation → graph → result → precautions.

Student signature: __________________ Faculty signature: __________________

EXPERIMENT 9 · MODULE 2

Voltage-divider circuit

Date: __________________ Batch: __________ Roll No.: __________

Aim

Construct a two-resistor voltage divider and compare calculated and measured output under unloaded and loaded conditions.

Objectives

Identify the relevant equipment and components, make correct connections, take systematic observations, interpret the result, and work safely.

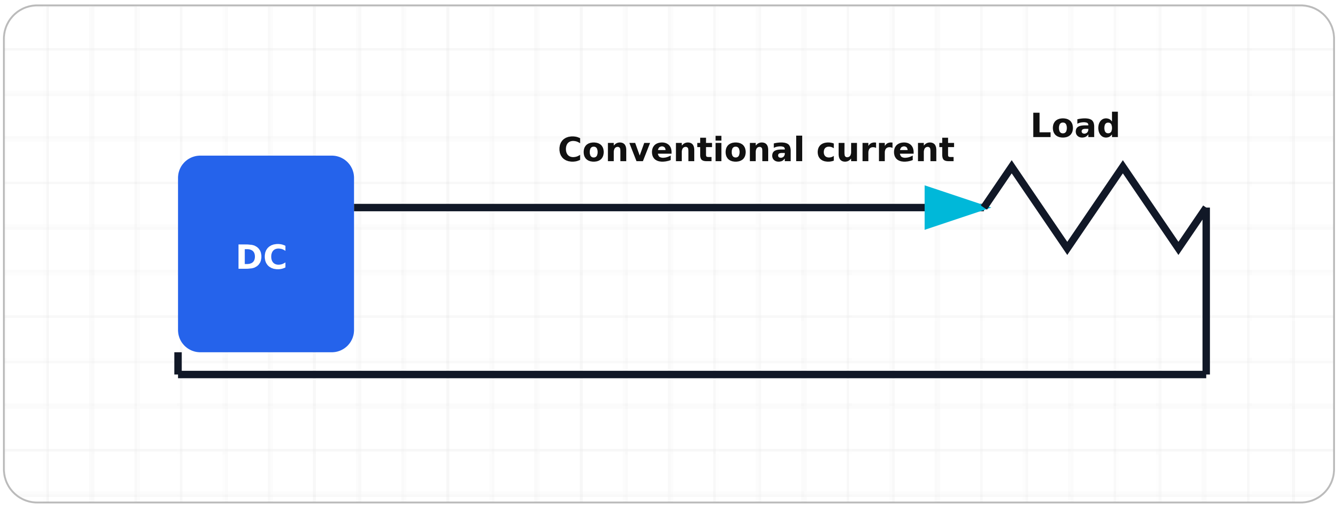
Apparatus

Breadboard, DC supply, DMM.

Components

R1 = 2.2 kΩ, R2 = 3.3 kΩ and a suitable load resistor.

Circuit / connection diagram



Voltage-divider circuit connection. Generic safe low-voltage arrangement. Follow the specific approved circuit and instructor direction for the experiment.

Theory

For an unloaded divider, Vout = VinR2/(R1+R2). A load in parallel with R2 reduces the effective lower resistance and output voltage.

Procedure

1. Measure resistor values.
2. Set supply to a safe voltage with current limit.
3. Construct divider and verify connections before enabling.
4. Measure Vin and unloaded Vout.
5. Connect the specified load across R2 and measure loaded Vout.
6. Calculate predicted values using actual measured resistances.

Observation table

Vin	R1	R2	Load	Calculated Vout	Measured Vout	% difference

Calculations / graph

Show formulas, substitutions, units, percentage difference and graph axes where applicable.

Result

Unloaded divider equation and loading effect were demonstrated.

Precautions: Disable output while changing connections; do not use divider as a power supply for heavy loads.

Common errors: Taking output across R1, ignoring load, wrong ground reference.

Troubleshooting

If output equals supply or zero, inspect for open/short or wrong breadboard nodes.

Viva questions

What causes loading error?

The load changes the effective resistance of the lower divider branch.

Can a divider regulate changing loads?

No, not without buffering or regulation.

Record-writing format

Write: date → experiment number/title → aim → apparatus/components → neat labelled circuit → brief theory/formula → numbered procedure → observation table → sample calculation → graph → result → precautions.

Student signature: ____________________ Faculty signature: ____________________

EXPERIMENT 10 · MODULE 2

RC timing circuit

Date: ____________________ Batch: ___________ Roll No.: ___________

Aim

Construct an RC charging/discharging circuit and verify the time constant.

Objectives

Identify the relevant equipment and components, make correct connections, take systematic observations, interpret the result, and work safely.

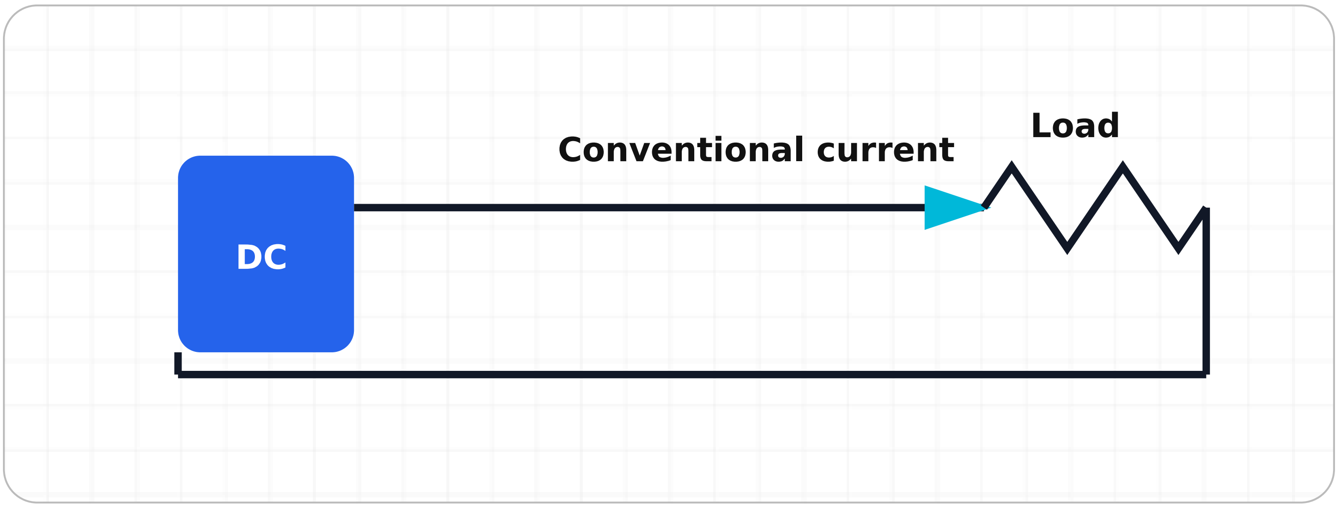
Apparatus

Breadboard, low-voltage DC supply or square-wave generator, DSO or DMM with stopwatch.

Components

Resistor and capacitor selected for a visible safe time constant.

Circuit / connection diagram



RC timing circuit connection. Generic safe low-voltage arrangement. Follow the specific approved circuit and instructor direction for the experiment.

Theory

$\tau = RC$. During charging capacitor voltage reaches about 63.2% of final value at one time constant and about 99.3% at five time constants.

Procedure

1. Measure R and C and calculate τ.
2. Construct the circuit with correct capacitor polarity.
3. Apply a step or square wave.
4. Observe capacitor voltage and identify the 63.2% point.
5. Measure charging and discharging times for several cycles.
6. Compare measured τ with calculated value.

Observation table

R	C	Calculated τ	Final voltage	63.2% voltage	Measured τ charge	Measured τ discharge

Calculations / graph

Show formulas, substitutions, units, percentage difference and graph axes where applicable.

Result

Exponential charging/discharging and the RC time constant were verified.

Precautions: Use safe voltage, observe capacitor polarity, discharge through resistor.

Common errors: Using kΩ and μF without unit conversion, measuring resistor voltage instead of capacitor voltage.

Troubleshooting

If curve is too fast/slow, select R and C appropriate to the instrument time base.

Viva questions

What is capacitor voltage at 1τ during charging?

Approximately 63.2% of the final value.

Why is 5τ used practically?

The capacitor is about 99.3% charged or discharged.

Record-writing format

Write: date → experiment number/title → aim → apparatus/components → neat labelled circuit → brief theory/formula → numbered procedure → observation table → sample calculation → graph → result → precautions.

Student signature: ________________ Faculty signature: ________________

EXPERIMENT 11 · MODULE 3

PN-junction diode identification and biasing

Date: ________________ Batch: _________ Roll No.: _________

Aim

Identify diode terminals and demonstrate forward and reverse bias safely.

Objectives

Identify the relevant equipment and components, make correct connections, take systematic observations, interpret the result, and work safely.

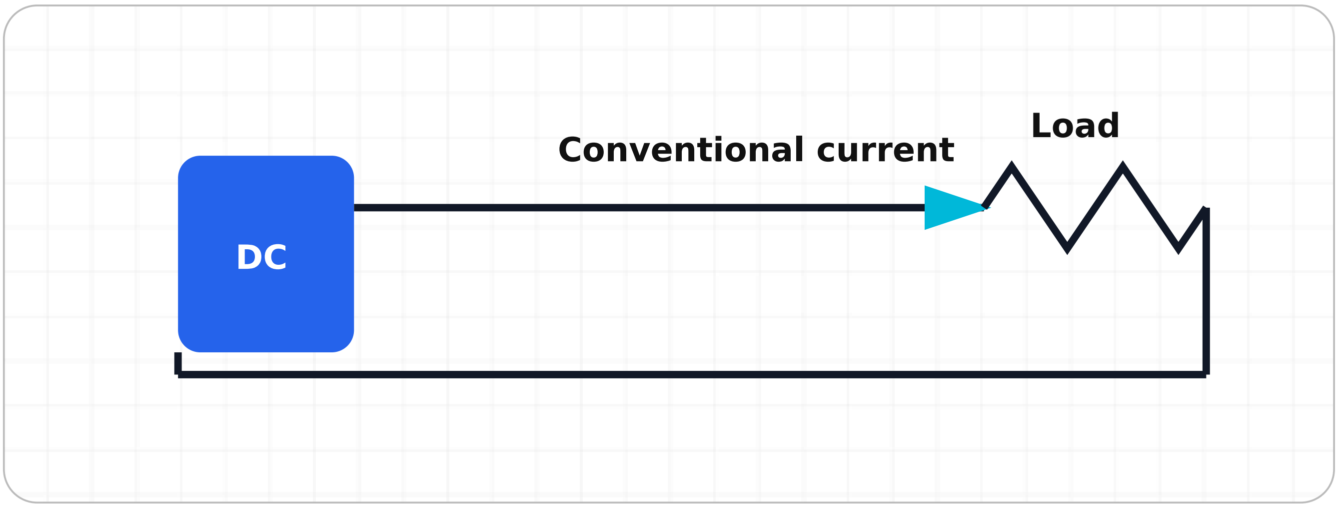
Apparatus

DMM diode mode, breadboard, current-limited DC supply.

Components

Silicon diode and 1 kΩ series resistor.

Circuit / connection diagram



PN-junction diode identification and biasing connection. Generic safe low-voltage arrangement. Follow the specific approved circuit and instructor direction for the experiment.

Theory

A DMM diode test shows forward voltage when red is on anode and black on cathode for most meters. Forward bias permits current; reverse bias blocks except leakage.

Procedure

1. Inspect package and identify cathode band.
2. Test both polarities in diode mode and record readings.
3. Construct forward-bias circuit with series resistor and low voltage.
4. Measure diode voltage and current.
5. Reverse the diode with supply off and repeat within rating.
6. Compare observations.

Observation table

Test/bias	Anode connection	Cathode connection	Diode voltage	Current	Observation

Calculations / graph

Show formulas, substitutions, units, percentage difference and graph axes where applicable.

Result

Diode terminals and forward/reverse behaviour were identified.

Precautions: Always use a series resistor and current limit; observe reverse-voltage rating.

Common errors: Direct supply connection, confusing DMM lead polarity, reversing ammeter connection.

Troubleshooting

If both directions conduct, isolate and retest; device may be shorted or another path may exist.

Viva questions

Which terminal has the package band?

Cathode on common axial diodes.

Why is a series resistor mandatory?

Diode current rises rapidly and must be limited.

Record-writing format

Write: date → experiment number/title → aim → apparatus/components → neat labelled circuit → brief theory/formula → numbered procedure → observation table → sample calculation → graph → result → precautions.

Student signature: ________________ Faculty signature: ________________

EXPERIMENT 12 · MODULE 3

PN-junction diode V-I characteristics

Date: ________________ Batch: _________ Roll No.: _________

Aim

Plot the forward and reverse V-I characteristics of a PN diode and determine static and dynamic resistance.

Objectives

Identify the relevant equipment and components, make correct connections, take systematic observations, interpret the result, and work safely.

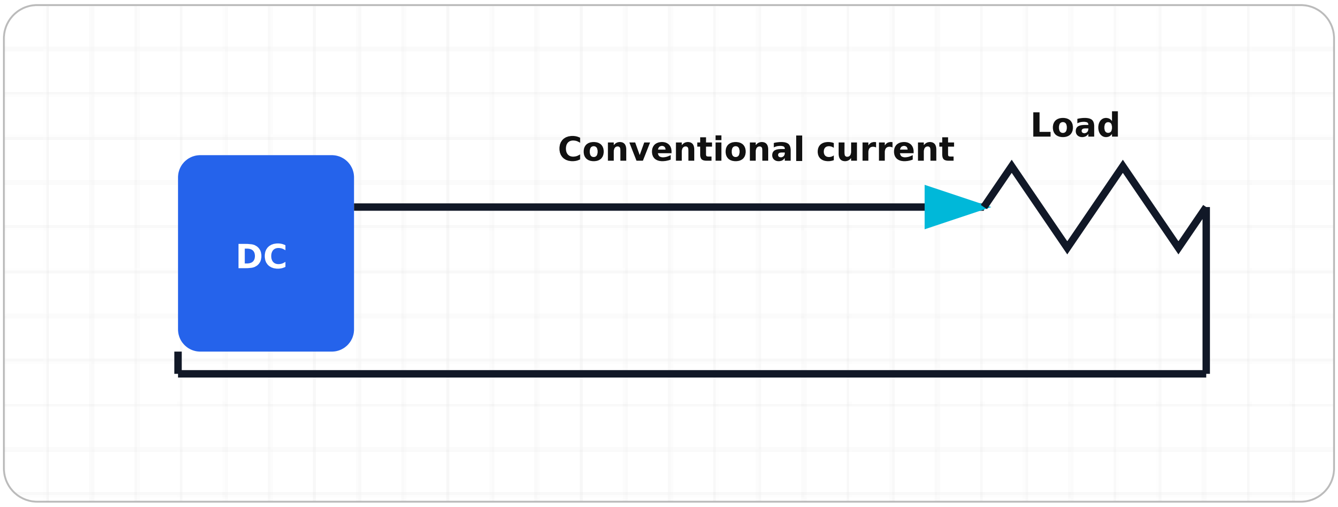
Apparatus

Regulated supplies or diode-characteristic trainer, DMMs, breadboard.

Components

Silicon diode, current-limiting resistors.

Circuit / connection diagram



PN-junction diode V-I characteristics connection. Generic safe low-voltage arrangement. Follow the specific approved circuit and instructor direction for the experiment.

Theory

Forward current rises nonlinearly after the knee region. Static resistance is V/I at a point; dynamic resistance is ΔV/ΔI between nearby points.

Procedure

1. Construct the approved forward-bias circuit and verify polarity.
2. Increase supply in small steps, recording diode voltage and current without exceeding rating.
3. Plot IF versus VF.
4. Construct the approved reverse-bias circuit and record small reverse current within safe voltage.
5. Select an operating point for RDC and two nearby points for rac.
6. Calculate and state units.

Observation table

Supply	Diode voltage	Current	Bias region	Remarks

Calculations / graph

Show formulas, substitutions, units, percentage difference and graph axes where applicable.

Result

The diode V-I curve was plotted and static/dynamic resistance were calculated.

Precautions: Use current limiting, do not approach uncontrolled breakdown, and switch off before reconnection.

Common errors: Plotting supply voltage instead of diode voltage, mixing mA with A, using distant points for dynamic resistance.

Troubleshooting

No current may indicate wrong polarity, open resistor, wrong meter jack or breadboard error.

Viva questions

What is static resistance?

V/I at the operating point.

What is dynamic resistance?

The local slope $\Delta V/\Delta I$ around an operating point.

Record-writing format

Write: date → experiment number/title → aim → apparatus/components → neat labelled circuit → brief theory/formula → numbered procedure → observation table → sample calculation → graph → result → precautions.

Student signature: __________________ Faculty signature: __________________

EXPERIMENT 13 · MODULE 3

Zener diode V-I characteristics

Date: __________________ Batch: __________ Roll No.: __________

Aim

Plot reverse Zener characteristics and identify the breakdown region under current-limited conditions.

Objectives

Identify the relevant equipment and components, make correct connections, take systematic observations, interpret the result, and work safely.

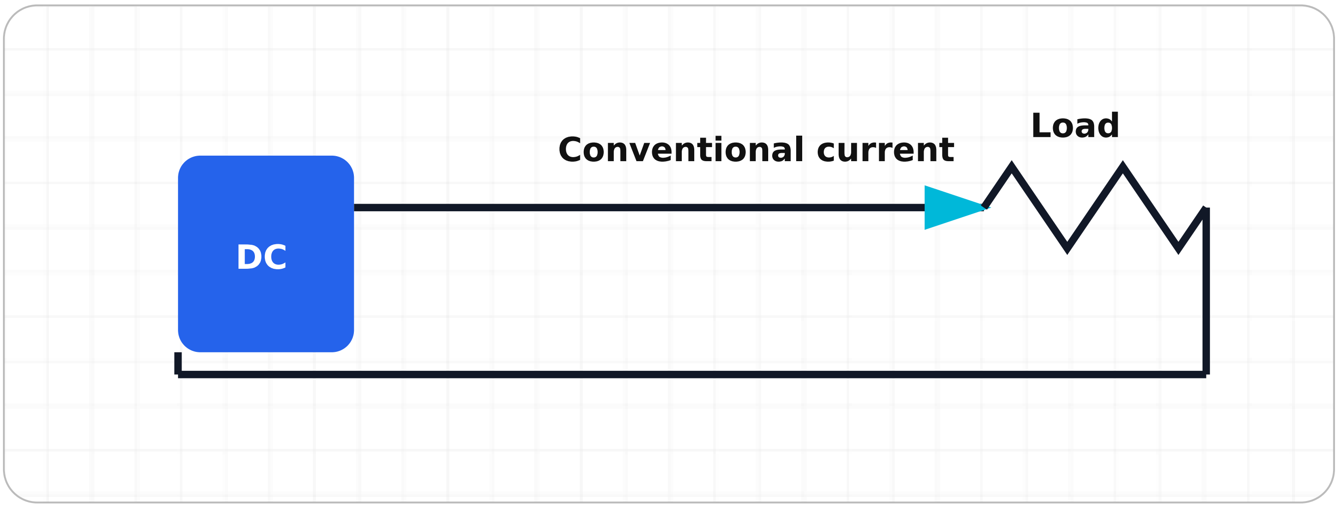
Apparatus

Regulated DC supply, DMMs, breadboard.

Components

Zener diode of known rating and calculated series resistor.

Circuit / connection diagram



Zener diode V-I characteristics connection. Generic safe low-voltage arrangement. Follow the specific approved circuit and instructor direction for the experiment.

Theory

A Zener diode operates in reverse breakdown near its rated voltage when current is limited between specified minimum and maximum values.

Procedure

1. Confirm Zener rating and calculate a safe series resistor.
2. Connect the Zener reverse-biased with current limit.
3. Increase supply gradually while recording Zener voltage and current.
4. Stop below maximum power and current ratings.
5. Plot IZ versus VZ and identify the knee/regulation region.
6. Optionally record forward characteristic at safe current.

Observation table

Supply voltage	Zener voltage	Zener current	Zener power	Region

Calculations / graph

Show formulas, substitutions, units, percentage difference and graph axes where applicable.

Result

Controlled Zener breakdown and voltage regulation region were observed.

> **Precautions:** Never omit the series resistor; monitor PZ = VZIZ; avoid overheating.

> **Common errors:** Connecting Zener forward for reverse test, calculating resistor without worst-case supply, exceeding power.

Troubleshooting

If VZ does not stabilize, check polarity, rating, current range and resistor.

Viva questions

Why is current limiting essential?

Breakdown current can otherwise rise destructively.

Is Zener voltage perfectly constant?

No, it varies with current, tolerance and temperature.

Record-writing format

Write: date → experiment number/title → aim → apparatus/components → neat labelled circuit → brief theory/formula → numbered procedure → observation table → sample calculation → graph → result → precautions.

Student signature: ________________ Faculty signature: ________________

EXPERIMENT 14 · MODULE 4

PCB type and material identification

Date: ________________ Batch: _________ Roll No.: _________

Aim

Identify PCB types, layers, materials, tracks, pads, vias, markings and assembly features.

Objectives

Identify the relevant equipment and components, make correct connections, take systematic observations, interpret the result, and work safely.

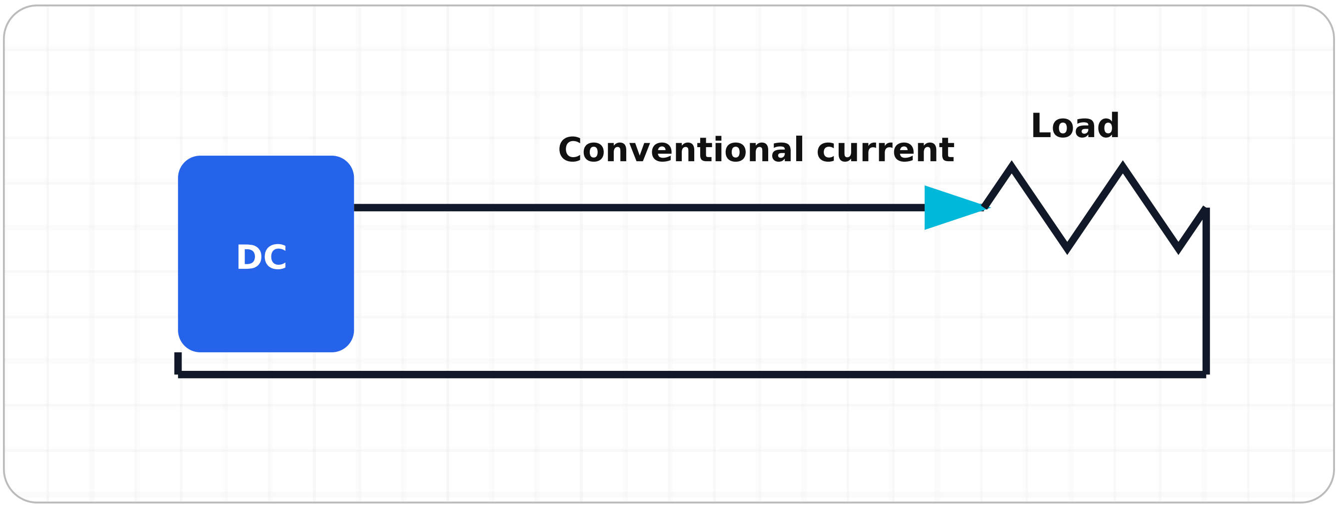
Apparatus

Magnifier, DMM continuity mode, sample unpowered PCBs, drawings if available.

Components

Single-sided, double-sided and multilayer board samples where available.

Circuit / connection diagram



PCB type and material identification connection. Generic safe low-voltage arrangement. Follow the specific approved circuit and instructor direction for the experiment.

Theory

Visible copper and plated holes help distinguish board construction, but internal layers cannot be confirmed by appearance alone.

Procedure

1. Ensure all boards are isolated and capacitors discharged.
2. Identify substrate, solder mask, silkscreen, tracks, pads, vias and footprints.
3. Classify boards using visible evidence and documentation.
4. Trace selected nets visually and verify continuity.
5. Record component-side and solder-side orientation.
6. Identify at least five workmanship or design features.

Observation table

Board ID	Type	Material	Layers/evidence	Features	Observed defects

Calculations / graph

Show formulas, substitutions, units, percentage difference and graph axes where applicable.

Result

PCB structures and materials were identified and documented.

> **Precautions:** Avoid sharp edges, charged boards and destructive probing; do not drill or scrape multilayer boards.

> **Common errors:** Assuming all green boards are FR-4, confusing plated holes with ordinary holes, mirrored orientation.

Troubleshooting

If a track is not visible, consult drawings; it may be internal.

Viva questions

What is a via?

A plated connection between copper layers.

Why are power tracks often wider?

To reduce resistance, voltage drop and heating.

Record-writing format

Write: date → experiment number/title → aim → apparatus/components → neat labelled circuit → brief theory/formula → numbered procedure → observation table → sample calculation → graph → result → precautions.

Student signature: ________________ Faculty signature: ________________

EXPERIMENT 15 · MODULE 4

Soldering tool identification, wire tinning and through-hole soldering

Date: ________________ Batch: _________ Roll No.: _________

Aim

Use soldering tools safely to tin wire and produce reliable through-hole joints.

Objectives

Identify the relevant equipment and components, make correct connections, take systematic observations, interpret the result, and work safely.

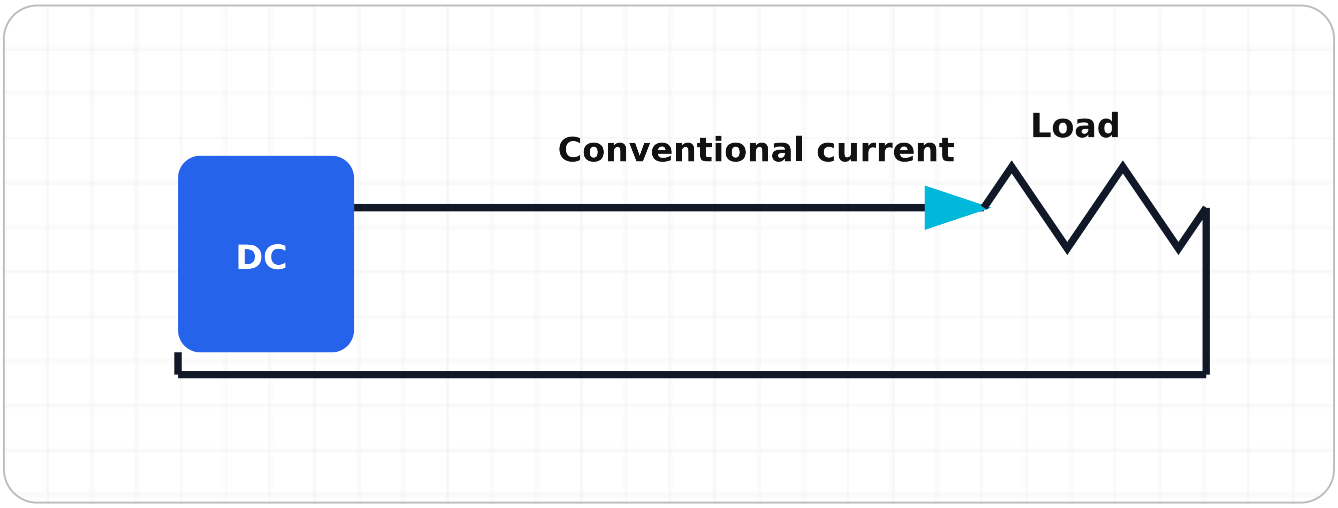
Apparatus

Temperature-controlled iron, stand, brass wool/sponge, extraction, cutters, pliers, eye protection.

Components

Approved solder and flux, practice PCB, resistors, insulated wire.

Circuit / connection diagram



Soldering tool identification, wire tinning and through-hole soldering connection. Generic safe low-voltage arrangement. Follow the specific approved circuit and instructor direction for the experiment.

Theory

A good joint requires clean surfaces, correct heat transfer, wetting and controlled cooling. The pad and lead are heated together.

Procedure

1. Check ventilation, PPE, stand, iron cable, tip condition and approved temperature.
2. Clean and lightly tin the tip.
3. Strip wire without nicking strands; twist and apply a controlled thin tin coating.
4. Insert and mechanically support the component without stressing leads.
5. Heat pad and lead together, feed solder to the joint, remove solder, then iron.
6. Allow to cool without movement; trim leads and inspect under magnification.

Observation table

Joint ID	Temperature	Heating time	Wetting	Fillet shape	Defect/rework	Status

Calculations / graph

Show formulas, substitutions, units, percentage difference and graph axes where applicable.

Result

Wire tinning and reliable through-hole solder joints were completed and inspected.

Precautions: Use fume extraction and eye protection; return iron to stand; wash hands; keep hot tip away from insulation.

Common errors: Feeding solder only to tip, excessive solder, movement during cooling, overheating pad.

Troubleshooting

Poor wetting: stop, clean, flux, restore tip condition and reheat correctly.

Viva questions

Why heat pad and lead together?

Both surfaces must reach soldering temperature for proper wetting.

Why tin the tip?

To prevent oxidation and improve heat transfer.

Record-writing format

Write: date → experiment number/title → aim → apparatus/components → neat labelled circuit → brief theory/formula → numbered procedure → observation table → sample calculation → graph → result → precautions.

Student signature: ________________ Faculty signature: ________________

Desoldering, joint inspection and continuity testing

Date: ________________ Batch: _________ Roll No.: _________

Aim

Remove a through-hole component without damaging the PCB and verify the repaired connection.

Objectives

Identify the relevant equipment and components, make correct connections, take systematic observations, interpret the result, and work safely.

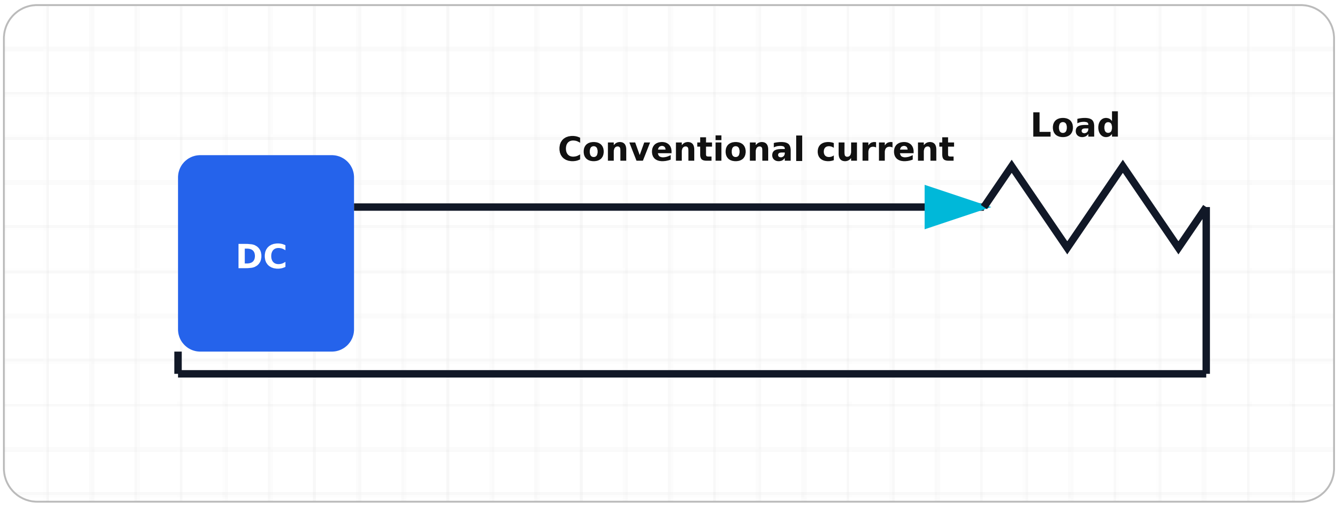
Apparatus

Iron, pump, wick, flux, extraction, magnifier, DMM.

Components

Practice PCB with sacrificial component and replacement part.

Circuit / connection diagram



Desoldering, joint inspection and continuity testing connection. Generic safe low-voltage arrangement. Follow the specific approved circuit and instructor direction for the experiment.

Theory

Controlled desoldering removes solder while minimizing thermal and mechanical stress. Electrical verification follows visual inspection.

Procedure

1. Record component orientation and isolate the board.
2. Add a small amount of fresh solder/flux if needed.
3. Heat and remove solder using pump or wick without levering the pad.
4. Confirm every lead is free before lifting component.
5. Clean and inspect pad, barrel and track.
6. Install replacement, solder, clean as required, inspect and perform continuity/short checks.

Observation table

Pad/joint	Visual condition before	Desolder method	Condition after	Continuity	Short to adjacent	Final status

Calculations / graph

Show formulas, substitutions, units, percentage difference and graph axes where applicable.

Result

The component was removed and replaced with PCB integrity verified.

Precautions: Limit dwell time, use extraction, avoid force, and confirm capacitors are discharged.

Common errors: Pulling before all leads are free, repeated overheating, using oversized tip or braid.

Troubleshooting

For a stubborn plated hole, add fresh solder and use a correctly sized tool; do not drill unless an approved repair procedure permits.

Viva questions

Why add fresh solder before desoldering?

Fresh flux and improved heat transfer can help old solder reflow.

What checks follow repair?
Visual inspection, continuity, isolation from adjacent nets and controlled functional test.

Record-writing format

Write: date → experiment number/title → aim → apparatus/components → neat labelled circuit → brief theory/formula → numbered procedure → observation table → sample calculation → graph → result → precautions.

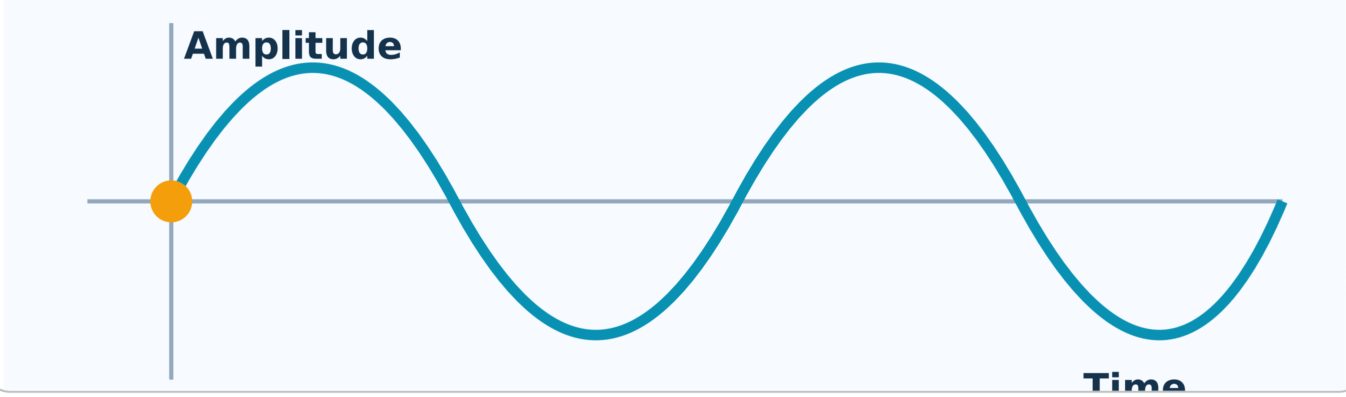
Student signature: ____________________ Faculty signature: ____________________

EDUCATIONAL ANIMATIONS

Controlled Concept Visualisations

Use Play, Pause, Reset and Speed controls. Reduced-motion mode stops motion. Print/PDF shows a labelled static frame.

AC waveform generation



Observe how one revolution produces one complete sine-wave cycle.

Malayalam explanation: AC signal □□□□ □□□□□□□□□□ magnitude-□□□ direction-□□□ □□□□□□□□. Frequency, Period, Amplitude, Phase □□□□□□ waveform □□□□□□□□□ □□□□□□□□□□□□□ □□□□□□ terms □□□.

Conventional current flow

The loop must be complete; the return path is part of the circuit.

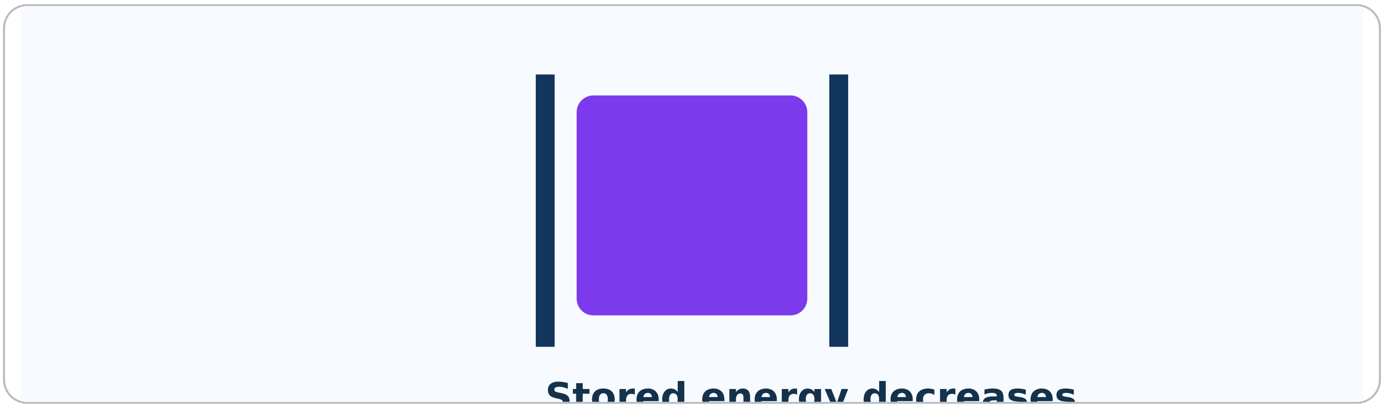
Malayalam explanation: Current □□□□□□□ charge flow rate □□□. Ammeter □□□□□□□□□ circuit-□ series □□□ connect □□□□□□□□; source-□□ across □□□ connect □□□□□□□□□□□.

Capacitor charging

At one time constant, capacitor voltage is about 63.2% of final value.

Malayalam explanation: Capacitor charge store □□□□□□□□□□□□. Charging/Discharging speed-□□ RC Time Constant □□□□□□□□□□□□□□□□□□□□; electrolytic capacitor-□□□□ polarity□□□ voltage rating-□□□ □□□□□□□□□□□□□ □□□□□□□□□□□.

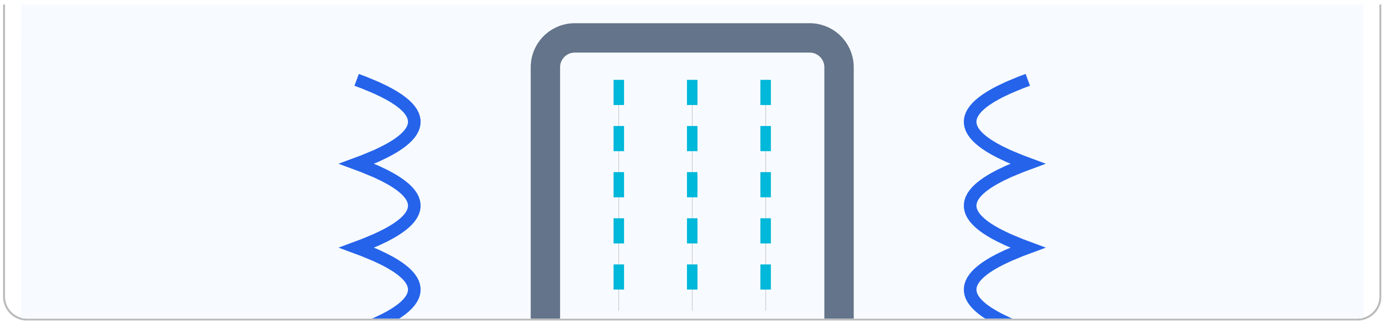
Capacitor discharging



Use a resistor for controlled discharge.

Malayalam explanation: Capacitor charge store □□□□□□□□□□□□. Charging/Discharging speed-□□ RC Time Constant □□□□□□□□□□□□□□□□□□□□; electrolytic capacitor-□□□□ polarity□□□ voltage rating-□□□ □□□□□□□□□□□□□ □□□□□□□□□□□.

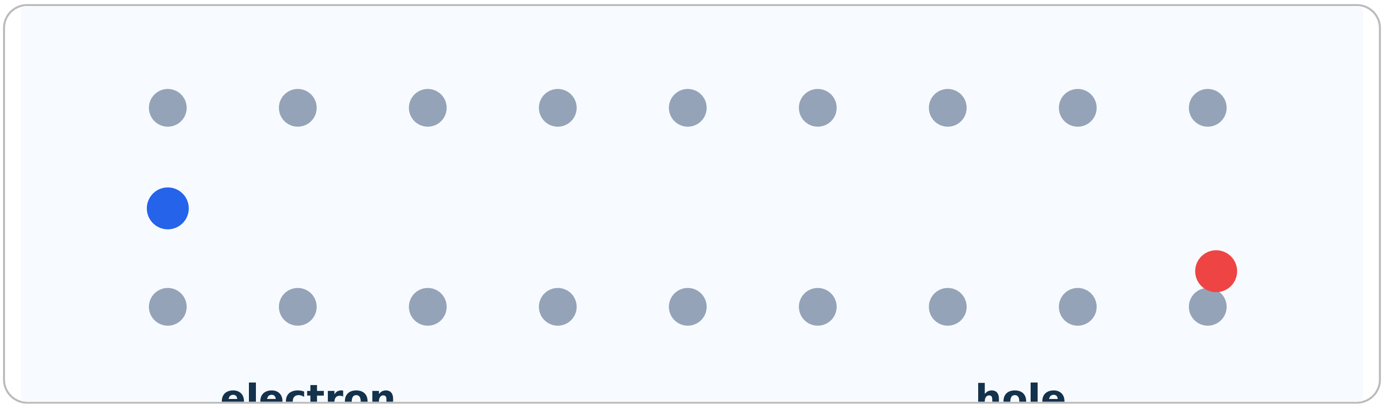
Transformer magnetic flux



Steady DC does not sustain transformer action.

Malayalam explanation: Transformer changing magnetic flux □□□□□□□□□□□ voltage transfer □□□□□□□□□□□. Steady DC transformer action □□□□□□□; turns ratio □□□□□□□□□□□□.

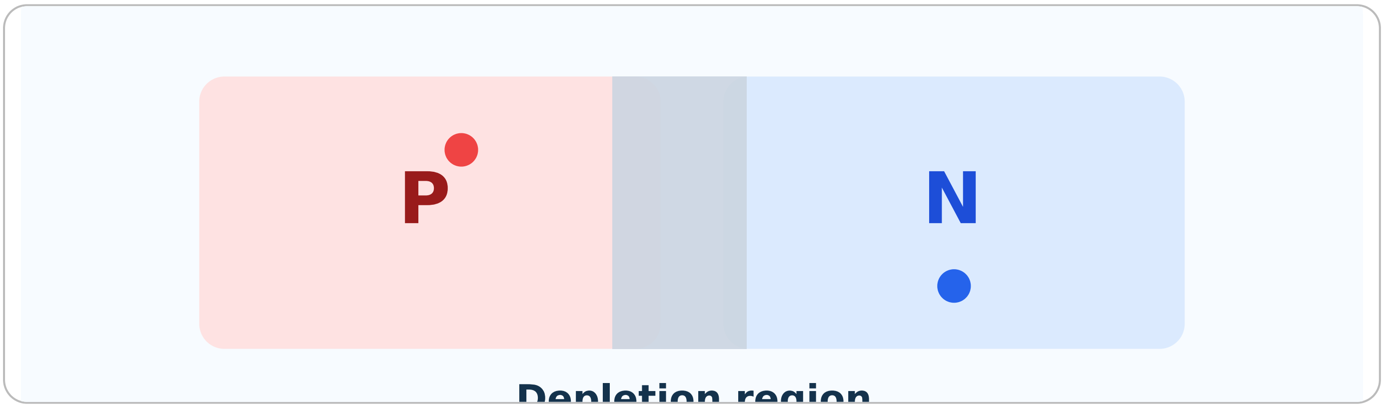
Semiconductor carrier movement



Distinguish mobile carriers from fixed ionized dopants.

Malayalam explanation: Semiconductor conductivity doping, temperature, electrons, holes □□□□□□□□□ □□□□□□□□□□□□□□. Mobile carriers-□□ fixed dopant ions-□□ □□□□□□□□□□□ □□□□□□□□□□□□□.

PN-junction formation

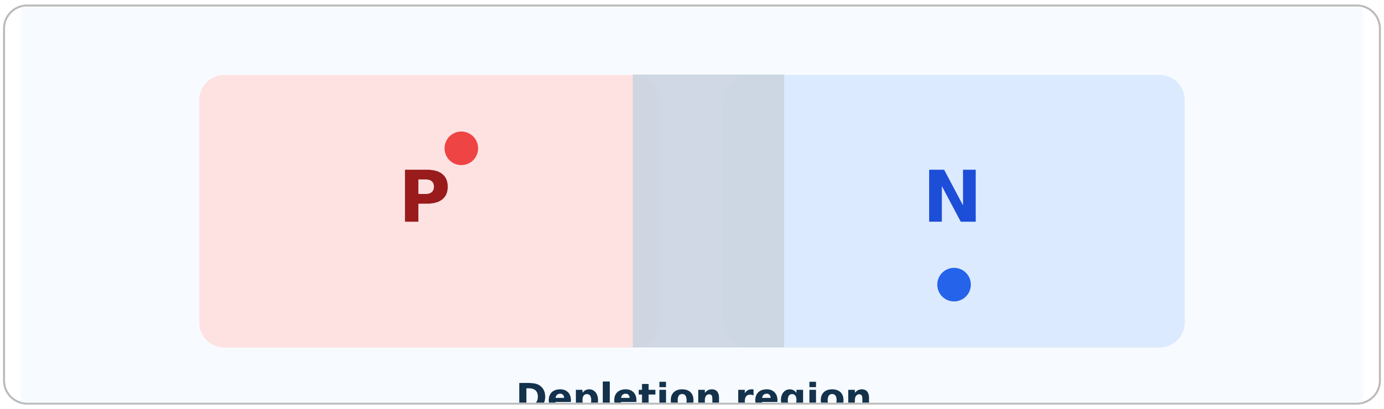


The internal field opposes further majority-carrier diffusion.

Malayalam explanation: PN Junction □□□□□□□□□□□□□ Depletion Region□□□ Barrier

Potential□□□ □□□□□□□□. Applied Bias □□□□□□□□□□ barrier width □□□□□□□□.

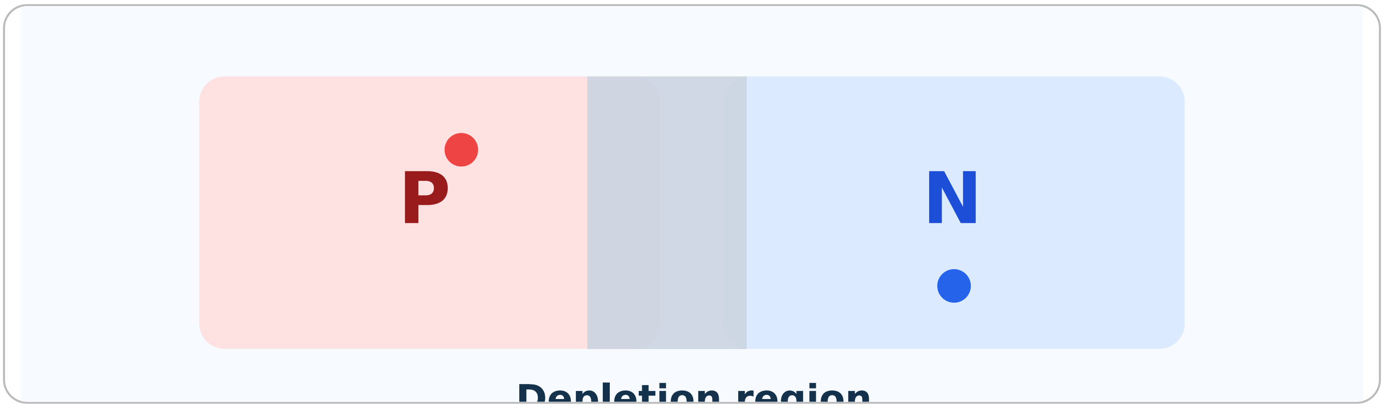
Forward bias



Current must always be limited by a resistor or controlled source.

Malayalam explanation: Forward Bias junction barrier □□□□□□□ current flow □□□□□□□□□□□□□□□. Diode current safe value-□ limit □□□□□□□.

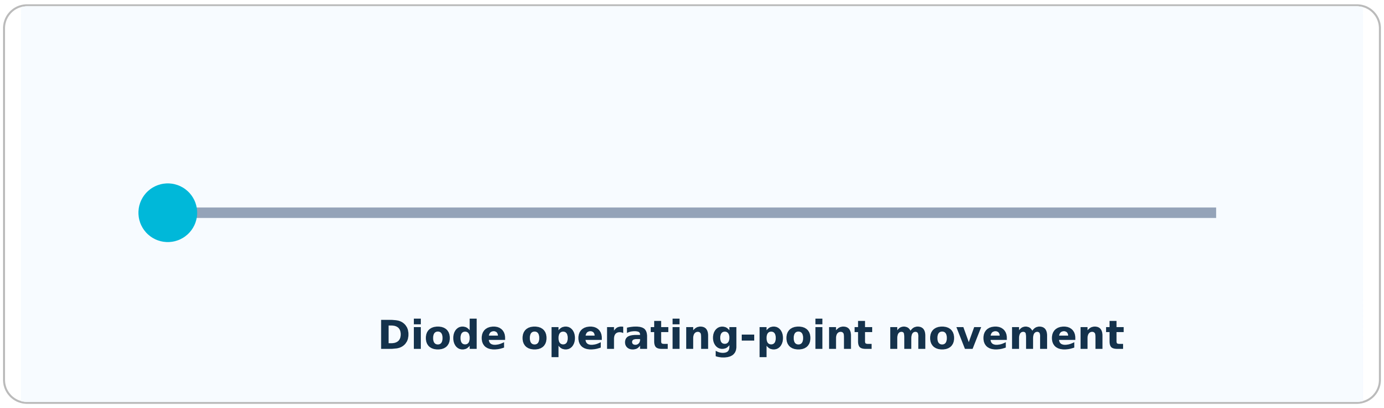
Reverse bias



Only small leakage flows before breakdown.

Malayalam explanation: Reverse Bias depletion region □□□□□□□□□□□□□□□□ majority-carrier current □□□□□□□□ □□□□□□□□□□. Breakdown rating exceed □□□□□□□□□.

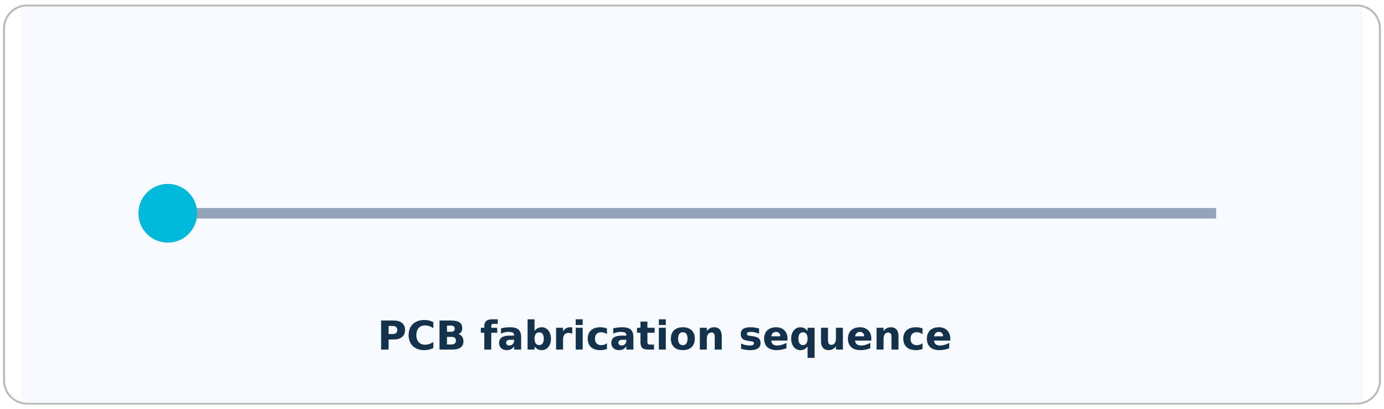
Diode operating-point movement



Static and dynamic resistance are different at the same operating region.

Malayalam explanation: Diode □□□□□□□□□□□ one-direction conduction □□□□□□□ semiconductor device □□□. Anode, Cathode, polarity, forward drop, current limit □□□□□□ □□□□□□□□□□□□.

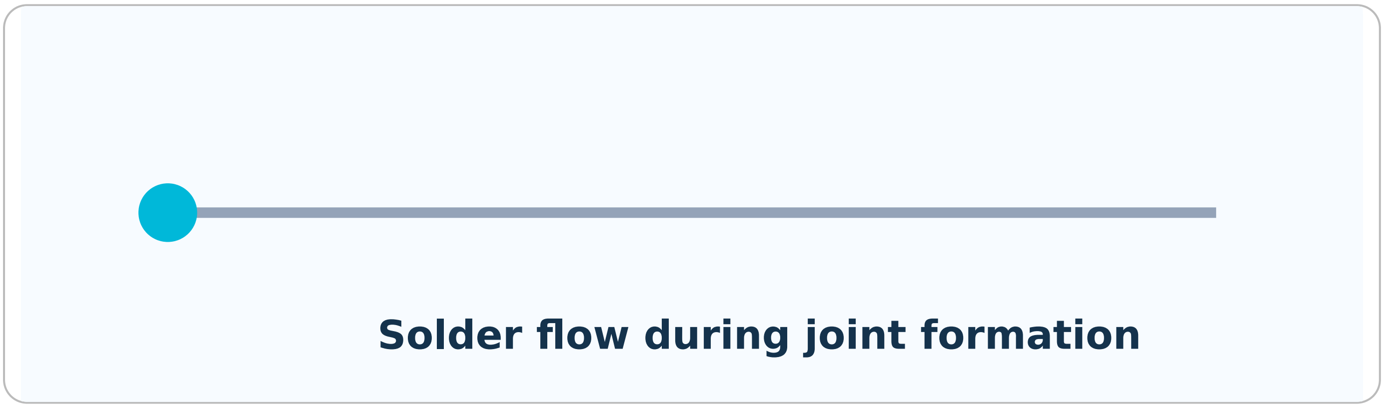
PCB fabrication sequence



Inspection between stages prevents defect multiplication.

Malayalam explanation: PCB-□□□□ Track current path □□□; Pad component connection □□□; Via layers □□□□□□ connection □□□□□□□□. Layout-□ current, noise, heat, clearance □□□□□□ □□□□□□□□□□□.

Solder flow during joint formation



Solder must flow to the joint, not merely melt on the iron tip.

Malayalam explanation: Good solder joint lead-□□□ pad-□□□ □□□□□□□ wet □□□□□□ smooth concave fillet □□□□□□□□□□. Excess heat, insufficient flux, movement □□□□□□ dry joint □□□□□□□□□□ lifted pad □□□□□□□□□□.

FORMULA BANK

Core Equations, Symbols and Practical Use

Ohm’s law

$$V = IR$$

Symbols: V: volt (V), I: ampere (A), R: ohm (Ω)

Use: Find one electrical quantity when the other two are known.

Power

$$P = VI$$

Symbols: P: watt (W), V: volt (V), I: ampere (A)

Use: Check source, load and component power.

Resistive power

$$P = I^2R$$

Symbols: I: ampere (A), R: ohm (Ω)

Use: Find resistor heating from current.

Resistive power

$$P = V^2/R$$

Symbols: V: volt (V), R: ohm (Ω)

Use: Find resistor heating from voltage.

Electrical energy

$$E = Pt$$

Symbols: E: joule (J) when P in W and t in s; watt-hour when t in h

Use: Estimate consumption or stored-energy use.

Frequency

$$f = 1/T$$

Symbols: f: hertz (Hz), T: second (s)

Use: Convert measured period to frequency.

Time period

$$T = 1/f$$

Symbols: T: second (s), f: hertz (Hz)

Use: Convert frequency to one-cycle time.

Angular frequency

$$\omega = 2\pi f$$

Symbols: ω: rad/s, f: Hz

Use: Write sinusoidal equations.

Series resistance

$$Req = R1 + R2 + \ldots$$

Symbols: All resistances in the same unit

Use: Calculate a series network.

Parallel resistance

$$1/Req = 1/R1 + 1/R2 + \ldots$$

Symbols: All resistances in the same unit

Use: Calculate a parallel network.

Series capacitance

$$1/Ceq = 1/C1 + 1/C2 + \ldots$$

Symbols: All capacitances in the same unit

Use: Calculate series capacitor combinations.

Parallel capacitance

$$Ceq = C1 + C2 + \ldots$$

Symbols: All capacitances in the same unit

Use: Calculate parallel capacitor combinations.

Voltage divider

$$Vout = Vin \times R2/(R1+R2)$$

Symbols: Unloaded two-resistor divider; output across R2

Use: Predict divider output.

RC time constant

$$\tau = RC$$

Symbols: R in Ω, C in F, τ in s

Use: Estimate charging/discharging speed.

Transformer ratio

$$Vp/Vs = Np/Ns$$

Symbols: V: winding voltage, N: turns; ideal transformer

Use: Relate primary and secondary voltage.

Diode static resistance

$$RDC = V/I$$

Symbols: Use V and I from the same operating point

Use: DC operating-point resistance.

Diode dynamic resistance

$$rac = \Delta V/\Delta I$$

Symbols: Use nearby V-I points and consistent units

Use: Small-signal slope resistance.

LED resistor

$$R = (VS - VF)/I$$

Symbols: VS supply, VF LED forward voltage, I desired current

Use: Limit LED current safely.

> **Engineering rule:** convert all prefixes before substitution, write the formula first, carry units through the calculation, and check whether the answer is physically reasonable.

INTERACTIVE TOOLS

Electronics Formula Calculators

Calculators support learning; they do not replace units, ratings, tolerance or engineering judgement.

Resistor colour-code calculator

Band 1

Black

Band 2

Black

Multiplier exponent

Tolerance %

Enter values and calculate.

Resistance tolerance calculator

Nominal resistance (Ω)

Tolerance (%)

Enter values and calculate.

Series-resistance calculator

R1 (Ω)

R2 (Ω)

R3 (Ω, optional)

Enter values and calculate.

Parallel-resistance calculator

R1 (Ω)

R2 (Ω)

R3 (Ω, optional)

Enter values and calculate.

Capacitor-code decoder

Three-digit code

Enter values and calculate.

Series-capacitance calculator

C1 (µF)

C2 (µF)

C3 (µF, optional)

Enter values and calculate.

Parallel-capacitance calculator

C1 (µF)

C2 (µF)

C3 (µF, optional)

Enter values and calculate.

Transformer-ratio calculator

Primary voltage Vp (V)

Primary turns Np

Secondary turns Ns

Enter values and calculate.

Voltage-divider calculator

Input voltage Vin (V)

R1 (Ω)

R2 (Ω)

Enter values and calculate.

RC time-constant calculator

Resistance R (Ω)

Capacitance C (F)

Enter values and calculate.

LED resistor calculator

Supply voltage VS (V)

LED forward voltage VF (V)

Desired current (mA)

Enter values and calculate.

SOLVED EXAMPLES

Checked Step-by-Step Numerical Problems

Example 1: Electric current from charge

Given: Q = 24 C flows in 6 s.

Required: Current I.

Formula: $I = Q/t$.

Substitution and calculation: I = 24/6 = 4 A.

Final answer: 4 A

Common mistake: Do not divide time by charge.

Example 2: Voltage from work and charge

Given: 60 J of energy moves 5 C of charge.

Required: Voltage V.

Formula: $V = W/Q$.

Substitution and calculation: V = 60/5 = 12 V.

Final answer: 12 V

Common mistake: J/C is volt.

Example 3: Resistance from voltage and current

Given: A resistor carries 0.25 A at 10 V.

Required: Resistance R.

Formula: $R = V/I$.

Substitution and calculation: R = 10/0.25 = 40 Ω.

Final answer: 40 Ω

Common mistake: Convert mA to A before division.

Example 4: Electrical power

Given: A load operates at 24 V and 1.5 A.

Required: Power P.

Formula: $P = VI$.

Substitution and calculation: P = 24 × 1.5 = 36 W.

Final answer: 36 W

Common mistake: Do not report watt-hours; this is power.

Example 5: Electrical energy

Given: A 60 W lamp runs for 5 h.

Required: Energy in Wh and kWh.

Formula: $E = Pt$.

Substitution and calculation: E = 60 × 5 = 300 Wh = 0.300 kWh.

Final answer: 300 Wh or 0.300 kWh

Common mistake: Use hours for Wh; use seconds for joules.

Example 6: Unit conversion

Given: Convert 4.7 kΩ to Ω and 220 nF to µF.

Required: Converted values.

Formula: $1\ \mathrm{k\Omega} = 1000\ \Omega;\ 1000\ \mathrm{nF} = 1\ \mu\mathrm{F}$.

Substitution and calculation: 4.7 × 1000 = 4700 Ω; 220/1000 = 0.22 µF.

Final answer: 4700 Ω and 0.22 µF

Common mistake: Prefix errors produce factors of 1000.

Example 7: Frequency from period

Given: A waveform period is 2 ms.

Required: Frequency f.

Formula: $f = 1/T$.

Substitution and calculation: $T = 2 \times 10^{-3}$ s; $f = 1/0.002 = 500$ Hz.

Final answer: 500 Hz

Common mistake: Convert ms to s.

Example 8: Period from frequency

Given: A signal frequency is 20 kHz.

Required: Period T.

Formula: $T = 1/f$.

Substitution and calculation: $f = 20{,}000$ Hz; $T = 1/20{,}000 = 50 \times 10^{-6}$ s.

Final answer: 50 µs

Common mistake: Convert kHz to Hz.

Example 9: Angular frequency

Given: A sine wave has $f = 50$ Hz.

Required: Angular frequency ω.

Formula: $\omega = 2\pi f$.

Substitution and calculation: $\omega = 2\pi \times 50 = 314.16$ rad/s.

Final answer: Approximately 314 rad/s

Common mistake: Do not write the unit as Hz.

Example 10: Four-band resistor

Given: Bands are brown–black–red–gold.

Required: Nominal resistance and tolerance.

Formula: Digits 1,0; multiplier 10^2; gold ±5%.

Substitution and calculation: $10 \times 100 = 1000\ \Omega = 1\ k\Omega$; range 950 Ω to 1050 Ω.

Final answer: 1 kΩ ±5%

Common mistake: Gold is tolerance, not a significant digit.

Example 11: Tolerance range

Given: A 4.7 kΩ resistor has ±5% tolerance.

Required: Minimum and maximum.

Formula: $\mathbf{Rmin = R(1-0.05); Rmax = R(1+0.05).}$

Substitution and calculation: 4700×0.95 = 4465 Ω; 4700×1.05 = 4935 Ω.

Final answer: 4.465 kΩ to 4.935 kΩ

Common mistake: Apply tolerance to the nominal value.

Example 12: Series resistance

Given: R1 = 1 kΩ, R2 = 2.2 kΩ, R3 = 3.3 kΩ.

Required: Equivalent resistance.

Formula: $\mathbf{Req = R1+R2+R3.}$

Substitution and calculation: Req = 1 + 2.2 + 3.3 = 6.5 kΩ.

Final answer: 6.5 kΩ

Common mistake: Keep units consistent.

Example 13: Parallel resistance

Given: R1 = 6 Ω and R2 = 3 Ω.

Required: Equivalent resistance.

Formula: $\mathbf{Req = R1R2/(R1+R2).}$

Substitution and calculation: Req = 6×3/(6+3) = 18/9 = 2 Ω.

Final answer: 2 Ω

Common mistake: Parallel result must be below 3 Ω.

Example 14: Capacitor code

Given: A capacitor is marked 104.

Required: Capacitance.

Formula: **First two digits 10; multiplier 10^4 pF.**

Substitution and calculation: 10×10^4 pF = 100,000 pF = 100 nF = 0.1 μF.

Final answer: 100 nF or 0.1 μF

Common mistake: 104 does not mean 104 pF.

Example 15: Series capacitance

Given: C1 = 6 μF and C2 = 3 μF.

Required: Equivalent capacitance.

Formula: **Ceq = C1C2/(C1+C2).**

Substitution and calculation: Ceq = 6×3/(6+3) = 18/9 = 2 μF.

Final answer: 2 μF

Common mistake: Series result is below the smallest capacitor.

Example 16: Parallel capacitance

Given: C1 = 2.2 μF, C2 = 4.7 μF and C3 = 1 μF.

Required: Equivalent capacitance.

Formula: **Ceq = C1+C2+C3.**

Substitution and calculation: Ceq = 2.2+4.7+1 = 7.9 μF.

Final answer: 7.9 μF

Common mistake: Parallel capacitances add directly.

Example 17: Transformer turns ratio

Given: Np = 1000, Ns = 200, Vp = 230 V.

Required: Secondary voltage.

Formula: **Vp/Vs = Np/Ns.**

Substitution and calculation: Vs = Vp×Ns/Np = 230×200/1000 = 46 V.

Final answer: 46 V ideal no-load

Common mistake: Real output differs due to regulation and losses.

Example 18: Voltage-divider output

Given: Vin = 12 V, R1 = 2 kΩ, R2 = 4 kΩ.

Required: Unloaded Vout across R2.

Formula: $V_{out} = V_{in}R_2/(R_1+R_2)$.

Substitution and calculation: Vout = 12×4/(2+4) = 48/6 = 8 V.

> **Final answer:** 8 V

> **Common mistake:** Output location matters.

Example 19: RC time constant

Given: R = 100 kΩ and C = 10 μF.

Required: Time constant τ.

Formula: $\tau = RC$.

Substitution and calculation: R = 100,000 Ω; C = 10×10^{-6} F; τ = 1 s.

> **Final answer:** 1 s

> **Common mistake:** kΩ×μF is not used blindly without conversion.

Example 20: Diode static resistance

Given: A diode has 0.72 V at 12 mA.

Required: Static resistance.

Formula: $R_{DC} = V/I$.

Substitution and calculation: I = 0.012 A; RDC = 0.72/0.012 = 60 Ω.

> **Final answer:** 60 Ω

> **Common mistake:** Use A, not mA, unless units are handled consistently.

Example 21: Diode dynamic resistance

Given: Current rises from 8 mA to 12 mA while voltage rises from 0.68 V to 0.72 V.

Required: Dynamic resistance.

Formula: $r_{ac} = \Delta V/\Delta I$.

Substitution and calculation: ΔV=0.04 V; ΔI=0.004 A; rac=0.04/0.004=10 Ω.

> **Final answer:** 10 Ω

Common mistake: Use nearby points and differences.

Example 22: LED current-limiting resistor

Given: A 5 V source drives an LED with VF = 2 V at 10 mA.

Required: Series resistor and approximate power.

Formula: $R=(VS-VF)/I$; $P=I^2R$.

Substitution and calculation: $R=(5-2)/0.01=300\ \Omega$; $P=0.01^2\times300=0.03$ W. Choose 330 Ω, ≥0.125 W.

Final answer: 330 Ω practical choice

Common mistake: Choose the next safe standard value and check supply tolerance.

COMPONENT ATLAS

Identification, Ratings, Testing and Failure Modes

Zig-zag/rectangle

Fixed resistor

Physical appearance / construction

Axial or SMD body with resistive element

Terminal identification

Two non-polar terminals

Common values

Ω to MΩ; common 1/8 W to power types

Ratings

Resistance, tolerance, power, voltage, temperature coefficient

Polarity

Non-polar

Multimeter test

De-energize, isolate one lead where needed, measure resistance

Applications

Current limiting, biasing, dividers, pull-up/down

Common failure modes

Open, drift high, burnt, cracked

Safety: Power resistors can remain hot.

Three-terminal variable resistor

Potentiometer

Physical appearance / construction
Resistive track and movable wiper

Terminal identification
Outer terminals are full track; centre is wiper

Common values
Ω to MΩ; linear/log tapers

Ratings
Resistance, power, taper, mechanical life

Polarity
Non-polar

Multimeter test
Measure end-to-end, then wiper to each end while rotating

Applications
Level, setpoint and calibration control

Common failure modes
Worn/dirty track, intermittent wiper

Safety: Do not exceed wiper current.

Photoresistor symbol

LDR

Physical appearance / construction
Photoconductive material in serpentine pattern

Terminal identification
Two non-polar leads

Common values
High resistance in dark, lower in light

Ratings
Dark resistance, light resistance, spectral response

Polarity

Non-polar

Multimeter test

Measure resistance in dark and illuminated conditions

Applications

Light sensing and simple automatic lighting

Common failure modes

Contamination, ageing, open lead

> **Safety:** Not for fast precision measurement.

Varistor symbol

VDR

Physical appearance / construction

Metal-oxide nonlinear resistor disc/block

Terminal identification

Two non-polar leads

Common values

Voltage-dependent clamp values

Ratings

Varistor voltage, surge current, energy, diameter

Polarity

Non-polar

Multimeter test

Check visually and for abnormal low resistance; full test needs controlled equipment

Applications

Surge suppression

Common failure modes

Short, cracked, thermally damaged

> **Safety:** May fail violently under severe surge; use protection.

Temperature-dependent resistor

Thermistor

Physical appearance / construction

Semiconductor bead, disc or probe

Terminal identification
Two non-polar leads

Common values
NTC/PTC values specified at reference temperature

Ratings
R25, B value, dissipation, temperature range

Polarity
Non-polar

Multimeter test
Measure resistance and observe safe temperature response

Applications
Temperature sensing, compensation, inrush limiting

Common failure modes
Drift, crack, open/short

> **Safety:** Inrush NTCs can be hot.

Polarized capacitor

Electrolytic capacitor

Physical appearance / construction
Foil/electrolyte wound construction in can

Terminal identification
Positive and negative; negative stripe common

Common values
µF to thousands of µF

Ratings
Capacitance, voltage, ESR, ripple current, temperature

Polarity
Polarized

Multimeter test
Discharge, inspect, measure capacitance/ESR with suitable instrument

Applications
Filtering, energy storage, decoupling

Common failure modes

Dry-out, high ESR, leakage, bulging, short/open

Safety: Reverse polarity may vent or explode.

Two-plate capacitor

Ceramic capacitor

Physical appearance / construction

Ceramic dielectric; disc or multilayer chip

Terminal identification

Two non-polar terminals

Common values

pF to µF

Ratings

Capacitance, voltage, dielectric class, tolerance

Polarity

Non-polar

Multimeter test

Discharge and measure capacitance out of circuit where needed

Applications

Decoupling, filtering, timing, RF

Common failure modes

Crack, short, capacitance loss

Safety: Large MLCCs can crack under board flex.

Variable capacitor

Variable capacitor

Physical appearance / construction

Overlapping rotor/stator plates or trimmer structure

Terminal identification

Two or more section terminals

Common values

pF ranges common

Ratings

Capacitance range, voltage, Q, mechanical travel

Polarity
Usually non-polar

Multimeter test
Measure capacitance while rotating; check for shorts

Applications
Tuning and calibration

Common failure modes
Bent plates, dirty contact, cracked dielectric

> **Safety:** Use insulated alignment tool.

Coil symbol

Inductor

Physical appearance / construction
Wire coil with air, ferrite or other core

Terminal identification
Two terminals unless tapped/coupled

Common values
µH to H

Ratings
Inductance, current, DCR, saturation, Q

Polarity
Non-polar unless winding start matters

Multimeter test
Check continuity/DCR; inductance meter for value

Applications
Filtering, energy storage, chokes

Common failure modes
Open winding, shorted turns, cracked core

> **Safety:** Inductive kick can create high voltage.

Coupled coils

Transformer

Physical appearance / construction

Primary/secondary windings on magnetic core

Terminal identification
Identify windings from label/drawing

Common values
Application-specific ratios

Ratings
Primary/secondary voltage/current, frequency, insulation, power

Polarity
AC winding orientation matters

Multimeter test
De-energized continuity and resistance; apply only rated controlled AC

Applications
Isolation, step-up/down, coupling

Common failure modes
Open/shorted turns, insulation failure, overheating

> **Safety:** Primary may be mains hazardous.

Diode symbol

PN diode

Physical appearance / construction
PN junction in axial/SMD package

Terminal identification
Anode and cathode; band commonly cathode

Common values
Signal, rectifier and switching types

Ratings
Forward current, reverse voltage, power, recovery

Polarity
Polarized

Multimeter test
DMM diode mode in both directions

Applications
Rectification, protection, switching

Common failure modes

Open, short, leakage

> **Safety:** Use current limiting.

Zener symbol

Zener diode

Physical appearance / construction

Special PN junction designed for breakdown

Terminal identification

Anode/cathode; package band cathode

Common values

Common Zener voltages from a few volts upward

Ratings

VZ, tolerance, test current, power, dynamic resistance

Polarity

Polarized

Multimeter test

DMM basic diode test plus controlled reverse test with resistor

Applications

Reference, clamp, simple regulation

Common failure modes

Open, short, shifted voltage

> **Safety:** Never reverse-test without current limit.

Light-emitting diode

LED

Physical appearance / construction

Semiconductor junction in optical package

Terminal identification

Anode/cathode; markings vary

Common values

Colours and forward voltages vary

Ratings

Current, VF, reverse voltage, optical output

Polarity

Polarized

Multimeter test

DMM diode mode may glow; confirm polarity

Applications

Indicators, lighting, optical transmission

Common failure modes

Open, dim, colour shift, short

> **Safety:** Always use current limiting.

Light-sensitive diode

Photodiode

Physical appearance / construction

Optical window over semiconductor junction

Terminal identification

Anode/cathode

Common values

Spectral and area variants

Ratings

Responsivity, dark current, capacitance, reverse voltage

Polarity

Polarized

Multimeter test

Observe photocurrent/voltage change in safe circuit

Applications

Optical sensing, encoder, receiver

Common failure modes

Leakage, contamination, damaged window

> **Safety:** Protect from intense unknown optical sources.

Board/track representation

PCB

Physical appearance / construction

Insulating laminate with patterned copper and coatings

Terminal identification

Pads, connectors and test points define terminals

Common values

Single, double and multilayer

Ratings

Material, thickness, copper weight, spacing, finish

Polarity

Board orientation critical

Multimeter test

Visual inspection, continuity, isolation and controlled functional test

Applications

Mechanical support and interconnection

Common failure modes

Cracked tracks, lifted pads, corrosion, burnt substrate

> **Safety:** Stored energy and hazardous voltage may remain.

Tool set

Soldering tools

Physical appearance / construction

Temperature-controlled iron, stand, tip cleaner, solder, flux, pump/wick

Terminal identification

Tool-specific

Common values

Tips, power and temperature ranges

Ratings

Temperature stability, ESD grounding, approved alloy/flux

Polarity

Not applicable

Multimeter test

Inspect cable, earth/ESD, tip and temperature; test on practice joint

Applications

Assembly, repair and rework

Common failure modes

Worn tip, heater failure, contamination

Safety: Burn, fume, fire and chemical hazards.

OFFICIAL SELF-LEARNING ACTIVITIES

15 Guided Activities · 90 Hours

Each activity is planned for approximately six hours including preparation, execution, documentation and submission.

Week 1 · Module 1

Identify domestic electronic systems and list their purpose.

Objective: Identify domestic electronic systems and list their purpose.

Resources: Five domestic systems or photographs; notebook.

Safety: Do not open sealed or mains-powered equipment.

Procedure: List system, source, input, processing, output, purpose, and one safety feature.

Expected output: Completed comparison table and one block diagram.

Evaluation points: Five systems identified; functional blocks correct; safety noted.

- [ ] Submission checklist completed

Week 2 · Module 1

Generate a sine wave and compare function-generator and DSO frequency readings.

Objective: Generate a sine wave and compare function-generator and DSO frequency readings.

Resources: Function generator, DSO, approved leads.

Safety: Use low-voltage isolated signal and common reference.

Procedure: Set three frequencies, record generator display and DSO/cursor values, and calculate percentage error.

Expected output: Observation table with calculations and screenshots/sketches.

Evaluation points: Correct settings, units, calculation, and interpretation.

- [ ] Submission checklist completed

Week 3 · Module 1

Collect or observe at least three battery types and compare rated and measured voltage.

Objective: Collect or observe at least three battery types and compare rated and measured voltage.

Resources: Three safe batteries, DMM, datasheets.

Safety: Do not short, puncture, heat, charge, or mix damaged cells.

Procedure: Record chemistry, nominal voltage, measured no-load voltage, polarity, capacity marking, and use.

Expected output: Battery comparison table.

Evaluation points: Safe handling, correct meter use, meaningful comparison.

- [ ] Submission checklist completed

Week 4 · Module 1

Study one instrument manual or datasheet.

Objective: Study one instrument manual or datasheet.

Resources: Manual for DMM, DSO, function generator, or DC supply.

Safety: Do not reproduce unsafe procedures; note category and input limits.

Procedure: Extract purpose, main controls, ranges, accuracy, safety symbols, and one correct setup.

Expected output: Two-page technical summary.

Evaluation points: Source cited, limits understood, controls explained.

- [ ] Submission checklist completed

Week 5 · Module 2

Study one resistor and one capacitor datasheet.

Objective: Study one resistor and one capacitor datasheet.

Resources: Manufacturer datasheets.

Safety: Use authentic documents and identify package-specific ratings.

Procedure: Record value range, tolerance, power/voltage rating, package, temperature limits, and applications.

Expected output: Side-by-side component data sheet.

Evaluation points: Ratings distinguished and terms explained.

- [ ] Submission checklist completed

Week 6 · Module 2

Verify series and parallel capacitance using three capacitors.

Objective: Verify series and parallel capacitance using three capacitors.

Resources: Three non-polarised capacitors, DMM, breadboard.

Safety: Discharge capacitors and remain below ratings.

Procedure: Calculate both combinations, measure individual and combined values, and compare.

Expected output: Calculation and measurement table.

Evaluation points: Correct formula, unit conversion, and tolerance discussion.

- [ ] Submission checklist completed

Week 7 · Module 2

Observe a step-down transformer nameplate without dismantling.

Objective: Observe a step-down transformer nameplate without dismantling.

Resources: Approved charger/adapter or exposed training transformer nameplate.

Safety: Do not open sealed mains equipment.

Procedure: Record input voltage/frequency, output voltage/current, symbols, insulation class where shown, and application.

Expected output: Annotated nameplate study.

Evaluation points: Primary/secondary ratings interpreted correctly.

- [ ] Submission checklist completed

Week 8 · Module 2

Visit or document an electric-vehicle charging station.

Objective: Visit or document an electric-vehicle charging station.

Resources: Publicly visible nameplate or operator-provided information.

Safety: Do not touch connectors, open panels, block access, or photograph restricted information.

Procedure: Record charger type, input/output ratings, connector, power, protection markings, and operator instructions.

Expected output: One-page charger specification report.

Evaluation points: Accurate transcription, safety awareness, source/date noted.

- [ ] Submission checklist completed

Week 9 · Module 3

Compare incandescent, fluorescent, LED and halogen light sources.

Objective: Compare incandescent, fluorescent, LED and halogen light sources.

Resources: Product labels, reference data, optional safe samples.

Safety: Do not handle hot or broken lamps; avoid exposed mains.

Procedure: Compare operating principle, power, efficacy, life, heat, control gear, and disposal.

Expected output: Comparative chart.

Evaluation points: Clear technical differences and environmental notes.

- [ ] Submission checklist completed

Week 10 · Module 3

Prepare a chart on the evolution of semiconductor devices.

Objective: Prepare a chart on the evolution of semiconductor devices.

Resources: Library/web resources approved by faculty.

Safety: Use original wording and cite sources.

Procedure: Create a timeline from crystal detectors and diodes to transistors, ICs, microprocessors and modern power devices.

Expected output: A3/A4 chart with dates and significance.

Evaluation points: Chronology, technical relevance, readable design.

- [ ] Submission checklist completed

Week 11 · Module 3

Explore online or offline circuit-simulation tools and document one.

Objective: Explore online or offline circuit-simulation tools and document one.

Resources: Computer and chosen simulator.

Safety: Use only safe low-voltage educational circuits.

Procedure: Compare features, component library, instruments, export, limitations, and simulate a diode or resistor circuit.

Expected output: Feature review plus simulation output.

Evaluation points: Circuit correct, parameters stated, limitations acknowledged.

- [ ] Submission checklist completed

Week 12 · Module 4

Document mobile-phone hardware and software features.

Objective: Document mobile-phone hardware and software features.

Resources: Manufacturer specification and device settings.

Safety: Do not expose personal data, credentials or disassemble sealed device.

Procedure: List processor, memory, display, sensors, radios, battery, charging, OS and security features.

Expected output: Technical feature sheet.

Evaluation points: Hardware/software distinguished; source cited.

- [ ] Submission checklist completed

Week 13 · Module 4

Prepare a chart on types of soldering iron.

Objective: Prepare a chart on types of soldering iron.

Resources: Catalogues/manuals and workshop samples.

Safety: Do not energise unfamiliar tools.

Procedure: Compare uncontrolled, temperature-controlled, station, hot-air and specialised tools by power, temperature control, tip and use.

Expected output: Illustrated comparison chart.

Evaluation points: Correct tool selection and safety.

- [ ] Submission checklist completed

Week 14 · Module 4

Observe a PCB in a safe electronic product.

Objective: Observe a PCB in a safe electronic product.

Resources: Unpowered open training product or visible board; magnifier.

Safety: Do not open sealed mains/battery products unless authorised and safe.

Procedure: Identify board type, approximate size, tracks, pads, vias, mounted components and one workmanship feature.

Expected output: Annotated sketch or photo with table.

Evaluation points: Features correctly labelled; orientation and safety stated.

- [ ] Submission checklist completed

Week 15 · Module 4

Simulate an LED switched circuit.

Objective: Simulate an LED switched circuit.

Resources: Tinkercad or another approved simulator.

Safety: Set a safe supply and include current-limiting resistor.

Procedure: Build source-switch-resistor-LED circuit, verify polarity, calculate resistor, simulate on/off and record current.

Expected output: Circuit screenshot/sketch and calculation.

Evaluation points: Correct polarity, current limit, and explanation.

- [ ] Submission checklist completed

ASSESSMENT

Official Evaluation Pattern and Preparation Guide

Assessment methodology overview		
Mode	**Tentative schedule**	**Converted/final marks**
Attendance	End of semester	5
CA1 — average self-learning activities	By 4th week	5
CE — continuous evaluation	By 7th week	10
CA2 — practical test, first half	By 11th week	10
CA3 — practical test, second half	By 14th week	10
CA4 — series exam, two modules	8th week	10
CA5 — series exam, two modules	15th week	10
ESE written theory	End of semester	60
ESE lab examination	End of semester	40

Practical continuous evaluation — total 50			
No.	**Criterion**	**Description**	**Marks**
1	Preparation & Punctuality	Attendance, record submission, pre-lab preparation, experiment objective and theory	10
2	Experimental Setup & Procedure	Identify components/equipment, correct connections, systematic and safe execution	10
3	Observation & Data Recording	Accuracy, units, tabulation and systematic recording	5
4	Analysis & Result Interpretation	Calculations, graphs, verification, discussion and conclusion	10
5	Viva Voce / Concept Understanding	Theory, procedure and related questions	10
6	Workmanship, Discipline & Teamwork	Instrument care, discipline, teamwork, attitude and safety	5

Practical test — total 40		
Category	**Things evaluated**	**Marks**
Write-up / Procedure	Aim, circuit/flowchart and stepwise procedure	10
Experiment Setup & Execution	Correct setup, instrument handling and accuracy	10
Observation & Result / Output	Readings, calculations/graphs and correctness	8
Viva Voce	Conceptual and related theoretical knowledge	8
Record	Completion, neatness and faculty certification	4

Series examination pattern — theory

Duration: 2 hours. Any two modules are assessed. For each module, Part A contains two 1-mark questions, Part B contains three 3-mark questions, and Part C contains a 7-mark question set with choice. The two-module total is 50 marks.

End-semester theory pattern

Duration: 2.5 hours. Each of four modules contributes 15 marks: two 1-mark questions, two answered from three 3-mark questions, and one 7-mark question answered from a choice set. Overall total: 60 marks.

PRACTICE QUESTIONS · MODULE 1 · CO1

Introduction to Electronics, Signals and Measurements

These are internally created practice questions, not official SITTTR questions.

One-mark questions — 20

1. Define or state the meaning of Definition and purpose of electronics. M1 · CO1 · 1 mark

Answer: Electronics is the engineering discipline that controls electron or charge-carrier flow to sense, process, store, communicate, and act on information.

Marking: one correct technical statement — 1 mark.

2. Define or state the meaning of Applications of electronics. M1 · CO1 · 1 mark

Answer: Electronics enables sensing, amplification, conversion, control, computation, communication, display, and protection.

Marking: one correct technical statement — 1 mark.

3. Define or state the meaning of Industrial automation and control. M1 · CO1 · 1 mark

Answer: Industrial electronics acquires sensor data, executes control logic, and drives actuators through PLCs, controllers, drives, relays, and safety circuits.

Marking: one correct technical statement — 1 mark.

4. Define or state the meaning of Instrumentation and measurement. M1 · CO1 · 1 mark

Answer: Instrumentation converts a physical quantity into a usable signal, conditions it, displays or records it, and may feed it to a controller.

Marking: one correct technical statement — 1 mark.

5. Define or state the meaning of Consumer electronics. M1 · CO1 · 1 mark

Answer: Consumer products integrate power supplies, user interfaces, processors, memory, sensors, audio/video stages, and communication modules.

Marking: one correct technical statement — 1 mark.

6. Define or state the meaning of Computer and IT systems. M1 · CO1 · 1 mark

Answer: Computers use electronic switching devices to represent, process, store, and communicate binary information.

Marking: one correct technical statement — 1 mark.

7. Define or state the meaning of Electric charge. M1 · CO1 · 1 mark

Answer: Electric charge is a fundamental property of matter measured in coulombs (C).

Marking: one correct technical statement — 1 mark.

8. Define or state the meaning of Voltage. M1 · CO1 · 1 mark

Answer: Voltage is electric potential difference: energy transferred per unit charge.

Marking: one correct technical statement — 1 mark.

9. Define or state the meaning of Electrical power. M1 · CO1 · 1 mark

Answer: Electrical power is the rate of electrical energy conversion.

Marking: one correct technical statement — 1 mark.

10. Define or state the meaning of AC quantities. M1 · CO1 · 1 mark

Answer: Alternating quantities change magnitude and direction periodically.

Marking: one correct technical statement — 1 mark.

11. Define or state the meaning of AC versus DC. M1 · CO1 · 1 mark

Answer: AC reverses direction periodically; DC has a fixed direction.

Marking: one correct technical statement — 1 mark.

12. Define or state the meaning of Analog signals. M1 · CO1 · 1 mark

Answer: An analog signal can take continuously varying values within a range.

Marking: one correct technical statement — 1 mark.

13. Define or state the meaning of Sinusoidal AC waveform. M1 · CO1 · 1 mark

Answer: A sine wave is a smooth periodic waveform defined by amplitude, frequency, phase, and offset.

Marking: one correct technical statement — 1 mark.

14. Define or state the meaning of Time period. M1 · CO1 · 1 mark

Answer: Time period T is the time required for one complete cycle.

Marking: one correct technical statement — 1 mark.

15. Define or state the meaning of Angular frequency. M1 · CO1 · 1 mark

Answer: Angular frequency expresses the rate of phase change in radians per second.

Marking: one correct technical statement — 1 mark.

16. Define or state the meaning of Instantaneous value. M1 · CO1 · 1 mark

Answer: The instantaneous value is the value of a time-varying quantity at a specified instant.

Marking: one correct technical statement — 1 mark.

17. Define or state the meaning of Electronic measuring instruments. M1 · CO1 · 1 mark

Answer: Measuring instruments extend human observation into voltage, current, resistance, time, frequency, and waveform domains.

Marking: one correct technical statement — 1 mark.

18. Define or state the meaning of Ammeter. M1 · CO1 · 1 mark

Answer: An ammeter measures current and is connected in series.

Marking: one correct technical statement — 1 mark.

19. Define or state the meaning of DC power supply. M1 · CO1 · 1 mark

Answer: A laboratory DC supply provides adjustable regulated voltage and current limiting.

Marking: one correct technical statement — 1 mark.

20. Define or state the meaning of Digital Storage Oscilloscope. M1 · CO1 · 1 mark

Answer: A DSO displays voltage versus time and can measure amplitude, period, frequency, phase, rise time, and transients.

Marking: one correct technical statement — 1 mark.

Three-mark questions — 15

1. Explain Communication electronics with its principle and one practical point. M1 · CO1 · 3 marks

Model points: Communication systems convert information into electrical or electromagnetic signals, transmit it through a medium, and recover it at a receiver. Typical blocks are source, encoder/modulator, channel, receiver/demodulator, decoder, and destination. Identify microphone, transmitter, antenna/cable, receiver, and loudspeaker blocks in a public-address system.

Suggested marking: definition 1; principle 1; application/engineering point 1.

2. Explain Instrumentation and measurement with its principle and one practical point. M1 · CO1 · 3 marks

Model points: Instrumentation converts a physical quantity into a usable signal, conditions it, displays or records it, and may feed it to a controller. Accuracy, precision, resolution, range, sensitivity, loading, and calibration are separate performance terms. Before measuring, select the correct function, range, terminals, reference point, and expected magnitude.

Suggested marking: definition 1; principle 1; application/engineering point 1.

3. Explain Consumer electronics with its principle and one practical point. M1 · CO1 · 3 marks

Model points: Consumer products integrate power supplies, user interfaces, processors, memory, sensors, audio/video stages, and communication modules. Cost, compactness, electromagnetic compatibility, safety, and user experience drive design decisions. Block-diagram a television, induction cooker, or washing machine.

Suggested marking: definition 1; principle 1; application/engineering point 1.

4. Explain Renewable-energy systems with its principle and one practical point. M1 · CO1 · 3 marks

Model points: Renewable-energy electronics converts, controls, protects, and monitors energy from solar, wind, storage, and grid interfaces. Power electronics performs DC–DC conversion, inversion, maximum-power-point tracking, charging, and grid synchronization. Follow the energy path in a small solar system: panel → controller → battery → inverter/load.

Suggested marking: definition 1; principle 1; application/engineering point 1.

5. Explain Voltage with its principle and one practical point. M1 · CO1 · 3 marks

Model points: Voltage is electric potential difference: energy transferred per unit charge. $V = W/Q$, where voltage is in volts, energy in joules, and charge in coulombs. Measure voltage between two points with a voltmeter connected in parallel.

Suggested marking: definition 1; principle 1; application/engineering point 1.

6. Explain Electrical energy with its principle and one practical point. M1 · CO1 · 3 marks

Model points: Electrical energy equals power multiplied by time. $E = Pt$. Joule is the SI unit; watt-hour and kilowatt-hour are widely used practical units. Convert time units before substitution and state the final unit.

Suggested marking: definition 1; principle 1; application/engineering point 1.

7. Explain DC quantities with its principle and one practical point. M1 · CO1 · 3 marks

Model points: Direct quantities maintain one direction, though their magnitude may contain ripple. An ideal DC source is constant with time; practical supplies may include ripple, noise, or transient variation. Measure battery and regulated-supply outputs using the correct DC range.

Suggested marking: definition 1; principle 1; application/engineering point 1.

8. Explain Analog signals with its principle and one practical point. M1 · CO1 · 3 marks

Model points: An analog signal can take continuously varying values within a range. Analog circuits preserve amplitude information but are sensitive to noise, offset, and component tolerances. Temperature-sensor voltage and microphone output are common examples.

Suggested marking: definition 1; principle 1; application/engineering point 1.

9. Explain Cycle with its principle and one practical point. M1 · CO1 · 3 marks

Model points: A cycle is one complete repetition of a periodic waveform. Equivalent points one cycle apart have the same magnitude and direction of change. Use repeated landmarks such as positive-going zero crossings.

Suggested marking: definition 1; principle 1; application/engineering point 1.

10. Explain Frequency with its principle and one practical point. M1 · CO1 · 3 marks

Model points: Frequency f is the number of cycles per second. $f = 1/T$ and is measured in hertz (Hz). Compare function-generator display with DSO measurement and calculate percentage error.

Suggested marking: definition 1; principle 1; application/engineering point 1.

11. Explain Instantaneous value with its principle and one practical point. M1 · CO1 · 3 marks

Model points: The instantaneous value is the value of a time-varying quantity at a specified instant. For a sine wave, $v = V_m \sin(\omega t + \varphi)$. Mark the time point and evaluate the phase angle.

Suggested marking: definition 1; principle 1; application/engineering point 1.

12. Explain Voltmeter with its principle and one practical point. M1 · CO1 · 3 marks

Model points: A voltmeter measures potential difference and is connected in parallel. A high input resistance minimizes loading. Connect the common lead to the reference and the voltage lead to the test point.

Suggested marking: definition 1; principle 1; application/engineering point 1.

13. Explain DC power supply with its principle and one practical point. M1 · CO1 · 3 marks

Model points: A laboratory DC supply provides adjustable regulated voltage and current limiting. Constant-voltage mode controls voltage; constant-current mode limits current when the load demand exceeds the set limit. Set voltage and current limit before connecting the circuit.

Suggested marking: definition 1; principle 1; application/engineering point 1.

14. Explain Digital Storage Oscilloscope with its principle and one practical point. M1 · CO1 · 3 marks

Model points: A DSO displays voltage versus time and can measure amplitude, period, frequency, phase, rise time, and transients. Vertical scale, horizontal scale, trigger, coupling, probe ratio, and reference connection determine the displayed result. Compensate the probe and confirm the probe-ratio setting.

Suggested marking: definition 1; principle 1; application/engineering point 1.

15. Explain Applications of electronics with its principle and one practical point. M1 · CO1 · 3 marks

Model points: Electronics enables sensing, amplification, conversion, control, computation, communication, display, and protection. Most modern products combine analog sensing, digital processing, power conversion, and human-machine interfaces. Classify applications by function rather than by product name.

Suggested marking: definition 1; principle 1; application/engineering point 1.

Seven-mark questions — 10

1. Discuss Automobile electronics in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M1 · CO1 · 7 marks

Model answer framework: Vehicle electronics covers engine control, battery management, braking, lighting, safety, infotainment, diagnostics, and electric-drive systems. Sensors and electronic control units exchange data over robust communication networks. Inspect a vehicle fuse-box diagram to distinguish power distribution from low-current control. **Precaution/common error:** Automotive transients can be severe; laboratory assumptions about supply quality may not apply.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

2. Discuss Renewable-energy systems in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M1 · CO1 · 7 marks

Model answer framework: Renewable-energy electronics converts, controls, protects, and monitors energy from solar, wind, storage, and grid interfaces. Power electronics performs DC–DC conversion, inversion, maximum-power-point tracking, charging, and grid synchronization. Follow the energy path in a small solar system: panel → controller → battery → inverter/load. **Precaution/common error:** High DC voltage and stored battery energy remain hazardous even when the grid is disconnected.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

3. Discuss Resistance in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M1 · CO1 · 7 marks

Model answer framework: Resistance opposes current and converts electrical energy, commonly to heat. For an ohmic conductor at constant conditions, $R = V/I$. De-energize the circuit before measuring resistance with a multimeter. **Precaution/common error:** In-circuit resistance can be distorted by parallel paths.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

4. Discuss AC versus DC in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M1 · CO1 · 7 marks

Model answer framework: AC reverses direction periodically; DC has a fixed direction. Their generation, transmission, conversion, measurement, and applications differ. AC is conveniently transformed; DC is native to batteries and most electronic circuits. Identify whether the instrument is set to AC or DC before taking a reading. **Precaution/common error:** The same numeric voltage does not imply equal heating or peak stress unless waveform and RMS meaning are known.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

5. Discuss Sinusoidal AC waveform in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M1 · CO1 · 7 marks

Model answer framework: A sine wave is a smooth periodic waveform defined by amplitude, frequency, phase, and offset. $v(t) = V_m \sin(\omega t + \varphi)$. Measure peak-to-peak voltage and period on a DSO, then calculate amplitude and frequency. **Precaution/common error:** Peak, peak-to-peak, average, and RMS are different quantities.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

6. Discuss Angular frequency in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M1 · CO1 · 7 marks

Model answer framework: Angular frequency expresses the rate of phase change in radians per second. $\omega = 2\pi f$. Use angular frequency when writing sinusoidal equations or analysing reactive circuits. **Precaution/common error:** Angular frequency is not the same as ordinary frequency.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

7. Discuss Electronic measuring instruments in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable.

M1 · CO1 · 7 marks

Model answer framework: Measuring instruments extend human observation into voltage, current, resistance, time, frequency, and waveform domains. The instrument must have suitable range, bandwidth, category rating, input impedance, and calibration. Inspect leads and settings before connecting. **Precaution/common error:** Measurement safety is determined by the entire setup, not just the instrument body.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

8. Discuss Function generator in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M1 · CO1 · 7 marks

Model answer framework: A function generator produces test waveforms such as sine, square, and triangle waves with adjustable frequency, amplitude, and offset. Its output amplitude can depend on load termination. Verify the actual output on a DSO rather than relying only on the panel display. **Precaution/common error:** Excessive offset or amplitude can damage the circuit under test.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

9. Discuss Applications of electronics in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable.

M1 · CO1 · 7 marks

Model answer framework: Electronics enables sensing, amplification, conversion, control, computation, communication, display, and protection. Most modern products combine analog sensing, digital processing, power conversion, and human-machine interfaces. Classify applications by function rather than by product name. **Precaution/common error:** Do not assume every electronic system is digital; sensors and power stages are often analog.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

10. Discuss Instrumentation and measurement in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable.

M1 · CO1 · 7 marks

Model answer framework: Instrumentation converts a physical quantity into a usable signal, conditions it, displays or records it, and may feed it to a controller. Accuracy, precision, resolution, range, sensitivity, loading, and calibration are separate performance terms. Before measuring, select the correct function, range, terminals, reference point, and expected magnitude. **Precaution/common error:** A digital display does not guarantee correctness; incorrect setup can produce a precise-looking wrong value.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

Multiple-choice questions — 15

1. Which term best matches this statement: "Electronics progressed from vacuum tubes to discrete semiconductor devices, integrated circuits, microprocessors, embedded systems, and highly integrated system-on-chip devices." M1 · CO1 · MCQ

A. Historical evolution of electronics
B. Industrial automation and control
C. Defence and security electronics
D. Voltage

Answer: A — Historical evolution of electronics.

2. Which term best matches this statement: "Industrial electronics acquires sensor data, executes control logic, and drives actuators through PLCs, controllers, drives, relays, and safety circuits." M1 · CO1 · MCQ

A. Automobile electronics
B. Electric charge
C. Electrical energy
D. Industrial automation and control

Answer: D — Industrial automation and control.

3. Which term best matches this statement: "Instrumentation converts a physical quantity into a usable signal, conditions it, displays or records it, and

may feed it to a controller.” M1 · CO1 · MCQ

A. Voltage
B. DC quantities
C. Instrumentation and measurement
D. Defence and security electronics

Answer: C — Instrumentation and measurement.

4. Which term best matches this statement: “Defence and security systems use sensing, communication, navigation, surveillance, identification, and resilient control.” M1 · CO1 · MCQ

A. Analog signals
B. Defence and security electronics
C. Electric charge
D. Electrical energy

Answer: B — Defence and security electronics.

5. Which term best matches this statement: “Electric charge is a fundamental property of matter measured in coulombs (C).” M1 · CO1 · MCQ

A. Electric charge
B. Resistance
C. AC versus DC
D. Cycle

Answer: A — Electric charge.

6. Which term best matches this statement: “Resistance opposes current and converts electrical energy, commonly to heat.” M1 · CO1 · MCQ

A. AC quantities
B. Digital signals
C. Angular frequency
D. Resistance

Answer: D — Resistance.

7. Which term best matches this statement: “Electrical energy equals power

multiplied by time.” M1 · CO1 · MCQ

A. Cycle
B. Instantaneous value
C. Electrical energy
D. AC versus DC

Answer: C — Electrical energy.

8. Which term best matches this statement: “AC reverses direction periodically; DC has a fixed direction.” M1 · CO1 · MCQ

A. Voltmeter
B. AC versus DC
C. Digital signals
D. Angular frequency

Answer: B — AC versus DC.

9. Which term best matches this statement: “A digital signal uses discrete states, commonly binary LOW and HIGH levels.” M1 · CO1 · MCQ

A. Digital signals
B. Time period
C. Peak value
D. DC power supply

Answer: A — Digital signals.

10. Which term best matches this statement: “A cycle is one complete repetition of a periodic waveform.” M1 · CO1 · MCQ

A. Angular frequency
B. Voltmeter
C. Digital Storage Oscilloscope
D. Cycle

Answer: D — Cycle.

11. Which term best matches this statement: “Angular frequency expresses the rate of phase change in radians per second.” M1 · CO1 · MCQ

A. DC power supply

B. Applications of electronics

C. Angular frequency

D. Peak value

Answer: C — Angular frequency.

12. Which term best matches this statement: "Peak value is the maximum magnitude reached relative to the zero or reference level." M1 · CO1 · MCQ

A. Medical and healthcare electronics

B. Peak value

C. Ammeter

D. Definition and purpose of electronics

Answer: B — Peak value.

13. Which term best matches this statement: "An ammeter measures current and is connected in series." M1 · CO1 · MCQ

A. Ammeter

B. Function generator

C. Communication electronics

D. Consumer electronics

Answer: A — Ammeter.

14. Which term best matches this statement: "A laboratory DC supply provides adjustable regulated voltage and current limiting." M1 · CO1 · MCQ

A. Definition and purpose of electronics

B. Medical and healthcare electronics

C. Computer and IT systems

D. DC power supply

Answer: D — DC power supply.

15. Which term best matches this statement: "Electronics is the engineering discipline that controls electron or charge-carrier flow to sense, process, store, communicate, and act on information." M1 · CO1 · MCQ

A. Consumer electronics

B. Electric current

C. Definition and purpose of electronics

D. Communication electronics

Answer: C — Definition and purpose of electronics.

Fill in the blanks — 10

1. Electric current is the rate of flow of ______. M1 · CO1 · Fill

Answer: charge

2. The SI unit of voltage is the ______. M1 · CO1 · Fill

Answer: volt

3. Frequency is the reciprocal of ______. M1 · CO1 · Fill

Answer: time period

4. Angular frequency is measured in ______. M1 · CO1 · Fill

Answer: radians per second

5. A voltmeter is connected in ______. M1 · CO1 · Fill

Answer: parallel

6. An ammeter is connected in ______. M1 · CO1 · Fill

Answer: series

7. A DSO plots voltage against ______. M1 · CO1 · Fill

Answer: time

8. A continuously varying signal is called an ______ signal. M1 · CO1 · Fill

Answer: analog

9. A laboratory supply operating in current-limit mode is in ______-current mode. M1 · CO1 · Fill

Answer: constant

10. For a centered sine wave, Vpp = _____ × Vpeak. M1 · CO1 · Fill

Answer: 2

True or false — 10

1. Conventional current in a metal is in the same direction as electron drift. M1 · CO1 · True/False

Answer: False. Conventional current is opposite to electron drift.

2. A voltmeter is normally connected in parallel. M1 · CO1 · True/False

Answer: True. It measures potential difference between two points.

3. An ammeter has high internal resistance. M1 · CO1 · True/False

Answer: False. It has low internal resistance.

4. A 1 kHz waveform has a period of 1 ms. M1 · CO1 · True/False

Answer: True. T = 1/1000 s = 1 ms.

5. Digital signals can only exist at exactly 0 V and 5 V. M1 · CO1 · True/False

Answer: False. Logic levels are defined voltage ranges and vary by logic family.

6. A DSO trigger helps stabilise a repetitive waveform. M1 · CO1 · True/False

Answer: True. Triggering starts acquisition at a repeatable event.

7. AC always has zero RMS value. M1 · CO1 · True/False

Answer: False. A sine wave has non-zero RMS despite zero average over a full cycle.

8. The DMM current jack can be left connected for voltage measurement. M1 · CO1 · True/False

Answer: False. The lead must be returned to the voltage jack.

9. Power is measured in watts. M1 · CO1 · True/False

Answer: True. Watt is joule per second.

10. A line tester alone proves safe isolation. M1 · CO1 · True/False

Answer: False. An approved verification method is required.

Viva questions — 10

1. What is the most important practical point when dealing with Medical and healthcare electronics? M1 · CO1 · Viva

Answer: Study why isolation and leakage-current limits are critical in patient-connected equipment. Also remember: Never connect general laboratory instruments directly to a patient or body.

2. What is the most important practical point when dealing with Defence and security electronics? M1 · CO1 · Viva

Answer: Compare normal commercial requirements with mission-critical requirements. Also remember: Avoid discussing or attempting restricted weapon-control details.

3. What is the most important practical point when dealing with Electric current? M1 · CO1 · Viva

Answer: Measure current by opening the circuit and inserting an ammeter in series. Also remember: Never connect an ammeter directly across a voltage source; its low internal resistance can cause destructive current.

4. What is the most important practical point when dealing with AC quantities? M1 · CO1 · Viva

Answer: Observe polarity and frequency using a DSO. Also remember: Mains AC is hazardous; use approved isolated training sources and probes.

5. What is the most important practical point when dealing with Analog signals? M1 · CO1 · Viva

Answer: Temperature-sensor voltage and microphone output are common examples. Also remember: Sampling an analog signal does not make the original physical quantity digital.

6. What is the most important practical point when dealing with Time period? M1 · CO1 · Viva

Answer: Use oscilloscope time divisions multiplied by time/div. Also remember: Convert milliseconds or microseconds to seconds before calculating frequency.

7. What is the most important practical point when dealing with Instantaneous value? M1 · CO1 · Viva

Answer: Mark the time point and evaluate the phase angle. Also remember: An instantaneous value can be zero even while the source is active.

8. What is the most important practical point when dealing with Digital multimeter? M1 · CO1 · Viva

Answer: Use diode mode to identify PN-junction polarity. Also remember: Continuity mode is not a substitute for measuring insulation resistance.

9. What is the most important practical point when dealing with Definition and purpose of electronics? M1 · CO1 · Viva

Answer: Trace any familiar system—mobile charger, escalator controller, medical monitor—from source to input, processing, output, and feedback. Also remember: Do not confuse electronics with electrical power engineering; the two overlap, but the design objectives and signal levels differ.

10. What is the most important practical point when dealing with Industrial automation and control? M1 · CO1 · Viva

Answer: In an escalator, speed sensing, safety-chain monitoring, VFD control, braking, and annunciation form coordinated electronic subsystems. Also remember: Never treat a safety circuit as ordinary control logic; safety architecture requires validated fail-safe behaviour.

Numerical problems — 5

1. A charge of 18 C passes in 3 s. Find current. M1 · CO1 · Numerical

Solution: $I=Q/t=18/3=6$ A.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

2. A 24 V load draws 0.5 A. Find power. M1 · CO1 · Numerical

Solution: $P = VI = 24 \times 0.5 = 12$ W.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

3. A 100 W appliance runs for 2.5 h. Find energy. M1 · CO1 · Numerical

Solution: $E = Pt = 250$ Wh $= 0.25$ kWh.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

4. Find frequency when T=4 ms. M1 · CO1 · Numerical

Solution: $T = 0.004$ s; $f = 1/T = 250$ Hz.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

5. Find angular frequency for f=1 kHz. M1 · CO1 · Numerical

Solution: $\omega = 2\pi \times 1000 \approx 6283$ rad/s.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

Diagram-based questions — 5

1. Draw and label a sine wave showing peak, period and zero crossings. M1 · CO1 · Diagram

Expected labels: Axes, peak labels, one full cycle, T and direction.

2. Show correct voltmeter connection. M1 · CO1 · Diagram

Expected labels: Voltmeter in parallel with correct reference.

3. Show correct ammeter connection. M1 · CO1 · Diagram

Expected labels: Circuit opened and ammeter inserted in series.

4. Sketch analog and digital signals. M1 · CO1 · Diagram

Expected labels: Continuous curve and discrete-level waveform with labels.

5. Draw a DSO measurement setup. M1 · CO1 · Diagram

Expected labels: Generator output, channel input, common reference, probe ratio.

Application-based questions — 5

1. An escalator speed sensor gives unstable pulses. List an initial measurement plan. M1 · CO1 · Application

Model answer: Check supply, reference, sensor gap, waveform amplitude, frequency, shielding, connector and DSO grounding.

2. A DC supply enters current limit when a board is connected. What should be checked? M1 · CO1 · Application

Model answer: Disconnect, inspect polarity/shorts, verify current limit, resistance to ground, connectors and component heating.

3. A frequency reading differs between generator and DSO. Explain checks. M1 · CO1 · Application

Model answer: Verify units, probe ratio, trigger, cursor placement, generator load setting and calibration.

4. A battery reads correct voltage unloaded but fails in use. Why? M1 · CO1 · Application

Model answer: Internal resistance or low capacity can cause loaded voltage collapse.

5. Why is a high-input-resistance voltmeter preferred? M1 · CO1 · Application

Model answer: It reduces loading and disturbance of the measured circuit.

PRACTICE QUESTIONS · MODULE 2 · CO2

Passive Electronic Components

These are internally created practice questions, not official SITTTR questions.

One-mark questions — 20

1. Define or state the meaning of Resistor symbol and construction. M2 · CO2 · 1 mark

Answer: A resistor is represented by a zig-zag or rectangular symbol and is built from a resistive element with two terminals.

Marking: one correct technical statement — 1 mark.

2. Define or state the meaning of Resistance parameters. M2 · CO2 · 1 mark

Answer: Important parameters include nominal resistance, tolerance, power rating, temperature coefficient, maximum working voltage, noise, and stability.

Marking: one correct technical statement — 1 mark.

3. Define or state the meaning of Power rating. M2 · CO2 · 1 mark

Answer: Power rating is the maximum continuous dissipation under specified ambient and mounting conditions.

Marking: one correct technical statement — 1 mark.

4. Define or state the meaning of Carbon-film resistor. M2 · CO2 · 1 mark

Answer: Carbon-film resistors use a carbon film deposited on a ceramic body, often trimmed helically.

Marking: one correct technical statement — 1 mark.

5. Define or state the meaning of Preset. M2 · CO2 · 1 mark

Answer: A preset is a small adjustable resistor intended for infrequent calibration.

Marking: one correct technical statement — 1 mark.

6. Define or state the meaning of LDR. M2 · CO2 · 1 mark

Answer: A light-dependent resistor decreases resistance as light intensity increases.

Marking: one correct technical statement — 1 mark.

7. Define or state the meaning of Thermistor. M2 · CO2 · 1 mark

Answer: A thermistor changes resistance strongly with temperature; NTC types decrease and PTC types increase.

Marking: one correct technical statement — 1 mark.

8. Define or state the meaning of Five-band resistor colour code. M2 · CO2 · 1 mark

Answer: Five bands represent three significant digits, multiplier, and tolerance.

Marking: one correct technical statement — 1 mark.

9. Define or state the meaning of Parallel resistors. M2 · CO2 · 1 mark

Answer: Parallel resistors share the same voltage and branch currents add.

Marking: one correct technical statement — 1 mark.

10. Define or state the meaning of Capacitor charging and discharging. M2 · CO2 · 1 mark

Answer: A capacitor voltage changes exponentially through a resistor.

Marking: one correct technical statement — 1 mark.

11. Define or state the meaning of Electrolytic capacitor. M2 · CO2 · 1 mark

Answer: Electrolytic capacitors provide high capacitance and are usually polarized.

Marking: one correct technical statement — 1 mark.

12. Define or state the meaning of Film or paper capacitor. M2 · CO2 · 1 mark

Answer: Film capacitors use plastic film dielectric and offer low loss and good pulse performance.

Marking: one correct technical statement — 1 mark.

13. Define or state the meaning of Ganged capacitor. M2 · CO2 · 1 mark

Answer: A ganged capacitor mechanically varies two or more capacitor sections together.

Marking: one correct technical statement — 1 mark.

14. Define or state the meaning of Trimmer. M2 · CO2 · 1 mark

Answer: A trimmer capacitor provides fine adjustment during alignment.

Marking: one correct technical statement — 1 mark.

15. Define or state the meaning of Numeric capacitor coding. M2 · CO2 · 1 mark

Answer: A three-digit code gives two significant digits and a multiplier in picofarads; 104 means 10×10^4 pF = 100 nF.

Marking: one correct technical statement — 1 mark.

16. Define or state the meaning of Parallel capacitors. M2 · CO2 · 1 mark

Answer: Parallel capacitors share the same voltage and capacitances add.

Marking: one correct technical statement — 1 mark.

17. Define or state the meaning of Transformer construction. M2 · CO2 · 1 mark

Answer: A transformer has primary and secondary windings coupled by a magnetic core.

Marking: one correct technical statement — 1 mark.

18. Define or state the meaning of Step-up transformer. M2 · CO2 · 1 mark

Answer: A step-up transformer has more secondary turns than primary turns and increases voltage while reducing available current ideally.

Marking: one correct technical statement — 1 mark.

19. Define or state the meaning of Isolation transformer. M2 · CO2 · 1 mark

Answer: An isolation transformer often has approximately 1:1 turns ratio and galvanically separates circuits.

Marking: one correct technical statement — 1 mark.

20. Define or state the meaning of RC timing circuit. M2 · CO2 · 1 mark

Answer: An RC circuit creates a predictable exponential voltage change.

Marking: one correct technical statement — 1 mark.

Three-mark questions — 15

1. Explain Tolerance with its principle and one practical point. M2 · CO2 · 3 marks

Model points: Tolerance states the permitted percentage deviation from nominal value. Rmin = R(1 − tolerance) and Rmax = R(1 + tolerance). Calculate the expected resistance range before judging a measured component.

Suggested marking: definition 1; principle 1; application/engineering point 1.

2. Explain Carbon-film resistor with its principle and one practical point. M2 · CO2 · 3 marks

Model points: Carbon-film resistors use a carbon film deposited on a ceramic body, often trimmed helically. They provide better stability and lower noise than carbon-composition types. Identify the spiral-cut construction in educational illustrations.

Suggested marking: definition 1; principle 1; application/engineering point 1.

3. Explain Preset with its principle and one practical point. M2 · CO2 · 3 marks

Model points: A preset is a small adjustable resistor intended for infrequent calibration. Its wiper divides the resistive track and may be single-turn or multi-turn. Mark the initial position before adjustment during servicing.

Suggested marking: definition 1; principle 1; application/engineering point 1.

4. Explain VDR with its principle and one practical point. M2 · CO2 · 3 marks

Model points: A voltage-dependent resistor or varistor has resistance that falls sharply above a threshold voltage. It clamps transient overvoltage by conducting surge current. Inspect surge protectors and power inputs for MOV/VDR devices.

Suggested marking: definition 1; principle 1; application/engineering point 1.

5. Explain Five-band resistor colour code with its principle and one practical point. M2 · CO2 · 3 marks

Model points: Five bands represent three significant digits, multiplier, and tolerance. Five-band coding supports precision values. Distinguish 4-band and 5-band parts before decoding.

Suggested marking: definition 1; principle 1; application/engineering point 1.

6. Explain Capacitor construction with its principle and one practical point.

M2 · CO2 · 3 marks

Model points: A capacitor contains two conductors separated by a dielectric. Geometry and dielectric determine capacitance, voltage rating, losses, leakage, and stability. Identify plates, dielectric, terminals, and enclosure in a cross-section.

Suggested marking: definition 1; principle 1; application/engineering point 1.

7. Explain Voltage rating with its principle and one practical point.

M2 · CO2 · 3 marks

Model points: Voltage rating is the maximum safe voltage under specified conditions. Applied DC plus ripple and transients must remain within the rating. Use margin and consider derating.

Suggested marking: definition 1; principle 1; application/engineering point 1.

8. Explain Ceramic capacitor with its principle and one practical point.

M2 · CO2 · 3 marks

Model points: Ceramic capacitors are non-polarized and cover values from pF to µF. Different ceramic classes have different stability and voltage dependence. Use stable types for timing or frequency-sensitive circuits.

Suggested marking: definition 1; principle 1; application/engineering point 1.

9. Explain Ganged capacitor with its principle and one practical point.

M2 · CO2 · 3 marks

Model points: A ganged capacitor mechanically varies two or more capacitor sections together. It allows simultaneous tuning of multiple resonant circuits. Identify rotor, stator, and multiple sections.

Suggested marking: definition 1; principle 1; application/engineering point 1.

10. Explain Direct capacitor marking with its principle and one practical point.

M2 · CO2 · 3 marks

Model points: Large capacitors often print capacitance and voltage directly, such as 47 µF 25 V. Polarity and temperature rating may also be marked. Read all markings, not only capacitance.

Suggested marking: definition 1; principle 1; application/engineering point 1.

11. Explain Parallel capacitors with its principle and one practical point.

M2 · CO2 · 3 marks

Model points: Parallel capacitors share the same voltage and capacitances add. $C_{eq} = C_1 + C_2 + \dots$ Use parallel parts to increase capacitance or improve frequency response.

Suggested marking: definition 1; principle 1; application/engineering point 1.

12. Explain Transformer construction with its principle and one practical point.

M2 · CO2 · 3 marks

Model points: A transformer has primary and secondary windings coupled by a magnetic core. Insulation system, core material, winding arrangement, and enclosure determine performance and safety. Identify primary and secondary from nameplate and resistance measurements only when de-energized.

Suggested marking: definition 1; principle 1; application/engineering point 1.

13. Explain Step-down transformer with its principle and one practical point.

M2 · CO2 · 3 marks

Model points: A step-down transformer has fewer secondary turns and reduces voltage. $N_s < N_p$ and $V_s < V_p$. Observe common adapters and control transformers.

Suggested marking: definition 1; principle 1; application/engineering point 1.

14. Explain RC timing circuit with its principle and one practical point.

M2 · CO2 · 3 marks

Model points: An RC circuit creates a predictable exponential voltage change. The time constant $\tau = RC$; about 63.2% charging at 1τ and 99.3% at 5τ. Plot capacitor voltage against time.

Suggested marking: definition 1; principle 1; application/engineering point 1.

15. Explain Resistance parameters with its principle and one practical point.

M2 · CO2 · 3 marks

Model points: Important parameters include nominal resistance, tolerance, power rating, temperature coefficient, maximum working voltage, noise, and stability. The nominal value alone is not sufficient for engineering selection. Compare a resistor datasheet with the requirements of a voltage-divider circuit.

Suggested marking: definition 1; principle 1; application/engineering point 1.

Seven-mark questions — 10

1. Discuss Metal-film resistor in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M2 · CO2 · 7 marks

Model answer framework: Metal-film resistors use a thin metal layer and offer good accuracy, stability, and low noise. They are common in measurement and precision signal circuits. Choose metal film for stable sensor scaling and reference dividers. **Precaution/common error:** Do not infer wattage from body colour alone.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

2. Discuss VDR in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M2 · CO2 · 7 marks

Model answer framework: A voltage-dependent resistor or varistor has resistance that falls sharply above a threshold voltage. It clamps transient overvoltage by conducting surge current. Inspect surge protectors and power inputs for MOV/VDR devices. **Precaution/common error:** A degraded varistor can fail short or show thermal damage; replace with correct rating.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

3. Discuss Series resistors in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M2 · CO2 · 7 marks

Model answer framework: Series resistors carry the same current and their voltage drops add. $R_{eq} = R1 + R2 + \ldots$ Use series combinations for voltage division and obtaining non-standard values. **Precaution/common error:** Power dissipation must be checked for each resistor, not only the total.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

4. Discuss Voltage rating in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M2 · CO2 · 7 marks

Model answer framework: Voltage rating is the maximum safe voltage under specified conditions. Applied DC plus ripple and transients must remain within the rating. Use margin and consider derating. **Precaution/common error:** Exceeding the rating can cause dielectric breakdown or explosion.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

5. Discuss Mica capacitor in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M2 · CO2 · 7 marks

Model answer framework: Mica capacitors offer high stability, low loss, and good high-frequency performance. The mica dielectric is physically and electrically stable. Recognize use in RF and precision circuits. **Precaution/common error:** Do not substitute with a general capacitor where Q and stability are critical.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

6. Discuss Numeric capacitor coding in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M2 · CO2 · 7 marks

Model answer framework: A three-digit code gives two significant digits and a multiplier in picofarads; 104 means 10×10^4 pF = 100 nF. Letter codes may specify tolerance. Decode to pF first, then convert to nF or µF. **Precaution/common error:** Do not read 104 as 104 pF.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

7. Discuss Air-core inductor in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M2 · CO2 · 7 marks

Model answer framework: An air-core inductor uses no ferromagnetic core and avoids core saturation. It is useful at high frequency but needs more turns for a given inductance. Observe coil geometry and spacing. **Precaution/common error:** Nearby metal and coils can alter inductance.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

8. Discuss Isolation transformer in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M2 · CO2 · 7 marks

Model answer framework: An isolation transformer often has approximately 1:1 turns ratio and galvanically separates circuits. It reduces direct conductive connection but does not make all contact safe. Use only approved isolation procedures in supervised labs. **Precaution/common error:** Touching both isolated secondary conductors can still cause shock.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

9. Discuss Resistor operation in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M2 · CO2 · 7 marks

Model answer framework: A resistor limits current, drops voltage, biases devices, terminates signals, and converts energy to heat. For an ohmic resistor, current is proportional to voltage at constant temperature. Use Ohm's law and power equations together when selecting a resistor. **Precaution/common error:** A resistor does not 'consume current'; it establishes current for a given voltage.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

10. Discuss Carbon-film resistor in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M2 · CO2 · 7 marks

Model answer framework: Carbon-film resistors use a carbon film deposited on a ceramic body, often trimmed helically. They provide better stability and lower noise than carbon-composition types. Identify the spiral-cut construction in educational illustrations. **Precaution/common error:** Film damage can cause open circuit or value increase.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

Multiple-choice questions — 15

1. Which term best matches this statement: “A resistor limits current, drops voltage, biases devices, terminates signals, and converts energy to heat.”

M2 · CO2 · MCQ

A. Resistor operation
B. Power rating
C. Preset
D. Four-band resistor colour code

Answer: A — Resistor operation.

2. Which term best matches this statement: “Power rating is the maximum continuous dissipation under specified ambient and mounting conditions.”

M2 · CO2 · MCQ

A. Metal-film resistor
B. VDR
C. Parallel resistors
D. Power rating

Answer: D — Power rating.

3. Which term best matches this statement: “Metal-film resistors use a thin metal layer and offer good accuracy, stability, and low noise.” M2 · CO2 · MCQ

A. Five-band resistor colour code
B. Capacitance
C. Metal-film resistor
D. Potentiometer

Answer: C — Metal-film resistor.

4. Which term best matches this statement: “A potentiometer has three terminals and produces an adjustable output voltage from a resistive track.” M2 · CO2 · MCQ

A. Ceramic capacitor
B. Potentiometer
C. Thermistor
D. Capacitor construction

Answer: B — Potentiometer.

5. Which term best matches this statement: “A thermistor changes resistance strongly with temperature; NTC types decrease and PTC types increase.” M2 · CO2 · MCQ

A. Thermistor
B. Series resistors
C. Voltage rating
D. Ganged capacitor

Answer: A — Thermistor.

6. Which term best matches this statement: “Series resistors carry the same current and their voltage drops add.” M2 · CO2 · MCQ

A. Capacitor charging and discharging
B. Film or paper capacitor
C. Direct capacitor marking
D. Series resistors

Answer: D — Series resistors.

7. Which term best matches this statement: “A capacitor voltage changes exponentially through a resistor.” M2 · CO2 · MCQ

A. Padder
B. Parallel capacitors
C. Capacitor charging and discharging
D. Electrolytic capacitor

Answer: C — Capacitor charging and discharging.

8. Which term best matches this statement: “Voltage rating is the maximum safe voltage under specified conditions.” M2 · CO2 · MCQ

A. Air-core inductor
B. Voltage rating
C. Film or paper capacitor
D. Direct capacitor marking

Answer: B — Voltage rating.

9. Which term best matches this statement: “Film capacitors use plastic film dielectric and offer low loss and good pulse performance.” M2 · CO2 · MCQ

A. Film or paper capacitor
B. Padder
C. Parallel capacitors
D. Step-up transformer

Answer: A — Film or paper capacitor.

10. Which term best matches this statement: “A padder is a small adjustable capacitor commonly used for lower-frequency alignment.” M2 · CO2 · MCQ

A. Numeric capacitor coding
B. Transformer construction
C. Voltage-divider circuit
D. Padder

Answer: D — Padder.

11. Which term best matches this statement: “A three-digit code gives two significant digits and a multiplier in picofarads; 104 means 10×10^4 pF = 100 nF.” M2 · CO2 · MCQ

A. Step-down transformer
B. Resistor operation
C. Numeric capacitor coding
D. Inductors

Answer: C — Numeric capacitor coding.

12. Which term best matches this statement: “An inductor stores energy in a magnetic field and opposes change in current.” M2 · CO2 · MCQ

A. Power rating
B. Inductors
C. Transformer operation
D. RC timing circuit

Answer: B — Inductors.

13. Which term best matches this statement: “Alternating current in the primary produces changing magnetic flux that induces voltage in the secondary.” M2 · CO2 · MCQ

A. Transformer operation
B. Isolation transformer
C. Resistance parameters
D. Metal-film resistor

Answer: A — Transformer operation.

14. Which term best matches this statement: “An isolation transformer often has approximately 1:1 turns ratio and galvanically separates circuits.” M2 · CO2 · MCQ

A. Resistor symbol and construction
B. Carbon-composition resistor
C. Potentiometer
D. Isolation transformer

Answer: D — Isolation transformer.

15. Which term best matches this statement: “A resistor is represented by a zig-zag or rectangular symbol and is built from a resistive element with two terminals.” M2 · CO2 · MCQ

A. Wire-wound resistor
B. Thermistor
C. Resistor symbol and construction
D. Tolerance

Answer: C — Resistor symbol and construction.

Fill in the blanks — 10

1. The last band of a common four-band resistor indicates ______. M2 · CO2 · Fill

Answer: tolerance

2. Series resistances are ______ together. M2 · CO2 · Fill

Answer: added

3. Parallel equivalent resistance is less than the ______ branch resistance. M2 · CO2 · Fill

Answer: smallest

4. Capacitance is measured in ______. M2 · CO2 · Fill

Answer: farads

5. Code 104 represents ______ nF. M2 · CO2 · Fill

Answer: 100

6. An electrolytic capacitor is normally ______. M2 · CO2 · Fill

Answer: polarized

7. The time constant of an RC circuit is ______. M2 · CO2 · Fill

Answer: RC

8. A transformer requires changing magnetic ______. M2 · CO2 · Fill

Answer: flux

9. A step-down transformer has fewer ______ turns than primary turns. M2 · CO2 · Fill

Answer: secondary

10. An LDR changes resistance with ______. M2 · CO2 · Fill

Answer: light

True or false — 10

1. The equivalent resistance of parallel resistors is greater than the largest resistor. M2 · CO2 · True/False

Answer: False. It is less than the smallest branch resistance.

2. A capacitor can retain charge after power is removed. M2 · CO2 · True/False

Answer: True. Stored energy may remain.

3. Code 104 means 104 pF. M2 · CO2 · True/False

Answer: False. It means 100,000 pF = 100 nF.

4. Electrolytic capacitors are usually polarized. M2 · CO2 · True/False

Answer: True. Polarity must be observed.

5. A transformer operates correctly from steady DC. M2 · CO2 · True/False

Answer: False. It needs changing magnetic flux.

6. A wire-wound resistor can have significant inductance. M2 · CO2 · True/False

Answer: True. The winding behaves inductively.

7. At one RC time constant a charging capacitor reaches about 63.2%.

M2 · CO2 · True/False

Answer: True. This follows the exponential charging law.

8. Series capacitor equivalent capacitance exceeds the largest capacitor.
M2 · CO2 · True/False

Answer: False. It is below the smallest capacitor.

9. A potentiometer has three terminals. M2 · CO2 · True/False

Answer: True. Two track ends and a wiper.

10. Resistance may be measured safely in an energized circuit. M2 · CO2 · True/False

Answer: False. De-energize and discharge first.

Viva questions — 10

1. What is the most important practical point when dealing with Carbon-composition resistor? M2 · CO2 · Viva

Answer: Recognize older carbon-composition parts by construction and context. Also remember: Do not replace a pulse-rated part solely by matching resistance and wattage.

2. What is the most important practical point when dealing with Potentiometer? M2 · CO2 · Viva

Answer: Verify end-to-end resistance and wiper continuity with a DMM. Also remember: A dirty or worn wiper causes intermittent output.

3. What is the most important practical point when dealing with Four-band resistor colour code? M2 · CO2 · Viva

Answer: Decode the bands, then verify by meter. Also remember: Brown can be confused with red under poor lighting.

4. What is the most important practical point when dealing with Capacitor charging and discharging? M2 · CO2 · Viva

Answer: Measure voltage at 1τ, 3τ, and 5τ. Also remember: Discharge capacitors safely through a suitable resistor, not by shorting.

5. What is the most important practical point when dealing with Ceramic capacitor? M2 · CO2 · Viva

Answer: Use stable types for timing or frequency-sensitive circuits. Also remember: High-capacitance ceramic values can fall significantly under DC bias.

6. What is the most important practical point when dealing with Trimmer? M2 · CO2 · Viva

Answer: Record the original position before adjustment. Also remember: Avoid excessive force and use low-capacitance tools for RF circuits.

7. What is the most important practical point when dealing with Parallel capacitors? M2 · CO2 · Viva

Answer: Use parallel parts to increase capacitance or improve frequency response. Also remember: Ripple current sharing may be unequal if ESR differs.

8. What is the most important practical point when dealing with Step-up transformer? M2 · CO2 · Viva

Answer: Check insulation and output hazards. Also remember: Higher voltage remains dangerous even at lower current rating.

9. What is the most important practical point when dealing with RC timing circuit? M2 · CO2 · Viva

Answer: Plot capacitor voltage against time. Also remember: Meter input resistance and capacitor leakage affect long time constants.

10. What is the most important practical point when dealing with Power rating? M2 · CO2 · Viva

Answer: Select a resistor with practical margin above calculated dissipation. Also remember: Operating continuously at the nameplate maximum reduces reliability.

Numerical problems — 5

1. Find series equivalent of 1 kΩ, 2.2 kΩ and 4.7 kΩ. M2 · CO2 · Numerical

Solution: R_{eq}=7.9 kΩ.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

2. Find parallel equivalent of 10 Ω and 15 Ω. M2 · CO2 · Numerical

Solution: $R_{eq}=10\times15/(10+15)=6$ Ω.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

3. Decode brown-black-orange-gold. M2 · CO2 · Numerical

Solution: 10×10^3 Ω=10 kΩ ±5%.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

4. Find Vout for Vin=15 V, R1=5 kΩ and R2=10 kΩ. M2 · CO2 · Numerical

Solution: $V_{out}=15\times10/(5+10)=10$ V.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

5. Find τ for R=47 kΩ and C=10 μF. M2 · CO2 · Numerical

Solution: $\tau=47000\times10\times10^{-6}=0.47$ s.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

Diagram-based questions — 5

1. Draw a four-band resistor and label each band function. M2 · CO2 · Diagram

Expected labels: Two digits, multiplier, tolerance.

2. Draw series and parallel resistor circuits. M2 · CO2 · Diagram

Expected labels: Correct nodes and current/voltage relationships.

3. Draw capacitor construction. M2 · CO2 · Diagram

Expected labels: Two plates, dielectric, terminals.

4. Draw transformer construction. M2 · CO2 · Diagram

Expected labels: Primary, secondary, core and flux path.

5. Draw an RC timing circuit. M2 · CO2 · Diagram

Expected labels: Source, resistor, capacitor and measured capacitor voltage.

Application-based questions — 5

1. A resistor repeatedly burns in a divider. What must be reviewed? M2 · CO2 · Application

Model answer: Resistance, actual voltage, power dissipation, rating/derating, ambient heat, fault conditions and creepage.

2. An RC timer is much slower than calculated. Give causes. M2 · CO2 · Application

Model answer: Capacitor tolerance/leakage, resistor tolerance, meter loading, wrong units, wiring or component value.

3. A potentiometer output is intermittent. How is it tested? M2 · CO2 · Application

Model answer: Check end-to-end track and smooth wiper resistance while rotating; inspect contamination and solder joints.

4. A transformer hums and overheats on DC. Explain. M2 · CO2 · Application

Model answer: Steady DC does not provide transformer action and can saturate the core, causing excessive current.

5. A 104 capacitor was replaced by 104 µF. Identify the error. M2 · CO2 · Application

Model answer: 104 is a code for 100 nF, not 104 µF.

PRACTICE QUESTIONS · MODULE 3 · CO3

Active Electronic Components

These are internally created practice questions, not official SITTTR questions.

One-mark questions — 20

1. Define or state the meaning of Energy-band theory. M3 · CO3 · 1 mark

Answer: Energy bands describe allowed electron-energy ranges in solids and explain electrical conduction.

Marking: one correct technical statement — 1 mark.

2. Define or state the meaning of Insulators. M3 · CO3 · 1 mark

Answer: Insulators have a large energy gap and very few free carriers under normal conditions.

Marking: one correct technical statement — 1 mark.

3. Define or state the meaning of Intrinsic semiconductor. M3 · CO3 · 1 mark

Answer: An intrinsic semiconductor is ideally pure, with equal electron and hole concentrations.

Marking: one correct technical statement — 1 mark.

4. Define or state the meaning of Covalent bonding. M3 · CO3 · 1 mark

Answer: Neighbouring semiconductor atoms share valence electrons through covalent bonds.

Marking: one correct technical statement — 1 mark.

5. Define or state the meaning of Doping. M3 · CO3 · 1 mark

Answer: Doping intentionally adds controlled impurity atoms to change carrier concentration.

Marking: one correct technical statement — 1 mark.

6. Define or state the meaning of Acceptor impurities. M3 · CO3 · 1 mark

Answer: Acceptor atoms create holes by accepting electrons into incomplete bonds.

Marking: one correct technical statement — 1 mark.

7. Define or state the meaning of P-type semiconductor. M3 · CO3 · 1 mark

Answer: P-type material has holes as majority carriers and electrons as minority carriers.

Marking: one correct technical statement — 1 mark.

8. Define or state the meaning of Minority carriers. M3 · CO3 · 1 mark

Answer: Minority carriers are present in smaller concentration, mainly through thermal generation.

Marking: one correct technical statement — 1 mark.

9. Define or state the meaning of Diffusion. M3 · CO3 · 1 mark

Answer: Diffusion is carrier movement from high concentration to low concentration.

Marking: one correct technical statement — 1 mark.

10. Define or state the meaning of Depletion region. M3 · CO3 · 1 mark

Answer: The depletion region contains fixed ionized donors and acceptors with few mobile carriers.

Marking: one correct technical statement — 1 mark.

11. Define or state the meaning of Diffusion current. M3 · CO3 · 1 mark

Answer: Diffusion current results from concentration-gradient-driven carrier motion.

Marking: one correct technical statement — 1 mark.

12. Define or state the meaning of Anode and cathode. M3 · CO3 · 1 mark

Answer: The anode is the P-side terminal and the cathode is the N-side terminal of a basic PN diode.

Marking: one correct technical statement — 1 mark.

13. Define or state the meaning of Forward bias. M3 · CO3 · 1 mark

Answer: Forward bias connects P to positive and N to negative, reducing the barrier and narrowing the depletion region.

Marking: one correct technical statement — 1 mark.

14. Define or state the meaning of PN diode V-I characteristics. M3 · CO3 · 1 mark

Answer: The V-I curve shows low forward current below the knee, rapid current rise in forward conduction, and small reverse current before breakdown.

Marking: one correct technical statement — 1 mark.

15. Define or state the meaning of Reverse saturation current. M3 · CO3 · 1 mark

Answer: Reverse saturation current is a small minority-carrier current under reverse bias before breakdown.

Marking: one correct technical statement — 1 mark.

16. Define or state the meaning of Diode ratings. M3 · CO3 · 1 mark

Answer: Key ratings include forward current, repetitive peak reverse voltage, power dissipation, surge current, junction temperature, and recovery time.

Marking: one correct technical statement — 1 mark.

17. Define or state the meaning of Zener diode. M3 · CO3 · 1 mark

Answer: A Zener diode is designed to operate safely in reverse breakdown with controlled current.

Marking: one correct technical statement — 1 mark.

18. Define or state the meaning of LED. M3 · CO3 · 1 mark

Answer: A light-emitting diode emits photons when forward-biased carriers recombine.

Marking: one correct technical statement — 1 mark.

19. Define or state the meaning of LED current-limiting resistor. M3 · CO3 · 1 mark

Answer: The resistor sets LED current: R = (VS − VF)/I.

Marking: one correct technical statement — 1 mark.

20. Define or state the meaning of Photodiode applications. M3 · CO3 · 1 mark

Answer: Photodiodes are used in optical communication, encoders, smoke detectors, light meters, remote receivers, and safety sensors.

Marking: one correct technical statement — 1 mark.

Three-mark questions — 15

1. Explain Semiconductors with its principle and one practical point.
M3 · CO3 · 3 marks

Model points: Semiconductors have conductivity between conductors and insulators and can be controlled by doping, temperature, light, and electric fields. Silicon is widely used because of material properties and manufacturing maturity. Relate controllable conductivity to diodes, transistors, sensors, and ICs.

Suggested marking: definition 1; principle 1; application/engineering point 1.

2. Explain Covalent bonding with its principle and one practical point.
M3 · CO3 · 3 marks

Model points: Neighbouring semiconductor atoms share valence electrons through covalent bonds. The crystal lattice is stable until energy frees an electron, leaving a hole. Use a lattice diagram to track one electron–hole pair.

Suggested marking: definition 1; principle 1; application/engineering point 1.

3. Explain Doping with its principle and one practical point. M3 · CO3 · 3 marks

Model points: Doping intentionally adds controlled impurity atoms to change carrier concentration. Pentavalent donors produce N-type material; trivalent acceptors produce P-type material. Identify the purpose of doping before discussing junctions.

Suggested marking: definition 1; principle 1; application/engineering point 1.

4. Explain N-type semiconductor with its principle and one practical point.
M3 · CO3 · 3 marks

Model points: N-type material has electrons as majority carriers and holes as minority carriers. It remains electrically neutral overall because fixed ionized donors balance mobile charge. Use carrier symbols and fixed-ion labels in diagrams.

Suggested marking: definition 1; principle 1; application/engineering point 1.

5. Explain Minority carriers with its principle and one practical point.
M3 · CO3 · 3 marks

Model points: Minority carriers are present in smaller concentration, mainly through thermal generation. They contribute to reverse saturation current and temperature behaviour. Relate increased temperature to increased minority-carrier concentration.

Suggested marking: definition 1; principle 1; application/engineering point 1.

6. Explain Recombination with its principle and one practical point.
M3 · CO3 · 3 marks

Model points: Recombination occurs when a conduction electron fills a hole, releasing energy. It reduces mobile carrier population. Mark recombination near the junction during formation.

Suggested marking: definition 1; principle 1; application/engineering point 1.

7. Explain Drift current with its principle and one practical point. M3 · CO3 · 3 marks

Model points: Drift current results from carriers moving under an electric field. In a PN junction it is associated strongly with minority carriers across the depletion region. Show carrier movement due to the built-in field.

Suggested marking: definition 1; principle 1; application/engineering point 1.

8. Explain Diode symbol and construction with its principle and one practical point. M3 · CO3 · 3 marks

Model points: A PN-junction diode has an anode connected to P-type material and a cathode connected to N-type material. The symbol indicates the two terminals and conventional current direction in forward conduction. Identify cathode band on common packages.

Suggested marking: definition 1; principle 1; application/engineering point 1.

9. Explain Forward bias with its principle and one practical point. M3 · CO3 · 3 marks

Model points: Forward bias connects P to positive and N to negative, reducing the barrier and narrowing the depletion region. Current rises rapidly after the applied voltage reaches the practical conduction region. Use a series resistor to limit current.

Suggested marking: definition 1; principle 1; application/engineering point 1.

10. Explain Knee voltage with its principle and one practical point.
M3 · CO3 · 3 marks

Model points: Knee voltage is the approximate forward-voltage region where current begins increasing rapidly. It is a useful teaching approximation rather than a sharp physical switch point. Estimate from a plotted curve.

Suggested marking: definition 1; principle 1; application/engineering point 1.

11. Explain Diode ratings with its principle and one practical point.

M3 · CO3 · 3 marks

Model points: Key ratings include forward current, repetitive peak reverse voltage, power dissipation, surge current, junction temperature, and recovery time. Ratings interact and require derating. Check datasheet limits before substitution.

Suggested marking: definition 1; principle 1; application/engineering point 1.

12. Explain Zener diode with its principle and one practical point. M3 · CO3 · 3 marks

Model points: A Zener diode is designed to operate safely in reverse breakdown with controlled current. It provides voltage reference or regulation in suitable circuits. Always use a series resistor to limit current.

Suggested marking: definition 1; principle 1; application/engineering point 1.

13. Explain LED polarity with its principle and one practical point. M3 · CO3 · 3 marks

Model points: Common through-hole LEDs often have a longer anode lead, shorter cathode lead, flat package edge on cathode side, and larger internal cathode electrode. DMM diode mode can confirm polarity and may produce a faint glow. Verify package-specific markings.

Suggested marking: definition 1; principle 1; application/engineering point 1.

14. Explain Photodiode applications with its principle and one practical point.

M3 · CO3 · 3 marks

Model points: Photodiodes are used in optical communication, encoders, smoke detectors, light meters, remote receivers, and safety sensors. A transimpedance amplifier commonly converts photocurrent to voltage. Relate an optical escalator entry sensor to emitter, receiver, alignment, and contamination.

Suggested marking: definition 1; principle 1; application/engineering point 1.

15. Explain Insulators with its principle and one practical point. M3 · CO3 · 3 marks

Model points: Insulators have a large energy gap and very few free carriers under normal conditions. They resist current but can break down under excessive electric field, temperature, contamination, or damage. Inspect insulation rating and condition.

Suggested marking: definition 1; principle 1; application/engineering point 1.

Seven-mark questions — 10

1. Discuss Thermal generation of carriers in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M3 · CO3 · 7 marks

Model answer framework: Heat can provide enough energy to break covalent bonds and create electron–hole pairs. Carrier concentration increases strongly with temperature. Relate temperature rise to increased leakage current. **Precaution/common error:** Excess temperature can cause thermal runaway in semiconductor circuits.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

2. Discuss N-type semiconductor in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M3 · CO3 · 7 marks

Model answer framework: N-type material has electrons as majority carriers and holes as minority carriers. It remains electrically neutral overall because fixed ionized donors balance mobile charge. Use carrier symbols and fixed-ion labels in diagrams. **Precaution/common error:** N-type does not mean the entire material has a net negative charge.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

3. Discuss PN-junction formation in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M3 · CO3 · 7 marks

Model answer framework: When P-type and N-type regions join, carriers diffuse across the boundary and recombine, leaving fixed ions. The exposed ions create a depletion region and electric field that opposes further diffusion. Follow electrons from N to P and holes from P to N during formation. **Precaution/common error:** The depletion region is depleted of mobile carriers, not of all charge.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

4. Discuss Drift current in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M3 · CO3 · 7 marks

Model answer framework: Drift current results from carriers moving under an electric field. In a PN junction it is associated strongly with minority carriers across the depletion region. Show carrier movement due to the built-in field. **Precaution/common error:** Drift direction depends on carrier charge and field direction.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

5. Discuss Silicon and germanium diodes in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable.

M3 · CO3 · 7 marks

Model answer framework: Silicon diodes typically show higher forward drop and lower leakage than germanium diodes. Material properties affect knee behaviour, reverse leakage, temperature sensitivity, and application. Compare measured diode-mode readings. **Precaution/common error:** DMM readings depend on test current and are not universal constants.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

6. Discuss Reverse saturation current in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable.

M3 · CO3 · 7 marks

Model answer framework: Reverse saturation current is a small minority-carrier current under reverse bias before breakdown. It increases strongly with temperature. Use proper instruments and ranges when measuring small current. **Precaution/common error:** Leakage can be too small for a basic lab DMM to resolve accurately.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

7. Discuss Dynamic resistance in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M3 · CO3 · 7 marks

Model answer framework: Dynamic or AC resistance is the local slope $\Delta V/\Delta I$ around an operating point. It describes small-signal behaviour. Choose two nearby points on the curve and use consistent units. **Precaution/common error:** Using widely separated points gives an average, not true local dynamic resistance.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

8. Discuss LED current-limiting resistor in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable.

M3 · CO3 · 7 marks

Model answer framework: The resistor sets LED current: $R = (VS - VF)/I$. Resistor power is approximately I^2R. Choose the next standard value that keeps current within rating. **Precaution/common error:** Using nominal VF without considering supply tolerance can overdrive the LED.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

9. Discuss Conductors in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M3 · CO3 · 7 marks

Model answer framework: Conductors have many available charge carriers and overlapping or partially filled energy bands. Small applied electric fields produce significant current. Relate copper wiring to low resistance and current-carrying capacity. **Precaution/common error:** Good conductivity does not remove heating or voltage-drop limits.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

10. Discuss Covalent bonding in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M3 · CO3 · 7 marks

Model answer framework: Neighbouring semiconductor atoms share valence electrons through covalent bonds. The crystal lattice is stable until energy frees an electron, leaving a hole. Use a lattice diagram to track one electron–hole pair. **Precaution/common error:** A hole is a useful carrier model, not a positively charged particle detached from an atom.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

Multiple-choice questions — 15

1. Which term best matches this statement: “Conductors have many available charge carriers and overlapping or partially filled energy bands.” M3 · CO3 · MCQ

A. Conductors
B. Intrinsic semiconductor
C. Doping
D. Majority carriers

Answer: A — Conductors.

2. Which term best matches this statement: “An intrinsic semiconductor is ideally pure, with equal electron and hole concentrations.” M3 · CO3 · MCQ

A. Thermal generation of carriers
B. N-type semiconductor

C. Diffusion
D. Intrinsic semiconductor

Answer: D — Intrinsic semiconductor.

3. Which term best matches this statement: "Heat can provide enough energy to break covalent bonds and create electron–hole pairs." M3 · CO3 · MCQ

A. Minority carriers
B. Barrier potential
C. Thermal generation of carriers
D. Donor impurities

Answer: C — Thermal generation of carriers.

4. Which term best matches this statement: "Donor atoms contribute loosely bound electrons to the conduction process." M3 · CO3 · MCQ

A. Diode symbol and construction
B. Donor impurities
C. P-type semiconductor
D. Recombination

Answer: B — Donor impurities.

5. Which term best matches this statement: "P-type material has holes as majority carriers and electrons as minority carriers." M3 · CO3 · MCQ

A. P-type semiconductor
B. PN-junction formation
C. Drift current
D. Forward bias

Answer: A — P-type semiconductor.

6. Which term best matches this statement: "When P-type and N-type regions join, carriers diffuse across the boundary and recombine, leaving fixed ions." M3 · CO3 · MCQ

A. Depletion region
B. Anode and cathode
C. Knee voltage

D. PN-junction formation

Answer: D — PN-junction formation.

7. Which term best matches this statement: “The depletion region contains fixed ionized donors and acceptors with few mobile carriers.” M3 · CO3 · MCQ

A. Reverse bias
B. Diode ratings
C. Depletion region
D. Diffusion current

Answer: C — Depletion region.

8. Which term best matches this statement: “Drift current results from carriers moving under an electric field.” M3 · CO3 · MCQ

A. Dynamic resistance
B. Drift current
C. Anode and cathode
D. Knee voltage

Answer: B — Drift current.

9. Which term best matches this statement: “The anode is the P-side terminal and the cathode is the N-side terminal of a basic PN diode.” M3 · CO3 · MCQ

A. Anode and cathode
B. Reverse bias
C. Diode ratings
D. LED

Answer: A — Anode and cathode.

10. Which term best matches this statement: “Reverse bias connects P to negative and N to positive, increasing barrier and widening the depletion region.” M3 · CO3 · MCQ

A. Reverse saturation current
B. Zener diode
C. Photodiode
D. Reverse bias

Answer: D — Reverse bias.

11. Which term best matches this statement: "Reverse saturation current is a small minority-carrier current under reverse bias before breakdown." M3 · CO3 · MCQ

A. LED polarity
B. Conductors
C. Reverse saturation current
D. Static resistance

Answer: C — Reverse saturation current.

12. Which term best matches this statement: "Static or DC resistance at an operating point is V/I." M3 · CO3 · MCQ

A. Intrinsic semiconductor
B. Static resistance
C. Zener V-I characteristics
D. Photodiode applications

Answer: B — Static resistance.

13. Which term best matches this statement: "The Zener curve includes ordinary forward conduction and a reverse breakdown region near the rated Zener voltage." M3 · CO3 · MCQ

A. Zener V-I characteristics
B. LED current-limiting resistor
C. Insulators
D. Thermal generation of carriers

Answer: A — Zener V-I characteristics.

14. Which term best matches this statement: "The resistor sets LED current: R = (VS − VF)/I." M3 · CO3 · MCQ

A. Energy-band theory
B. Silicon and germanium
C. Donor impurities
D. LED current-limiting resistor

Answer: D — LED current-limiting resistor.

15. Which term best matches this statement: “Energy bands describe allowed electron-energy ranges in solids and explain electrical conduction.” M3 · CO3 · MCQ

A. Electron-hole pairs
B. P-type semiconductor
C. Energy-band theory
D. Semiconductors

Answer: C — Energy-band theory.

Fill in the blanks — 10

1. Electrons are majority carriers in _____-type material. M3 · CO3 · Fill

Answer: N

2. Holes are majority carriers in _____-type material. M3 · CO3 · Fill

Answer: P

3. The region with few mobile carriers is the _____ region. M3 · CO3 · Fill

Answer: depletion

4. Forward bias makes the depletion region _____. M3 · CO3 · Fill

Answer: narrower

5. The band on a common diode package marks the _____. M3 · CO3 · Fill

Answer: cathode

6. Static diode resistance equals _____. M3 · CO3 · Fill

Answer: V/I

7. Dynamic diode resistance equals _____. M3 · CO3 · Fill

Answer: $\Delta V/\Delta I$

8. A Zener diode is designed for controlled reverse ______. M3 · CO3 · Fill

Answer: breakdown

9. An LED requires a current-limiting ______. M3 · CO3 · Fill

Answer: resistor

10. A photodiode converts light into ______. M3 · CO3 · Fill

Answer: current

True or false — 10

1. N-type material has a net negative charge. M3 · CO3 · True/False

Answer: False. It is electrically neutral overall.

2. Forward bias widens the depletion region. M3 · CO3 · True/False

Answer: False. It narrows the depletion region.

3. The cathode band identifies the cathode on common axial diodes. M3 · CO3 · True/False

Answer: True. The band is a common package marking.

4. A diode can be connected directly across a power supply without a resistor. M3 · CO3 · True/False

Answer: False. Current must be limited.

5. Static resistance is V/I at an operating point. M3 · CO3 · True/False

Answer: True. It is the origin-to-point ratio.

6. Dynamic resistance uses ΔV/ΔI near an operating point. M3 · CO3 · True/False

Answer: True. It is the local slope.

7. Zener voltage is perfectly constant at all currents. M3 · CO3 · True/False

Answer: False. It varies with current, tolerance, and temperature.

8. LED colour depends mainly on semiconductor band gap. M3 · CO3 · True/False

Answer: True. Photon energy is linked to the band gap.

9. Photodiodes are only used in forward bias. M3 · CO3 · True/False

Answer: False. They are commonly used in reverse bias or zero bias.

10. Reverse saturation current generally increases with temperature. M3 · CO3 · True/False

Answer: True. Minority-carrier concentration increases.

Viva questions — 10

1. What is the most important practical point when dealing with Silicon and germanium? M3 · CO3 · Viva

Answer: Compare approximate diode knee voltages under similar current. Also remember: These values are approximate and depend on current and temperature.

2. What is the most important practical point when dealing with Donor impurities? M3 · CO3 · Viva

Answer: Show the fifth valence electron becoming available. Also remember: The ionized donor atom remains fixed in the lattice.

3. What is the most important practical point when dealing with Majority carriers? M3 · CO3 · Viva

Answer: Use majority-carrier direction to explain diode conduction. Also remember: Minority carriers still matter, especially in reverse current.

4. What is the most important practical point when dealing with Depletion region? M3 · CO3 · Viva

Answer: Compare width under forward and reverse bias. Also remember: The region can change width with applied voltage.

5. What is the most important practical point when dealing with Diode symbol and construction? M3 · CO3 · Viva

Answer: Identify cathode band on common packages. Also remember: Package markings vary; confirm uncertain devices with a meter and datasheet.

6. What is the most important practical point when dealing with PN diode V-I characteristics? M3 · CO3 · Viva

Answer: Plot measured current against diode voltage using safe steps. Also remember: Do not connect data points in a way that suggests unmeasured accuracy.

7. What is the most important practical point when dealing with Diode ratings? M3 · CO3 · Viva

Answer: Check datasheet limits before substitution. Also remember: A diode with sufficient current rating may still have inadequate reverse-voltage or speed rating.

8. What is the most important practical point when dealing with LED? M3 · CO3 · Viva

Answer: Use a series resistor selected from supply voltage, forward voltage, and desired current. Also remember: Never test an LED directly across a battery or power supply without current limiting.

9. What is the most important practical point when dealing with Photodiode applications? M3 · CO3 · Viva

Answer: Relate an optical escalator entry sensor to emitter, receiver, alignment, and contamination. Also remember: Do not stare into unknown infrared emitters or lasers.

10. What is the most important practical point when dealing with Intrinsic semiconductor? M3 · CO3 · Viva

Answer: Visualize covalent bonds and thermally released carriers. Also remember: At room temperature intrinsic conductivity is limited compared with doped material.

Numerical problems — 5

1. A diode has 0.7 V at 10 mA. Find static resistance. M3 · CO3 · Numerical

Solution: $R_{DC}=0.7/0.01=70\ \Omega$.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

2. Voltage changes 0.66 to 0.70 V while current changes 5 to 9 mA. Find dynamic resistance. M3 · CO3 · Numerical

Solution: rac=0.04/0.004=10 Ω.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

3. A 9 V source drives a 2 V LED at 15 mA. Find ideal resistor. M3 · CO3 · Numerical

Solution: R=(9−2)/0.015≈466.7 Ω; choose 470 Ω.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

4. A 5.1 V Zener carries 20 mA. Find Zener power. M3 · CO3 · Numerical

Solution: P=VZIZ=5.1×0.02=0.102 W.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

5. A 12 V supply, 560 Ω resistor and 0.7 V diode are in series. Estimate current. M3 · CO3 · Numerical

Solution: I=(12−0.7)/560≈20.2 mA.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

Diagram-based questions — 5

1. Draw energy-band diagrams of conductor, semiconductor and insulator. M3 · CO3 · Diagram

Expected labels: Valence band, conduction band and relative gap.

2. Draw N-type and P-type carrier models. M3 · CO3 · Diagram

Expected labels: Dopants, majority/minority carriers, neutrality.

3. Draw PN-junction formation. M3 · CO3 · Diagram

Expected labels: Diffusion, fixed ions, depletion region and electric field.

4. Draw forward and reverse bias connections. M3 · CO3 · Diagram

Expected labels: Correct source polarity and series resistor.

5. Sketch diode and Zener V-I curves. M3 · CO3 · Diagram

Expected labels: Axes, knee, leakage and breakdown labels.

Application-based questions — 5

1. An LED fails immediately on a 12 V supply. What is the likely design error? M3 · CO3 · Application

Model answer: Missing or undersized current-limiting resistor, wrong polarity, excessive reverse voltage or wrong rating.

2. A diode reads low in both DMM directions in circuit. What next? M3 · CO3 · Application

Model answer: Isolate one lead and retest; parallel paths may affect the reading.

3. A Zener regulator output is low. What should be checked? M3 · CO3 · Application

Model answer: Supply, series resistor, load current, Zener polarity, operating current, power and device value.

4. Why can a diode leak more when hot? M3 · CO3 · Application

Model answer: Minority-carrier concentration rises with temperature.

5. A photodiode sensor is noisy. Suggest checks. M3 · CO3 · Application

Model answer: Shielding, bias, amplifier, ambient light, alignment, contamination, grounding and bandwidth.

PRACTICE QUESTIONS · MODULE 4 · CO4

PCB and Soldering

These are internally created practice questions, not official SITTTR questions.

One-mark questions — 20

1. Define or state the meaning of Purpose of a PCB. M4 · CO4 · 1 mark

Answer: A printed circuit board mechanically supports components and electrically interconnects them with patterned copper conductors.

Marking: one correct technical statement — 1 mark.

2. Define or state the meaning of Copper layer. M4 · CO4 · 1 mark

Answer: Copper foil forms tracks, pads, planes, and shielding features.

Marking: one correct technical statement — 1 mark.

3. Define or state the meaning of Silkscreen. M4 · CO4 · 1 mark

Answer: Silkscreen identifies references, polarity, orientation, test points, warnings, and assembly information.

Marking: one correct technical statement — 1 mark.

4. Define or state the meaning of Vias. M4 · CO4 · 1 mark

Answer: Vias are plated holes that connect copper between layers.

Marking: one correct technical statement — 1 mark.

5. Define or state the meaning of Multilayer PCB. M4 · CO4 · 1 mark

Answer: A multilayer PCB contains internal copper layers laminated within the board.

Marking: one correct technical statement — 1 mark.

6. Define or state the meaning of FR-4. M4 · CO4 · 1 mark

Answer: FR-4 is a flame-retardant glass-reinforced epoxy laminate widely used for rigid PCBs.

Marking: one correct technical statement — 1 mark.

7. Define or state the meaning of Track width. M4 · CO4 · 1 mark

Answer: Track width is selected from current, copper thickness, temperature rise, length, and manufacturing limits.

Marking: one correct technical statement — 1 mark.

8. Define or state the meaning of Creepage. M4 · CO4 · 1 mark

Answer: Creepage is the shortest distance along an insulating surface between conductive parts.

Marking: one correct technical statement — 1 mark.

9. Define or state the meaning of Ground-return paths. M4 · CO4 · 1 mark

Answer: Current returns to its source through a complete loop; ground is not an infinite sink.

Marking: one correct technical statement — 1 mark.

10. Define or state the meaning of Layout preparation. M4 · CO4 · 1 mark

Answer: Layout preparation converts schematic connectivity and mechanical constraints into footprints, placement, routing, and fabrication data.

Marking: one correct technical statement — 1 mark.

11. Define or state the meaning of Cleaning. M4 · CO4 · 1 mark

Answer: After etching, resist and residues are removed and copper is cleaned for inspection and soldering.

Marking: one correct technical statement — 1 mark.

12. Define or state the meaning of Component mounting. M4 · CO4 · 1 mark

Answer: Mounting controls component height, heat clearance, mechanical support, and serviceability.

Marking: one correct technical statement — 1 mark.

13. Define or state the meaning of Soldering iron. M4 · CO4 · 1 mark

Answer: A soldering iron supplies controlled heat to the joint.

Marking: one correct technical statement — 1 mark.

14. Define or state the meaning of Iron stand. M4 · CO4 · 1 mark

Answer: An iron stand safely supports the hot tool and usually includes tip-cleaning media.

Marking: one correct technical statement — 1 mark.

15. Define or state the meaning of Desoldering wick. M4 · CO4 · 1 mark

Answer: Copper braid with flux absorbs molten solder by capillary action.

Marking: one correct technical statement — 1 mark.

16. Define or state the meaning of Component soldering. M4 · CO4 · 1 mark

Answer: A reliable joint is made by heating the pad and lead together, applying solder to the joint, removing solder, then removing heat.

Marking: one correct technical statement — 1 mark.

17. Define or state the meaning of Dry joint. M4 · CO4 · 1 mark

Answer: A dry joint has poor wetting, often caused by oxidation, contamination, insufficient heat, or movement.

Marking: one correct technical statement — 1 mark.

18. Define or state the meaning of Insufficient solder. M4 · CO4 · 1 mark

Answer: Insufficient solder leaves incomplete wetting or inadequate mechanical/electrical contact.

Marking: one correct technical statement — 1 mark.

19. Define or state the meaning of Lifted pad. M4 · CO4 · 1 mark

Answer: A lifted pad separates from the substrate due to excessive heat, force, repeated rework, or poor adhesion.

Marking: one correct technical statement — 1 mark.

20. Define or state the meaning of Soldering safety. M4 · CO4 · 1 mark

Answer: Soldering hazards include burns, fumes, sharp leads, splashes, chemicals, electricity, and fire.

Marking: one correct technical statement — 1 mark.

Three-mark questions — 15

1. Explain Copper layer with its principle and one practical point. M4 · CO4 · 3 marks

Model points: Copper foil forms tracks, pads, planes, and shielding features. Copper thickness and track geometry determine resistance, current capacity, and heat rise. Identify heavy-current tracks and compare width with signal tracks.

Suggested marking: definition 1; principle 1; application/engineering point 1.

2. Explain Pads with its principle and one practical point. M4 · CO4 · 3 marks

Model points: Pads provide solderable connection areas for component leads, terminals, vias, or test points. Pad diameter, hole size, annular ring, thermal relief, and mechanical stress determine reliability. Inspect for cracks and lifted pads under magnification.

Suggested marking: definition 1; principle 1; application/engineering point 1.

3. Explain Double-sided PCB with its principle and one practical point. M4 · CO4 · 3 marks

Model points: A double-sided PCB has copper on both sides with plated holes or vias connecting them. It supports denser routing and improved grounding. Inspect solder joints on both sides where accessible.

Suggested marking: definition 1; principle 1; application/engineering point 1.

4. Explain Copper-clad laminate with its principle and one practical point. M4 · CO4 · 3 marks

Model points: Copper-clad laminate is substrate bonded with copper foil before patterning. PCB fabrication selectively removes or builds copper to create the circuit. Handle edges carefully and prevent oxidation before processing.

Suggested marking: definition 1; principle 1; application/engineering point 1.

5. Explain Electrical clearance with its principle and one practical point. M4 · CO4 · 3 marks

Model points: Clearance is the shortest distance through air between conductive parts. Required clearance increases with voltage, pollution, transient category, and standards. Measure actual conductor-to-conductor distance, not only nominal grid spacing.

Suggested marking: definition 1; principle 1; application/engineering point 1.

6. Explain Ground-return paths with its principle and one practical point.

M4 · CO4 · 3 marks

Model points: Current returns to its source through a complete loop; ground is not an infinite sink. High-frequency return current follows the path of lowest impedance, often near the signal path. Identify signal and return together.

Suggested marking: definition 1; principle 1; application/engineering point 1.

7. Explain Artwork transfer with its principle and one practical point.

M4 · CO4 · 3 marks

Model points: Artwork transfer places the required copper pattern or resist image onto copper-clad material. Manual educational methods may use toner transfer or photoresist under controlled conditions. Inspect for breaks, pinholes, and unwanted bridges before etching.

Suggested marking: definition 1; principle 1; application/engineering point 1.

8. Explain Component insertion with its principle and one practical point.

M4 · CO4 · 3 marks

Model points: Components are inserted according to reference, value, orientation, polarity, and lead-forming rules. Proper lead spacing avoids stress on the component body and pads. Use the BOM and assembly drawing.

Suggested marking: definition 1; principle 1; application/engineering point 1.

9. Explain Soldering iron with its principle and one practical point.

M4 · CO4 · 3 marks

Model points: A soldering iron supplies controlled heat to the joint. Temperature control, tip size, power, grounding, and maintenance affect results. Select a tip that transfers heat efficiently without contacting adjacent pads.

Suggested marking: definition 1; principle 1; application/engineering point 1.

10. Explain Tip cleaner with its principle and one practical point. M4 · CO4 · 3 marks

Model points: Brass wool or a damp sponge removes residue from the soldering tip. Cleaning followed by light tinning preserves heat transfer and reduces oxidation. Clean only as needed; maintain a bright tinned surface.

Suggested marking: definition 1; principle 1; application/engineering point 1.

11. Explain Tip tinning with its principle and one practical point. M4 · CO4 · 3 marks

Model points: Tip tinning coats the iron tip with a thin layer of solder. It protects against oxidation and improves thermal contact. Tin the tip after cleaning and before storage.

Suggested marking: definition 1; principle 1; application/engineering point 1.

12. Explain Dry joint with its principle and one practical point. M4 · CO4 · 3 marks

Model points: A dry joint has poor wetting, often caused by oxidation, contamination, insufficient heat, or movement. It can produce high resistance or intermittent operation. Rework by cleaning, fluxing, and reheating correctly.

Suggested marking: definition 1; principle 1; application/engineering point 1.

13. Explain Excess solder with its principle and one practical point. M4 · CO4 · 3 marks

Model points: Excess solder obscures joint geometry, can hide poor wetting, and increases bridge risk. The objective is adequate—not maximum—solder. Remove excess and inspect the actual fillet.

Suggested marking: definition 1; principle 1; application/engineering point 1.

14. Explain ESD precautions with its principle and one practical point. M4 · CO4 · 3 marks

Model points: Electrostatic discharge can damage sensitive devices without visible evidence. Controls include grounded work surfaces, wrist straps, antistatic packaging, humidity management, and handling by edges. Verify the ESD workstation before opening protected components.

Suggested marking: definition 1; principle 1; application/engineering point 1.

15. Explain Substrate with its principle and one practical point. M4 · CO4 · 3 marks

Model points: The substrate provides mechanical support and electrical insulation. Common materials include FR-4 glass-reinforced epoxy; properties include dielectric strength, thermal rating, moisture resistance, and flame performance. Check board material and thickness where documented.

Suggested marking: definition 1; principle 1; application/engineering point 1.

Seven-mark questions — 10

1. Discuss Pads in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M4 · CO4 · 7 marks

Model answer framework: Pads provide solderable connection areas for component leads, terminals, vias, or test points. Pad diameter, hole size, annular ring, thermal relief, and mechanical stress determine reliability. Inspect for cracks and lifted pads under magnification. **Precaution/common error:** Repeated heating can detach a pad from the substrate.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

2. Discuss FR-4 in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M4 · CO4 · 7 marks

Model answer framework: FR-4 is a flame-retardant glass-reinforced epoxy laminate widely used for rigid PCBs. It offers good general mechanical and electrical performance. Recognize woven glass texture at cut edges where safe. **Precaution/common error:** FR-4 grade and temperature rating vary; use documentation.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

3. Discuss Electrical clearance in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M4 · CO4 · 7 marks

Model answer framework: Clearance is the shortest distance through air between conductive parts. Required clearance increases with voltage, pollution, transient category, and standards. Measure actual conductor-to-conductor distance, not only nominal grid spacing. **Precaution/common error:** Solder protrusions and component leads can reduce clearance.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

4. Discuss Layout preparation in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M4 · CO4 · 7 marks

Model answer framework: Layout preparation converts schematic connectivity and mechanical constraints into footprints, placement, routing, and fabrication data. Design-rule checks detect many but not all errors. Verify netlist, footprints, polarity, holes, board outline, and test points. **Precaution/common error:** A correct schematic does not guarantee correct footprint pin numbering.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

5. Discuss Component mounting in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M4 · CO4 · 7 marks

Model answer framework: Mounting controls component height, heat clearance, mechanical support, and serviceability. Heavy or hot components may require brackets, spacing, adhesive, or fasteners. Check connector seating and polarity before soldering. **Precaution/common error:** Solder should not be used as the only mechanical support for heavy parts.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

6. Discuss Iron stand in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M4 · CO4 · 7 marks

Model answer framework: An iron stand safely supports the hot tool and usually includes tip-cleaning media. It prevents burns, cable damage, and bench fires. Return the iron to the stand immediately after each joint. **Precaution/common error:** Never place a hot iron directly on the bench.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

7. Discuss Desoldering in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M4 · CO4 · 7 marks

Model answer framework: Desoldering removes components while preserving pads, barrels, tracks, and the component if required. Add fresh solder or flux, control heat, remove solder, and release all leads before lifting. Verify each terminal is free before applying force. **Precaution/common error:** Mechanical force while solder remains solid causes lifted pads and broken vias.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

8. Discuss Excess solder in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M4 · CO4 · 7 marks

Model answer framework: Excess solder obscures joint geometry, can hide poor wetting, and increases bridge risk. The objective is adequate—not maximum—solder. Remove excess and inspect the actual fillet. **Precaution/common error:** Large blobs make troubleshooting harder.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

9. Discuss Purpose of a PCB in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M4 · CO4 · 7 marks

Model answer framework: A printed circuit board mechanically supports components and electrically interconnects them with patterned copper conductors. PCBs improve repeatability, compactness, reliability, assembly speed, and service documentation. Trace power, ground, and signal paths on a simple board. **Precaution/common error:** A PCB drawing must distinguish copper side, component side, and viewing orientation.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

10. Discuss Tracks in detail with working principle, engineering relevance, precautions and a labelled sketch where suitable. M4 · CO4 · 7 marks

Model answer framework: Tracks are copper conductors connecting circuit nodes. Width, thickness, length, temperature, and allowable voltage drop determine current capability. Use continuity mode only on de-energized boards. **Precaution/common error:** A continuity beep does not prove low enough resistance for a power path.

Suggested marking: definition 1; principle 2; construction/characteristics 1; application 1; diagram 1; safety/limitation 1.

Multiple-choice questions — 15

1. Which term best matches this statement: "A PCB combines insulating substrate, copper conductors, holes or vias, protective coatings, markings, and component footprints." M4 · CO4 · MCQ

A. PCB structure

B. Solder mask

C. Single-sided PCB

D. Copper-clad laminate

Answer: A — PCB structure.

2. Which term best matches this statement: "Silkscreen identifies references, polarity, orientation, test points, warnings, and assembly information." M4 · CO4 · MCQ

A. Vias

B. FR-4

C. Electrical clearance

D. Silkscreen

Answer: D — Silkscreen.

3. Which term best matches this statement: “Vias are plated holes that connect copper between layers.” M4 · CO4 · MCQ

A. Track width
B. Ground-return paths
C. Vias
D. Multilayer PCB

Answer: C — Vias.

4. Which term best matches this statement: “PCB material selection affects electrical loss, temperature capability, mechanical strength, fire performance, and cost.” M4 · CO4 · MCQ

A. Artwork transfer
B. PCB materials
C. Component placement
D. PCB routing

Answer: B — PCB materials.

5. Which term best matches this statement: “Track width is selected from current, copper thickness, temperature rise, length, and manufacturing limits.” M4 · CO4 · MCQ

A. Track width
B. Creepage
C. Layout preparation
D. Component insertion

Answer: A — Track width.

6. Which term best matches this statement: “Creepage is the shortest distance along an insulating surface between conductive parts.” M4 · CO4 · MCQ

A. Power tracks
B. Cleaning
C. Soldering iron
D. Creepage

Answer: D — Creepage.

7. Which term best matches this statement: “Connectors must satisfy wiring access, keying, strain relief, creepage, current, and service orientation.” M4 · CO4 · MCQ

A. PCB inspection
B. Tip cleaner
C. Connector placement
D. Etching

Answer: C — Connector placement.

8. Which term best matches this statement: “After etching, resist and residues are removed and copper is cleaned for inspection and soldering.” M4 · CO4 · MCQ

A. Component soldering
B. Cleaning
C. Component mounting
D. Iron stand

Answer: B — Cleaning.

9. Which term best matches this statement: “Mounting controls component height, heat clearance, mechanical support, and serviceability.” M4 · CO4 · MCQ

A. Component mounting
B. Solder wire
C. Desoldering wick
D. Dry joint

Answer: A — Component mounting.

10. Which term best matches this statement: “Flux removes oxides and promotes wetting during soldering.” M4 · CO4 · MCQ

A. Desoldering pump
B. Good solder joint
C. Excess solder
D. Flux

Answer: D — Flux.

11. Which term best matches this statement: “A desoldering pump uses suction to remove molten solder.” M4 · CO4 · MCQ

A. Solder bridge
B. ESD precautions
C. Desoldering pump
D. Component soldering

Answer: C — Desoldering pump.

12. Which term best matches this statement: “Desoldering removes components while preserving pads, barrels, tracks, and the component if required.” M4 · CO4 · MCQ

A. Substrate
B. Desoldering
C. Cold joint
D. Damaged track

Answer: B — Desoldering.

13. Which term best matches this statement: “A solder bridge unintentionally connects adjacent conductive points.” M4 · CO4 · MCQ

A. Solder bridge
B. Lifted pad
C. PCB structure
D. Tracks

Answer: A — Solder bridge.

14. Which term best matches this statement: “A lifted pad separates from the substrate due to excessive heat, force, repeated rework, or poor adhesion.” M4 · CO4 · MCQ

A. Soldering safety
B. Solder mask
C. Single-sided PCB
D. Lifted pad

Answer: D — Lifted pad.

15. Which term best matches this statement: “A printed circuit board mechanically supports components and electrically interconnects them with patterned copper conductors.” M4 · CO4 · MCQ

A. Vias

B. FR-4

C. Purpose of a PCB

D. Copper layer

Answer: C — Purpose of a PCB.

Fill in the blanks — 10

1. The common rigid PCB substrate is ______. M4 · CO4 · Fill

Answer: FR-4

2. A plated hole connecting layers is a ______. M4 · CO4 · Fill

Answer: via

3. The protective coating over most copper is solder ______. M4 · CO4 · Fill

Answer: mask

4. The printed component legend is called ______. M4 · CO4 · Fill

Answer: silkscreen

5. The shortest distance through air is electrical ______. M4 · CO4 · Fill

Answer: clearance

6. The shortest path along an insulating surface is ______. M4 · CO4 · Fill

Answer: creepage

7. Flux promotes solder ______. M4 · CO4 · Fill

Answer: wetting

8. A short between adjacent pads made by solder is a solder ______. M4 · CO4 · Fill

Answer: bridge

9. A separated copper pad is called a ______ pad. M4 · CO4 · Fill

Answer: lifted

10. ESD stands for electrostatic ______. M4 · CO4 · Fill

Answer: discharge

True or false — 10

1. A via connects copper between PCB layers. M4 · CO4 · True/False

Answer: True. It is a plated interlayer connection.

2. Silkscreen is the main hazardous-voltage insulation. M4 · CO4 · True/False

Answer: False. It is a marking layer, not primary insulation.

3. Clearance and creepage are identical. M4 · CO4 · True/False

Answer: False. Clearance is through air; creepage is along a surface.

4. A good solder joint requires heating only the solder wire. M4 · CO4 · True/False

Answer: False. The pad and lead must be heated.

5. More solder always improves reliability. M4 · CO4 · True/False

Answer: False. Excess solder can hide defects and cause bridges.

6. A lifted pad can result from excessive heat or force. M4 · CO4 · True/False

Answer: True. Thermal and mechanical stress cause delamination.

7. Flux fumes should be extracted. M4 · CO4 · True/False

Answer: True. Fume control is required.

8. A continuity beep proves a track can carry rated current. M4 · CO4 · True/False

Answer: False. It only indicates a low-resistance path within meter threshold.

9. ESD damage can occur without visible evidence. M4 · CO4 · True/False

Answer: True. Latent or immediate damage may be invisible.

10. A hot iron may be placed on the bench for convenience. M4 · CO4 · True/False

Answer: False. It must be returned to an approved stand.

Viva questions — 10

1. What is the most important practical point when dealing with Silkscreen? M4 · CO4 · Viva

Answer: Verify component designators against the circuit diagram and BOM. Also remember: Silkscreen polarity can be wrong on poor-quality boards; confirm with design data.

2. What is the most important practical point when dealing with Multilayer PCB? M4 · CO4 · Viva

Answer: Use design files or X-ray methods for full visibility. Also remember: Do not assume an apparently unused hole or area has no internal connection.

3. What is the most important practical point when dealing with Track width? M4 · CO4 · Viva

Answer: Use a documented design rule rather than guessing. Also remember: Neck-down sections and connector pads can be the weakest point.

4. What is the most important practical point when dealing with Power tracks? M4 · CO4 · Viva

Answer: Check connector, fuse, track, and return-path ratings as one chain. Also remember: A wide positive track with a narrow return track is still a weak circuit.

5. What is the most important practical point when dealing with Drilling? M4 · CO4 · Viva

Answer: Drill perpendicular and inspect annular rings. Also remember: Fibreglass dust is hazardous; avoid inhalation and uncontrolled dry brushing.

6. What is the most important practical point when dealing with Solder wire? M4 · CO4 · Viva

Answer: Use only the approved alloy for the product and process. Also remember: Lead-containing solder requires strict hygiene and process control.

7. What is the most important practical point when dealing with Tip tinning? M4 · CO4 · Viva

Answer: Tin the tip after cleaning and before storage. Also remember: A black dry tip transfers heat poorly and should be restored according to manufacturer guidance.

8. What is the most important practical point when dealing with Solder bridge? M4 · CO4 · Viva

Answer: Remove with clean tip, wick, or controlled rework and re-inspect. Also remember: Power must remain off until the bridge and any hidden splashes are cleared.

9. What is the most important practical point when dealing with ESD precautions? M4 · CO4 · Viva

Answer: Verify the ESD workstation before opening protected components. Also remember: A wrist strap must use approved resistance and must not be used near exposed hazardous voltage.

10. What is the most important practical point when dealing with Solder mask? M4 · CO4 · Viva

Answer: Inspect for lifted or burnt mask around repaired joints. Also remember: Solder mask is not primary electrical insulation for hazardous spacing.

Numerical problems — 5

1. A round pad diameter is 2.4 mm and finished hole is 1.0 mm. Find annular-ring width. M4 · CO4 · Numerical

Solution: $(2.4-1.0)/2=0.7$ mm.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

2. A board has 200 joints and 4 defects. Find first-pass defect percentage. M4 · CO4 · Numerical

Solution: $4/200\times100=2\%$.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

3. A repaired 0.2 Ω path carries 1.5 A. Find voltage drop. M4 · CO4 · Numerical

Solution: $V=IR=1.5\times0.2=0.3$ V.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

4. A soldering station uses 60 W for 20 minutes. Find energy in Wh. M4 · CO4 · Numerical

Solution: $E=60\times(20/60)=20$ Wh.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

5. A 24 V test circuit is limited by 1.2 kΩ. Find maximum current ignoring load drop. M4 · CO4 · Numerical

Solution: $I=24/1200=0.02$ A=20 mA.

Suggested marking: formula 1; substitution 1; calculation 1; unit/final answer 1.

Diagram-based questions — 5

1. Draw and label PCB layers. M4 · CO4 · Diagram

Expected labels: Silkscreen, mask, copper, substrate and via.

2. Sketch a single-sided PCB track and pad. M4 · CO4 · Diagram

Expected labels: Component side/solder side orientation and copper path.

3. Draw a good through-hole solder joint. M4 · CO4 · Diagram

Expected labels: Lead, pad, hole and concave fillet.

4. Sketch three solder defects. M4 · CO4 · Diagram

Expected labels: Bridge, insufficient solder, dry/cold joint.

5. Draw the PCB fabrication flow. M4 · CO4 · Diagram

Expected labels: Layout, transfer, etch, clean, drill, mount, solder, inspect.

Application-based questions — 5

1. A repaired track opens again under load. What was probably missed? M4 · CO4 · Application

Model answer: Current rating, copper loss, mechanical support, root cause, secure wire link and thermal stress.

2. A solder joint looks shiny but is intermittent. What should inspection include? M4 · CO4 · Application

Model answer: Wetting to pad/lead, movement, cracks, contamination, barrel connection and continuity under gentle mechanical test.

3. A connector pad lifts during rework. State corrective action. M4 · CO4 · Application

Model answer: Stop heating/force, assess track/via damage, use approved repair, restore mechanical and electrical integrity, then verify.

4. Why may a multilayer board show continuity where no visible track exists? M4 · CO4 · Application

Model answer: The connection can run through a via and internal copper layer.

5. What should be checked before powering a newly soldered PCB? M4 · CO4 · Application

Model answer: Correct parts, polarity, bridges, values, connector orientation, continuity, shorts, supply current limit and safety spacing.

EXAMINATION PREPARATION

Practice Papers, Practical Tasks and Last-Minute Revision

PRACTICE MODEL PAPER — NOT AN OFFICIAL SITTTR QUESTION PAPER

Series Examination I · Modules I and II · 50 Marks · 2 Hours

Part A — 1 mark each (4 marks)

1. Define frequency.

2. State the correct connection of an ammeter.
3. What does the tolerance band of a resistor indicate?
4. Write the formula for RC time constant.

Part B — 3 marks each (18 marks)

1. Differentiate AC and DC with three points.
2. Explain the main controls of a DSO.
3. Calculate frequency from a period of 5 ms.
4. Explain four-band resistor colour coding.
5. Compare series and parallel resistor connections.
6. Explain transformer working principle.

Part C — 7 marks each; answer one from each pair (28 marks)

1. (a) Explain sinusoidal signal parameters with a labelled waveform. **OR** (b) Explain DMM, function generator and DSO use with safety precautions.
2. (a) Explain capacitor types, markings and series/parallel combinations. **OR** (b) Explain voltage-divider and RC timing circuits with formulae and diagrams.

Model-answer guide

Use the relevant module question-bank marking points. Numerical: T=0.005 s, so f=200 Hz. Every long answer must include definition, principle, labelled diagram, formula/application and safety or limitation.

PRACTICE MODEL PAPER — NOT AN OFFICIAL SITTTR QUESTION PAPER

Series Examination II · Modules III and IV · 50 Marks · 2 Hours

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Part A — 1 mark each (4 marks)

1. Define depletion region.
2. Name the majority carrier in P-type material.
3. What is a via?
4. State one cause of a cold solder joint.

Part B — 3 marks each (18 marks)

1. Explain N-type and P-type semiconductors.
2. Differentiate forward and reverse bias.
3. Explain static and dynamic diode resistance.
4. Classify PCB types.
5. Explain clearance and creepage.
6. List soldering tools and their uses.

Part C — 7 marks each; answer one from each pair (28 marks)

1. (a) Explain PN-junction formation and V-I characteristics. **OR** (b) Explain Zener diode, LED and photodiode with applications.

2. (a) Explain PCB fabrication from layout to inspection. **OR** (b) Explain reliable soldering, desoldering and five common defects.

Model-answer guide

Use carrier diagrams for semiconductor answers and a process flow for PCB/soldering answers. Include current limiting, polarity, ESD and workmanship points where relevant.

PRACTICE MODEL PAPER — NOT AN OFFICIAL SITTTR QUESTION PAPER

End-Semester Theory Examination · 60 Marks · 2.5 Hours

Part A — Answer all; 1 mark each (8 marks)

1. Define electric power.
2. Write the relation between f and T.
3. What is resistor tolerance?
4. State one application of an isolation transformer.
5. What is doping?
6. Identify the terminal marked by the band on an axial diode.
7. Define PCB pad.
8. What is a solder bridge?

Part B — Answer two of three from each module; 3 marks each (24 marks)

1. Explain analog and digital signals.
2. Explain DMM current-measurement precautions.
3. Calculate power and energy for a 24 V, 2 A load operating for 30 minutes.
4. Explain resistor colour code.
5. Compare capacitor series and parallel combinations.
6. Explain voltage-divider loading.
7. Explain intrinsic and extrinsic semiconductor.
8. Explain forward and reverse bias.
9. Explain LED current-limiting resistor selection.
10. Explain PCB layers.
11. Explain good and defective solder joints.
12. Explain ESD precautions.

Part C — Answer one from each choice set; 7 marks each (28 marks)

1. Explain electrical quantities and sinusoidal waveform parameters with solved examples. **OR** Explain electronic instruments and safe measurement practice.
2. Explain passive components and their testing. **OR** Explain transformer, voltage divider and RC timing circuits.
3. Explain PN-junction formation and diode V-I characteristics. **OR** Explain Zener diode, LED and photodiode.
4. Explain PCB fabrication and layout basics. **OR** Explain soldering tools, procedure, defects and repair checks.

Selected model solutions
Power/energy numerical: P=VI=24×2=48 W. Time=0.5 h. E=48×0.5=24 Wh, or 86,400 J.
LED resistor: state R=(VS−VF)/I, select safe standard value, verify resistor power and supply tolerance.
Long-answer rule: include accurate labelled diagram, principle, formula/characteristics, application, and safety/limitation.

Practical-test model tasks

Instrument setup task

Generate a stated sine wave and verify amplitude/frequency on DSO without grounding errors.

Evaluation: write-up, setup, observation/result, viva and record.

Resistor network task

Decode given resistors, construct a network and compare calculated/measured equivalent resistance.

Evaluation: write-up, setup, observation/result, viva and record.

RC task

Build a timing circuit and determine one time constant from a waveform.

Evaluation: write-up, setup, observation/result, viva and record.

Diode task

Identify terminals, construct forward-bias circuit and take safe V-I readings.

Evaluation: write-up, setup, observation/result, viva and record.

Zener task

Demonstrate reverse breakdown with calculated current limiting.

Evaluation: write-up, setup, observation/result, viva and record.

PCB identification task

Classify board, label layers/features and trace one net.

Evaluation: write-up, setup, observation/result, viva and record.

Soldering task

Produce two accepted joints, identify a defect sample and perform safe rework.

Evaluation: write-up, setup, observation/result, viva and record.

Important diagrams

Labelled sine wave; voltmeter/ammeter connections; resistor bands; series/parallel networks; capacitor and transformer construction; RC circuit; energy bands; N/P material; PN junction; forward/reverse bias; diode and Zener curves; LED circuit; PCB layers; PCB fabrication flow; good and defective solder joints.

Last-minute common-error list

- Wrong units or unconverted prefixes.
- Current measured in parallel or voltage measured in series.
- DMM lead left in current terminal.
- Peak confused with peak-to-peak or RMS.
- Resistance/capacitance measured in an energized circuit.
- Electrolytic or diode polarity reversed.
- No current-limiting resistor for diode, Zener or LED.
- Static and dynamic resistance confused.
- Clearance and creepage treated as identical.
- Solder quantity used to hide poor wetting.